Default Risk Surprises in Banks' Loan Pricing*

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ABSTRACT

In the loan market, similar firms pay different rates, even when they borrow the same loan type from the same bank. Do these differentials reflect market imperfections, or information banks have and the econometrician does not? Using two million French loans from 2006–2024 matched to firms' credit ratings assigned by Banque de France, I show that rate differentials predict subsequent changes in firms' credit risk. On average, firms whose default probability increases by 1 p.p. one year after loan issuance were charged an otherwise unexplained 0.04 p.p. rate premium at issuance. The premium doubles for younger and smaller firms, strengthens in crises and at longer maturities. The positive relationship is consistent with banks having information that the econometrician does not observe; yet, it can also emerge in a market with search frictions, when banks' pricing strategies induce partial separation. I bring these forces together in a search model of loan pricing with asymmetric information in which banks see an imperfect public signal of firms' risk types. Calibrating the model within each rating class, I find that banks are informed and separate borrowers in the safest classes but are uninformed and pool them in the riskiest, but search frictions explain most of the dispersion.

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1 Introduction

Credit markets exhibit substantial variation in interest rates across firms and banks. Much of this variation can be explained by macroeconomic conditions and observable characteristics at loan origination, including borrower credit risk, loan contract terms, and bank-firm relationships.[1] However, sizeable interest rate differentials persist even among otherwise similar firms and loan products. Studies across several countries document that at least 15% of interest rate variation remains unexplained.[2] Because these differentials persist after controlling for observable borrower risk, the literature typically attributes them to market imperfections rather than to differences in borrowers' fundamentals. However, these differences could also reflect information banks have and econometricians do not. This paper shows that market imperfections explain most of the variation, but rates still reflect this information.

The distinction matters for measurement and policy. If residual differentials reflect frictions, similar firms pay different prices for the same risk, and residual dispersion is a measure of market power. If instead they reflect information, banks are pricing risk that the econometrician does not observe, and measures of frictions based on residual dispersion are overstated. For policymakers, which mechanism dominates matters for assessing the value of public credit ratings and credit registries, for designing competition policy in local banking markets, and for firms' access to credit. Finally, it matters for firms: we do not know whether paying higher rates because of frictions harms their later financial health. I address this question through a combination of empirical analysis and theoretical modelling.

On the empirical side, I look at how these differentials relate to future changes in firms' credit risk, and what this relationship reveals about the roles of information and search frictions in loan pricing. I find a positive relationship between residual interest rates and future changes in credit risk. This relationship is substantially stronger for firms in safer credit-rating classes, for younger and smaller firms, and for loans with longer maturities. Three mechanisms could generate this pattern: banks' information about borrower quality, endogenous sorting driven by search frictions, or a causal effect of paying a higher rate on financial health. I rule out the last one empirically. Distinguishing between the first two requires a model, which can clarify what residual interest rate dispersion reveals about credit markets and what it implies for borrowers.

On the theoretical side, I develop a model of loan pricing that combines both mechanisms: search frictions and asymmetric information, with banks observing an imperfect signal of borrowers' risk types. Taking the model to French loan-level data, I find that both information and search frictions are needed to explain the empirical patterns, but their relative importance varies across credit rating classes: banks appear to distinguish between borrowers within safer rating classes, while pooling borrowers in riskier ones. Public ratings therefore seem to capture borrower risk to different extents across rating classes.

I now describe each step of the analysis in more detail, starting with the data. I construct a dataset covering 2006–2024 by merging granular administrative data on business loans with information on bank-firm relationships and on market competition at the department level, which I recover

[1] See Altavilla et al. (2024).

[2] Evidence from Portugal (Ferreira and Kozeniauskas, 2026), the US (Amiti et al., 2026), Italy (Bocola et al., 2026), and Brazil (Cavalcanti et al., 2023).

from observed relationships in the credit registry. For each loan, I observe a rich set of firm and contract characteristics at issuance. I regress loan rates on observables and isolate the unexplained component of interest rates. Residual dispersion is large, around 1 percentage point on average, varies over time and across firm characteristics, and is lower in more concentrated markets.[3]

To test whether residual rates reflect information about borrowers' creditworthiness, I merge these data with Banque de France credit ratings and use the evolution of each firm's default probability after issuance as a proxy for its true type at issuance. I map each rating category to a default probability, given by the historical three-year default frequency in that class. As firms move between rating categories over time, I measure the change in borrower risk as the difference between the default probabilities of the new and the old rating class.

Using public credit ratings has three advantages. First, Banque de France's ratings are a direct measure of firms' creditworthiness, based on balance-sheet data, payment history and annual interviews with managers. Other central banks either rate firms with statistical models based only on hard data, or have analysts rate them but with narrower coverage; the Banque de France has analysts rate more than 300,000 firms a year (Auria et al., 2021), so its ratings also reflect qualitative information that is not in firms' balance sheets. Second, firms are reviewed every year, so rating transitions provide far more variation than defaults alone, which are rare. Third, rating revisions are realized outcomes rather than banks' reported expectations. Studies that use banks' internal ratings to explain loan rates (Amiti et al., 2026) typically do not test whether those ratings predict default. Testing whether ratings predict default is an important validation exercise, as banks' internal assessments may be biased by regulatory considerations and capital requirements.

Using this measure, I find that loan rates predict future changes in creditworthiness. I regress the loan rate on the subsequent change in the firm's default probability, controlling for observables at issuance, so that the coefficient captures changes in risk that the econometrician could not have predicted. On average, a firm pays 0.04 percentage points more at issuance if its default probability is revised up by one percentage point a year later; the relationship weakens at longer horizons and disappears beyond seven years. Moreover, there is heterogeneity across market segments: the coefficient is positive and significant for firms in ex-ante safer rating classes and absent in riskier ones. It also almost doubles for firms aged 0–4 years relative to firms aged 10 years or more, it is higher for firms with fewer than 10 employees and at longer maturities, and it peaks in 2009–2012. These estimates are likely conservative: banks price expected risk, and only some of the firms they judge riskier are eventually downgraded. Overall, the estimates are strongest where there is more to learn about the borrower.

Of the three potential mechanism the last, a causal effect of paying a higher rate on financial health, can be directly tested for a rule out. I compare firms borrowing at variable versus fixed interest rates and examine whether firms with variable-rate loans are more likely to be downgraded three years later when market reference rates increase. I conduct this test both on the full sample and on the subsample of firms borrowing at the end of 2021, for which the 2022 increase in reference rates was unexpected. In neither case does exposure to higher rates increase the probability of a downgrade.

[3] As measured by the HHI. This last pattern is consistent with search models in which dispersion vanishes as markets approach monopoly.

Overall, this evidence suggests that banks' interest rates anticipate firms' outcomes: banks may hold information about borrowers that the econometrician does not observe, or search frictions may sort borrowers across rates in a way that correlates with risk. The reduced-form evidence cannot distinguish between them, which motivates the model.

I build a model of loan pricing that combines imperfect competition, modeled as search frictions in the manner of Burdett and Judd (1983), with asymmetric information, in which banks observe an imperfect public signal of firms' risk type. Search frictions generate dispersion even among identical borrowers: banks compete, but cannot perfectly undercut each other, and the equilibrium is in mixed strategies. Depending on the signal received and on the degree of adverse selection, banks either pool safe and risky borrowers at low rates, or induce self-selection of risky types by offering high rates only; when the signal is informative enough, banks sort types and charge safer types lower rates on average.

I solve and characterize the model equilibrium analytically. The first result from this exercise is that each mechanism alone can generate both residual dispersion and a positive correlation between rates and risk; yet neither is necessary, provided the other is at play. When banks observe an informative signal, they charge higher rates to firms receiving unfavorable signals, generating a positive relationship between rates and risk. In the absence of informative signals, search frictions can generate the same pattern through banks' equilibrium pricing strategies, which induce partial separation between borrower types. The second result is that information shapes how surplus is distributed between banks and borrowers. Greater information benefits banks and safer firms relative to riskier firms, whose access to credit depends on the severity of adverse selection. When adverse selection is mild and banks lack informative signals, safe borrowers and banks cross-subsidize risky borrowers in a pooling equilibrium. When adverse selection is severe, banks may instead offer only high rates, potentially pricing safe borrowers out of the market altogether.

These model results, in particular the first, frame the quantitative analysis. I calibrate the model separately within each Banque de France rating class to assess which equilibrium configurations are consistent with the empirical evidence and how the mechanisms of competition and information vary with borrower risk. Within a class, firms look identical to the econometrician, and to banks before they observe the signal. The data counterpart of the model's rate is the rate residual: the loan rate net of the controls and fixed effects of the baseline regression. The shares and default probabilities of safe and risky types, and the reservation rate of risky types, come directly from the data on firms whose rating subsequently improves or worsens. Three parameters are pinned down by matching three moments: the mean and variance of rate residuals, and the correlation between rates and subsequent risk. The parameters are the share of firms observing two offers, which governs the competitiveness of the market; the signal precision; and the reservation rate of good types, which contributes to the degree of adverse selection.

The calibration reveals that the degree of competition varies little across rating classes, whereas the informativeness of banks' signals varies sharply. The share of firms receiving two offers is stable, declining slightly in riskier classes. Signal precision is close to one in the safest classes, consistent with a separating equilibrium in which banks price risk that the rating does not capture. It is close to 0.5, that is, uninformative, in the riskiest classes, where adverse selection within the class is mild and banks pool borrowers. Both mechanisms are therefore needed to explain the data, and

their relative importance depends on borrower risk. The key to this result is that rating revisions in the data are small, implying that firms differ little in riskiness within a class. In the eyes of the model, adverse selection is mild, and banks prefer to pool borrowers when the signal is not informative. In this case, search frictions cannot alone generate the correlation, because every firm draws from the same rate distribution irrespectively of its type. The positive correlation observed in better rating classes therefore requires informed banks, and the calibration sets signal precision close to one in these classes.

I also find that, where banks are best informed, information accounts for at most 5% of rate dispersion relative to a benchmark without information. In fact, without search frictions, information alone could not match the variance of residuals. However, this is likely a lower bound. The model measures banks' information against realized outcomes: a firm counts as risky only if its rating actually deteriorates. Banks, however, price expected risk. Among firms they correctly judge to be riskier, only some will be downgraded, and the premium paid by the others is attributed to search frictions rather than to information.

Finally, as a robustness, I depart from the main model and ask whether these precisions are consistent with banks' incentives if information had to be acquired privately and at a cost, rather than observed for free. A signal that only sorts borrowers cannot sustain a fully informed equilibrium. Because search frictions make banks randomize over rates, a bank that stops screening and posts rates where the two types' distributions overlap earns the same expected profit as informed banks while saving the cost of the signal. The calibrated precisions can be sustained if information also lowers the cost of lending, a monitoring effect, provided that this benefit is at least nine times larger in the safest rating classes than in the riskiest.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. In Section 3 I show that interest rate differentials across similar firms borrowing similar loan products from the same bank contain information on future changes in firms' default risk. In Section 4 I develop a search model of loan pricing with asymmetric information in which banks see an imperfect public signal of firms' risk types. In Section 5 I calibrate the model separately for each rating class and show that the information embedded in loan rates is high in the best classes and low in the worst ones. Section 6 asks whether the estimated signal precision is what a bank would choose. Section 7 concludes.

2 Related Literature

This paper stands between the empirical work that measures residual dispersion in loan rates and the theoretical work on loan pricing with imperfect competition and asymmetric information. I set it against the closest paper in each: Amiti et al. (2026) for the empirics, Lester et al. (2019) for the model, and Crawford et al. (2018) for the joint estimation of information and market power.

Residual dispersion in loan rates. Recent papers leverage loan-level data to study the explained and unexplained components of interest rates. Altavilla et al. (2024) decompose Euro Area loan rates and attribute 80–85% of their variation to macroeconomic conditions, bank identity, and firm characteristics, and the remaining variation to the specific features of the contract. I explain a similar share using French loan-level data, and study the remainder. Several papers

in different settings and countries study unexplained dispersion: Cavalcanti et al. (2023) tie it to capital misallocation in Brazil, and the same residual appears in Portugal (Ferreira and Kozeniauskas, 2026), Italy (Bocola et al., 2026) and the United States (Amiti et al., 2026). In the French market, Mazet-Sonilhac (2025) documents search frictions directly. Mortgage markets deliver the same lesson with a direct measure of shopping.[4] Two of these papers examine the role of banks' private information and find that it accounts for only a small share of residual rate dispersion. Amiti et al. (2026) show that internal ratings explain little of the variation in interest rates among otherwise identical firms in a given quarter, and interpret the remaining dispersion as evidence of search frictions. Ferreira and Kozeniauskas (2026) reach a similar conclusion using default risk, interpreting the residual as reflecting heterogeneity in borrowers' search ability. My contribution differs along two dimensions: the measurement of private information and the interpretation of its limited contribution to rate dispersion. First, rather than relying on banks' internal ratings, I use the subsequent revision of borrowers' public credit ratings as a measure of realized credit risk. This measure is less directly affected by the regulatory considerations that may influence banks' internal ratings, including their implications for capital requirements. It also imposes a more demanding empirical test: although banks may anticipate a deterioration in a borrower's creditworthiness, only some borrowers ultimately experience a corresponding revision in their public rating. Second, I use this measure to assess the extent to which interest rates reflect credit risk and find that rates do predict subsequent rating revisions. I then develop a model that jointly accounts for search frictions and information asymmetries to quantify the contribution of information to rate dispersion. The estimated contribution remains small even when the public signal is highly informative, because search frictions determine the width of the equilibrium rate distribution. However, since my measure relies on realized rather than expected credit outcomes, this estimate should be interpreted as a lower bound on the contribution of banks' information. A limited contribution of information to aggregate rate dispersion does not imply an absence of screening: in the safest rating classes, where screening incentives are strongest, I estimate that banks' signal is almost perfect. Beyond these two dimensions, I also provide evidence on the consequences of rate dispersion for firms' outcomes. In a strategy that compares fixed- and variable-rate loans, I find no effect of the rate a firm pays on its future creditworthiness. I borrow the design that Gürkaynak et al. (2022) and Core et al. (2025) use for equity prices and for inflation, but to a different end: to rule out a causal effect of the rate paid on the borrower's own creditworthiness.

Price dispersion as an equilibrium outcome. My paper builds on the literature in which price dispersion arises as an equilibrium outcome even among identical agents. Diamond (1971) states the benchmark: with identical sellers and a common search cost, the only equilibrium is the monopoly price, so dispersion requires that buyers differ in what they see. Varian (1980) shows that when some buyers know every price and the rest know one, no price is a best response, so sellers randomize and identical goods sell at different prices. Burdett and Judd (1983) obtain this when buyers differ in how many quotes they sample, and Stahl (1989) under sequential search, with monopoly and competitive pricing as the two limits; Baye et al. (1992) characterize the mixed-

[4] Woodward and Hall (2012) and Bhutta et al. (2021) trace much of the dispersion in mortgage rates to the number of quotes a borrower collects and to financial sophistication. Coen et al. (2024) find that UK lenders price-discriminate through the menus they build rather than on borrower risk, which makes menu design a third source of dispersion next to search and screening. I compare loans with similar contract terms inside a lender and quarter, and I find no trace of the rate-quantity relation that menus imply (Table B.10), so I abstract from it.

strategy equilibria of this class.[5] I add to this structure two borrower types, which differ in risk and in their reservation rate, and a signal that allows banks to sort borrowers. The equilibrium is still dispersed, but with some differences. First, the bank's break-even rate varies with the type, so when banks meet multiple types the equal-profit condition that pins down the rate distribution depends on the average cost of the borrowers who accept each rate. The share of types met moves along the distribution: as good borrowers refuse rates above their reservation rate, the support splits at that rate. Below it banks lend to both types, above it only to bad types, and whether the lower part is offered to both depends on how severe adverse selection is. Second, the signal is binary, and its realization changes the share of types that the bank expects to meet, so banks post signal-specific rates. As a result, the equilibrium consists of two rate distributions, one for each signal, rather than one. I build on the approach of Lester et al. (2019), who add screening through menus to a market with search frictions and adverse selection. I model banks as sorting borrowers through the interest rates they offer, the instrument I observe in the data, and through a public signal that gives an imperfect indication of firms' types, rather than through price-quantity menus. This formulation allows me to incorporate asymmetric information while retaining a framework that can be taken directly to the data. My theoretical framework delivers the market failures typically associated with asymmetric information, such as mispricing and credit rationing (Akerlof, 1970; Rothschild and Stiglitz, 1976; Stiglitz and Weiss, 1981), but in a context of imperfect competition and price dispersion. Guerrieri et al. (2010) show that directed search can also produce price dispersion, when agents trade off the price they pay against the speed of trade. In the mortgage market, Agarwal et al. (2024) combine search and lender screening in a structural model of mortgages and call for the extension to business loans, which this paper takes up.

Asymmetric information and competition in banking. My work closely relates to Crawford et al. (2018). They estimate adverse selection and lender market power jointly in the Italian market for small business credit lines, and find that market power dampens the rise in rates due to adverse selection. Adverse selection is measured as the correlation between the unobserved drivers of borrowing (take-up and amount drawn) and of default, while the positive correlation between rates and default reveals moral hazard. Their objective is identifying firms' private information; to that end they control for firm fixed effects, which absorb banks' soft information in pricing. Banks' soft information is exactly the object of my paper. In my model, banks observe a public signal and use the rate as a screening device. The positive rate-risk correlation arises either when banks are informed or when banks induce self-selection of risky borrowers into high rates. I disentangle the two in the data, and I find that how much banks know varies with the ex-ante riskiness of the firm. I also show how the strength of the correlation varies in the cross-section, particularly for young and small firms, and how it varies over the time period, particularly during crises.

In Section 6 I ask whether the equilibrium I estimate can result from an information acquisition game, and find that market power is what gives banks a reason to screen, since they recover most of the surplus that adverse selection destroys. I treat the search friction as exogenous, while the incentive to screen is not: it depends on competition through expected profit. The trade-off behind

[5] The same framework is used well beyond bank credit: Duffie et al. (2005) show how search costs generate spreads and dealer rents in over-the-counter markets, Farboodi et al. (2025) that intermediaries earn rents because search gives them power over their clients, and Bech and Monnet (2015) apply random search to the dispersion of overnight rates in the interbank market.

Section 6 is theorized in this literature. Hauswald and Marquez (2006) show that competition lowers the return on screening technology and leads banks to under-screen, and Dell'Ariccia and Marquez (2006) that banks cannot recoup screening costs when rivals can poach the borrowers they identify, which loosens standards in lending booms. I ask the quantitative version of their condition: what a signal must be worth for the precision I estimate to be the precision a bank would buy. The empirical side of this literature establishes that banks hold information that is not public,[6] that such information is itself a source of market power,[7] and that market power has other sources as well.[8]

Dispersion and the pass-through of policy rates. Finally, this paper also hints at implications for monetary policy transmission. I show how the distribution of interest rate residuals responds to monetary policy shocks.[9] The model rationalizes why dispersion makes pass-through uneven across borrowers, by showing that a rise in the funding cost of banks shrinks the distribution, assuming that firms do not adjust the maximum rate they are willing to pay in the short term. Puglisi (2023) documents uneven pass-through in US auto loans, and Delis et al. (2025) that tighter conditions fall on weaker borrowers. In addition, the model predicts weaker transmission, as in the literature studying market power in deposits (Drechsler et al., 2017), in mortgage origination (Scharfstein and Sunderam, 2016) and in corporate lending (Wang et al., 2022). Therefore, the decomposition of the calibration is of direct relevance to policy making. I find that most of the residual comes from imperfect competition rather than from risk pricing, so these effects may run through a large part of the rate distribution. However, this paper also highlights the distributional consequences of information, and draws policy makers' attention to measuring the degree of adverse selection and banks' information.

3 Empirics

I have combined multiple administrative datasets on business loans, bank-firm relationships and public credit ratings assigned by the French central bank. The resulting loan-level dataset collects all loan terms, firm financial and demographic information and the evolution of default probability after loan issuance. I use this dataset to compute interest rate residuals after controlling for observables at loan issuance, and I show that they contain information about future unexpected changes in firms' default probability each year after loan issuance. This correlation is positive and

[6] Petersen and Rajan (1994) show that relationships improve credit availability and terms for small firms; Claessens et al. (2025) isolate the private-information component of internal ratings and find that it moves rates, collateral and maturity; Agarwal and Hauswald (2010) show that proximity to the borrower improves pricing accuracy; and Flanagan (2025) estimates that screening costs are embedded in rates and are larger for opaque borrowers.

[7] On the hold-up problem, Sharpe (1990) shows that a bank that learns about its borrower weakens the borrower's outside option, and Rajan (1992) studies the resulting choice between informed and arm's-length debt; Ioannidou and Ongena (2010) document discounts for firms that switch bank, and Cahn et al. (2024) that disclosing credit ratings attenuates the hold-up. Dell'Ariccia (2001) shows that the incumbent's informational advantage works as a barrier to entry, and Carletti et al. (2024) review the interaction between market power and information in banking.

[8] Concentrated deposit markets let banks keep spreads wide (Drechsler et al., 2017); Degryse and Ongena (2005) document spatial price discrimination against distant borrowers, who search less; and Petersen and Rajan (1995) and Boot and Thakor (2000) show that competition reduces relationship lending and the incentive to invest in it. For a survey, see Boot (2000); Berger and Udell (1995) and Berger and Udell (2002) give the foundational evidence on small firms.

[9] The shock series in Fig. B.9 is built from the high-frequency Euro Area surprises of Altavilla et al. (2019).

significant, particularly at shorter horizons. Moreover, it is heterogeneous across market segments, particularly strong for young, small firms, during financial crises and at longer maturities. I test whether the direction of causality runs from rates to rating worsening in a quasi-experimental setting. Overall, this evidence suggests that forward-looking information embedded in rates reflects information anticipated by banks at the time of loan issuance.

3.1 Data

Loan level data. The main dataset is a quarterly survey of all new loan contracts and credit lines to businesses in the first month of each quarter from 2006 Q1 to 2024 Q2 ("*Nouveaux Crédits*", provided by Banque de France). It includes anonymized bank identifiers, at the bank unit level, a wide range of loan terms (rate, amount, maturity, loan purpose, fixed or variable rate, pledged guarantees), firm-level identifiers and information on firms' demographics (age, sector, location, group affiliation), and financial and credit information (including balance-sheet size, debt structure, credit rating score, short-term due repayments, proportion of drawn and undrawn credit lines, revenues).

Bank-firm relationships. I recover information on bank-firm relationships from the French credit register ("Service Central des Risques", SCR). It collects monthly data on outstanding loans of at least €25,000 and credit lines granted by credit institutions and investment firms to companies. I measure the duration of an existing bank–firm relationship as the difference between the year of loan issuance and the year in which the firm first appears in the register. I also identify whether a firm maintains relationships with multiple banks in the month preceding loan issuance. Finally, I calculate concentration indexes at the department–quarter level using data from the end of the month preceding loan issuance.

Public credit ratings. The dataset "COTA" contains the credit rating assigned annually by Banque de France to firms and it represents an assessment of future creditworthiness over a three-year period. The dataset provides monthly observations of firms' most recently assigned credit ratings. Firms are reviewed when their turnover exceeds €750K (€1250K from 2024). As shown in Fig. A.1, the rating scale has 12 rating categories, including one entry for liquidation (P) and one for when there is not enough information to assign a rating (0). Ratings are based on the latest available balance-sheet data, information on payment defaults and delays (CIPE), sector-specific information, and qualitative assessments, including annual interviews with firm managers to evaluate the risks and challenges facing their businesses. This combination of quantitative and qualitative information is a distinctive feature of the Banque de France system. As documented by Auria et al. (2021), national credit assessment systems differ in both their coverage and their assessment methods: while Italy and Spain rely primarily on automated statistical models for the assessment of large numbers of firms, and Germany combines expert assessment with more limited coverage, the Banque de France applies expert evaluation on a large scale, assessing more than 300,000 companies annually. Its ratings therefore combine broad firm coverage with assessments that incorporate information beyond standard accounting measures. This feature is particularly valuable for my analysis, which examines whether loan interest rates reflect information about firms' future credit risk beyond that observable to the econometrician. The original rating scale was replaced by a more granular one in January 2022; to ensure consistency over time, I use Banque

de France's conversion table to map the new scale onto the old one. Table A.1 provides a summary of the three-year-ahead default outcome and rating transitions for firms with a rating in 2021. I will describe more in detail how I use information from the credit ratings in the empirical section.

Cleaning and sample. I focus on fixed-term loan products to private companies, so I exclude credit lines/other types of credit with undefined maturity; I exclude syndicated and regulated loans. I also recover whether a firm is granted multiple loans or other products in the same month. Finally, I set to missing loan-level observations for which total firm assets fall below the 0.1st or above the 99.9th percentile, and express every balance-sheet control and the loan-to-assets ratio as a percentage of total assets. After cleaning, the dataset has 1.9 million observations and 0.9 million unique firms. The average loan is around €1 million, with substantial heterogeneity between rating classes, and it represents between one fourth and one third of the firm's total bank debt.

3.2 Interest Rate Residuals

Baseline approach. I measure interest rate residuals by regressing each loan's interest rate on observables at loan issuance (see Altavilla et al. (2024), Amiti et al. (2026) for the closest approach): I gradually include fixed effects at the macro- (month, lender) and micro- (loan and firm characteristics) levels, and all observed loan and firm characteristics. For each firm-bank-loan-time observation $\text{Rate}_{i,b,l,t}$, I run:

$$r_{ilbt} \;=\; \alpha \;+\; \mathbf{x}'_{ilbt}\,\gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{1}$$

where t is the month, b the bank, i the firm and l the loan. The vector $\mathbf{x}_{ilbt}$ holds what the bank observes at issuance: log loan size, loan over assets, maturity, variable rate dummy, personal loan dummy, individual entrepreneur dummy, firm age and age squared, young firm dummy, new bank-firm relationship dummy, dummies for firms taking several products or several loans in the month, and, as a share of assets, short-term debt, turnover, pledged guarantees, undrawn and drawn credit, bank and non-bank debt. Fixed effects include Month × Bank × Loan type × 2-digit Sector, which captures the supply schedule for similar products for each bank; then I include region, size and rating fixed effects, and a group-membership dummy. The regression sample includes almost half a million observations and 120 thousand firms.

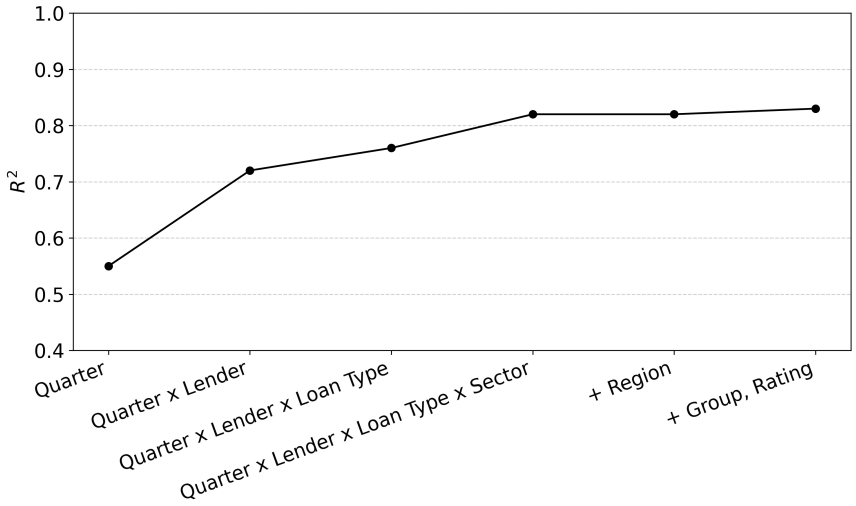


Figure 1: The figure shows the increase in R^2 of the baseline regression after gradually including fixed-effects interactions and the other observables. Observables explain up to 83% of the total variation in rates.

Fig. 1 summarizes the share of variation explained as I progressively add interacted fixed effects and other observables. Overall, observables explain 83% of the total variation in interest rates. Month-bank fixed effects alone explain 70%, while adding loan type times sector accounts for an additional 10 percentage points. The remaining observables contribute little additional explanatory power. The full regression table is reported in Table B.2 in the appendix. In Table B.3, I report a descriptive table of the main loan and firm characteristics for the regression sample, by rating class.

I run alternative specifications in Appendix B: I experiment with other regression models, to account for the fact that by construction there are no rates below zero (Tobit, truncated regression), but with worse explanatory performance (Appendix B.2). I explore two other dimensions: within-firm and within-bank. I include a specification with firm fixed effects (Table B.4 and Fig. B.4), which shows that some within-firm variation persists, although much lower than the comparison across firms; interestingly, the time-series in this case confirms increasing dispersion also within firm in crisis periods. I also run the baseline specification one bank at a time: conditional on observables and loan terms, the distribution of the R^2 of the regression of rates on observables is itself dispersed (Fig. B.5).

Time and cross-sectional distribution of rate residuals. I compute the standard deviation of residuals over time and across firm characteristics. Rate dispersion is substantial, 1pp on average over the time period - a meaningful gap in the context of loan pricing and external financing in general - and varies over the time period, as shown in Fig. B.2.

Rate residuals and competition. I study the distribution of unexplained rate residuals in relation to standard measures of competition. While higher concentration is associated with higher interest rates on average ((Crawford et al., 2018)), I examine how the dispersion of residual interest rates varies with concentration. Using credit registry data, I aggregate lending information across geographical areas and time. Specifically, I sum the amount of drawn credit for each firm at the bank-department-quarter level to assess the relative exposure of each bank in that market, and

I compute a standard Herfindahl-Hirschman Index (HHI) of concentration. Then I compute the standard deviation of residuals by quarter-department. I find that in more concentrated markets firms pay higher rates, but the distribution of rates is less dispersed (Fig. B.8).

Rate residuals and monetary policy. Using high-frequency monetary policy shocks, I show that overall dispersion decreases after a monetary policy tightening: the standard deviation of rate residuals across each lender's loans falls after a rate shock (Table B.5). Appendix B.5 describes how the shocks are recovered and the specification.

3.3 From public credit ratings to default probabilities

Throughout the sample, I observe firms' annual rating reviews, with firms staying in the same class or transitioning to a different one. When a firm is upgraded to a safer rating class, it moves to a class in which firms have a lower expected probability of default. The objective now is to turn a rating update into a change in default probability.

Approach. I compute historical 3-year ahead default frequencies by rating class and year: I take all firms in the same rating category at 01/01/x and compute the share that defaulted by 31/12/x+2. The resulting default frequencies are reported in Fig. 2, which shows two important facts. First, ratings do very well at discriminating across firms' default risk: safer classes have consistently lower default frequencies; they are a reliable measure of credit risk. Second, default frequencies feature time variation, but the relative distance in default frequencies between classes is stable over time. This stability allows me to focus on changes in default probabilities arising from rating transitions, rather than also incorporating changes due to time-varying macroeconomic conditions, by fixing the default probability associated with each rating class to a specific year (I take 2021, the most recent before the rating reform). Specifically, for each firm i, I assign a default probability $\mathbf{P_{i,t}}$ at time t as the default frequency associated with that class in 2021 and $\Delta P_{i,t+k} = \left(P_{i,t+k} - P_{i,t+k-1}\right)$ is the annual default probability change, in percentage points, of firm i between $k-1$ and k years after issuance.

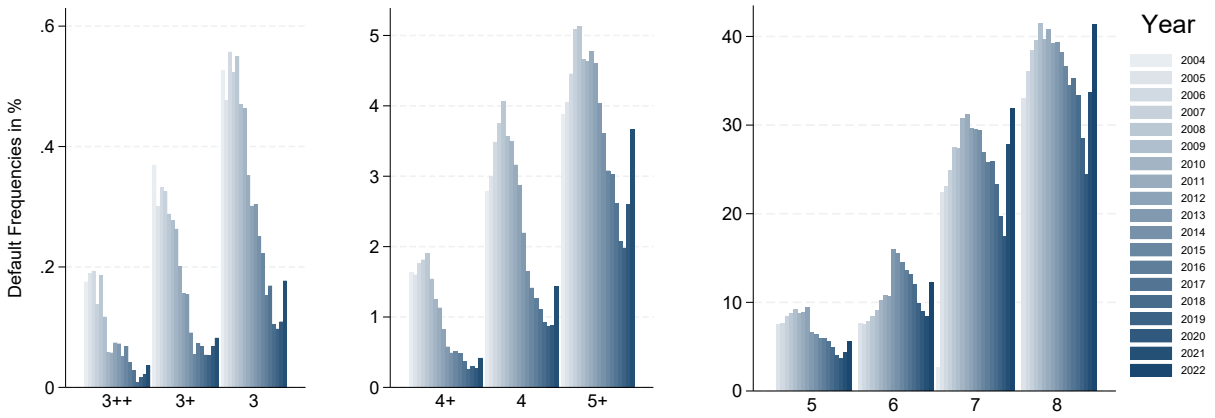


Figure 2: 3-year ahead default frequencies, by rating class and year (2004-2022).

Using rating transitions has three advantages over other measures of realized or perceived riskiness. First, it is an assessment of firms' default risk over a three-year horizon, a more direct measure of

creditworthiness than ex-post balance-sheet indicators or growth outcomes. Second, rating changes are far more frequent than defaults: firms are reviewed every year and can change class several times; for robustness, Appendix B.6.1 in the appendix repeats the test with defaults. Finally, this is a realized outcome rather than the reported subjective expectation of the bank. Other papers include banks' internal ratings to explain interest rates, but typically they do not test whether they effectively predict future default. This is an important validation exercise, as banks' internal assessments may be biased by regulatory considerations and capital requirements.

3.4 Rate residuals and changes in default probability

Consider the following regression:

$$r_{ilbt} \;=\; \alpha \;+\; \sum_{k=1}^{K} \beta_k \, \Delta P_{i,t+k} \;+\; \mathbf{x}'_{ilbt}\,\gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{2}$$

where r_{ilbt} is the rate on loan l granted by bank b to firm i in month t, $\mathbf{x}_{ilbt}$ and $\mu_{bt\tau s}$ are the same controls included when measuring rate residuals; $\Delta P_{i,t+k}$ is the annual default probability change, in percentage points, of firm i between k-1 and k years after issuance - this term is zero if the firm does not change rating class. I set the horizon $K = 10$ and use a balanced sample, so the same firms identify every β_k, and I exclude loans to firms that are already in default when the loan is granted ($P_{i,t} \neq 1$).

The main focus is to assess whether rates contain information on future changes in firms' default probability, as summarized by $\beta_k \neq 0$. Two features of Eq. (2) matter for the reading of β_k. All K surprises enter jointly, so β_k is the correlation with the revision at horizon k holding the revisions at every other horizon fixed. And since the controls and the fixed effects sit in the same regression, β_k equals the coefficient of a regression of the rate residual on the rating surprise residual: it is a statement about the part of the rate the observables do not explain.

I plot the β_k in Fig. 3: the coefficients are positive and significant up to seven years after loan issuance, and fading at longer horizons. An increase in the firms' three-year ahead default probability of one percentage point one year after loan issuance is associated, on average, with an otherwise unexplained 3.7bp rate premium at issuance. The coefficient is reasonable. Davydenko and Franks (2008) find a median bank recovery rate of 56% in French defaults, so let me assume a loss given default LGD = 0.44. With $T = 3$, a bank expecting an increase by one percentage point of the three-year default probability would raise the annual spread by 15 basis points to break even.[10] My estimated β_1 is 3.7 basis points (s.e. 0.9), which is 25% of that. The comparison does not depend on whether default probability is measured over three years or over one. Still assuming a constant one-year default probability, a one percentage point rise in the one-year default probability raises the three-year default probability by 2.97 points, so β_1 is about 11 basis points per point of one-year default probability, against a break-even spread of 44 basis points. The same benchmark predicts that β_k falls with the horizon k, as a revision in year k raises the expected loss only on the part of the loan still outstanding after year k, while the spread is paid on the whole loan from issuance. This evidence suggests that rates contain forward-looking information about the borrower's creditworthiness that is not captured by observable characteristics at the time of loan issuance. I repeat the exercise using actual defaults in Appendix B.6.1. I reverse the regression by putting the rating revision on the left and the rate on the right, focusing on predicting the annual change, with and without a balanced panel, Appendices B.6.2 and B.6.3, or the cumulative change from loan issuance, Appendix B.6.4.

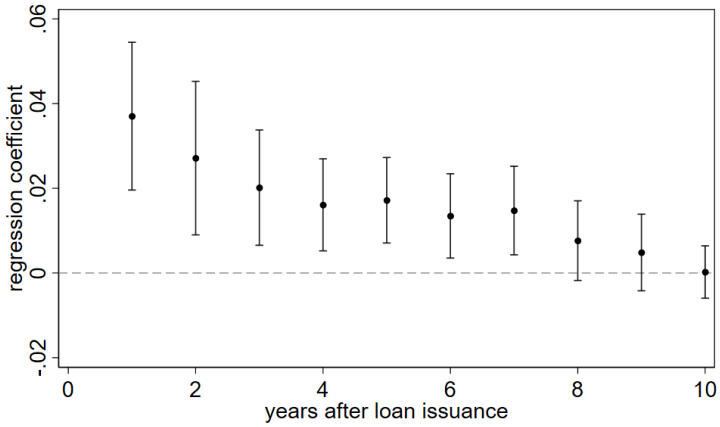


Figure 3: Correlation between rate residuals and unexpected changes in default probability, by year. Balanced sample.

[10] Consider a bank that loses a fraction LGD of the loan when the firm defaults. Let the firm default with a constant annual probability λ, the probability of default within a year for firms that have not yet defaulted, so that the T-year default probability is $P_T = 1 - (1-\lambda)^T$. Each year the bank earns the annual spread s over the funding cost if the loan survives that year and loses LGD if the firm defaults. With a one-year discount factor v and a loan of M years, the bank breaks even when $s \sum_{t=1}^{M} (1-\lambda)^t v^t = \text{LGD}\, \lambda \sum_{t=1}^{M} (1-\lambda)^{t-1} v^t$. The left-hand sum is $1-\lambda$ times the right-hand one, so $s = \text{LGD}\, \lambda/(1-\lambda)$, whatever the discount factor, the maturity and the repayment schedule: the bank breaks even when its annual spread equals about LGD times the one-year default probability. For small default probabilities $\lambda \approx P_T/T$ and $1-\lambda \approx 1$, so a T-year default probability P_T corresponds to a one-year default probability of about P_T/T, and $\partial s/\partial P_T \approx \text{LGD}/T$. At $P_3 = 1$ percentage point and LGD = 0.44, the exact formula gives 14.8 basis points and the approximation 14.7.

3.4.1 Correlation across firms' demographics

I further explore how the forward information contained in rates varies across borrowers' demographics. I separate firms by their Banque de France **rating** at loan issuance into three broad categories, and I document that the positive correlation is driven by firms with a relatively good rating, while it is not significant for relatively worse firms (see Fig. B.18). This finding might be counterintuitive, as riskier firms are closer to the default threshold.

One possibility is that interest rates may induce a worsening of the credit rating by mechanically increasing firm leverage, and in this case I should find a stronger effect when the loan is relatively large. By adding to Eq. (2) interaction terms between changes in default probabilities and **loan share of assets**, I find that the correlation coefficients when comparing the 10th and 90th percentiles of loans over assets are not significantly different (Fig. B.19). I repeat the exercise with the share of bank debt, in case the residual was informative about the rates on other loans, and results are robust. I move to **age interactions**, and I find that the correlation for younger firms (below the 10th percentile of age in my sample) is significantly larger than for older ones (above the 90th percentile), as shown in Fig. B.20. Age is both a proxy for risk/growth prospects (Haltiwanger et al., 2013) and a proxy for time in the credit market (Cloyne et al., 2023), so this finding suggests that the correlation may be stronger when information is more valuable.

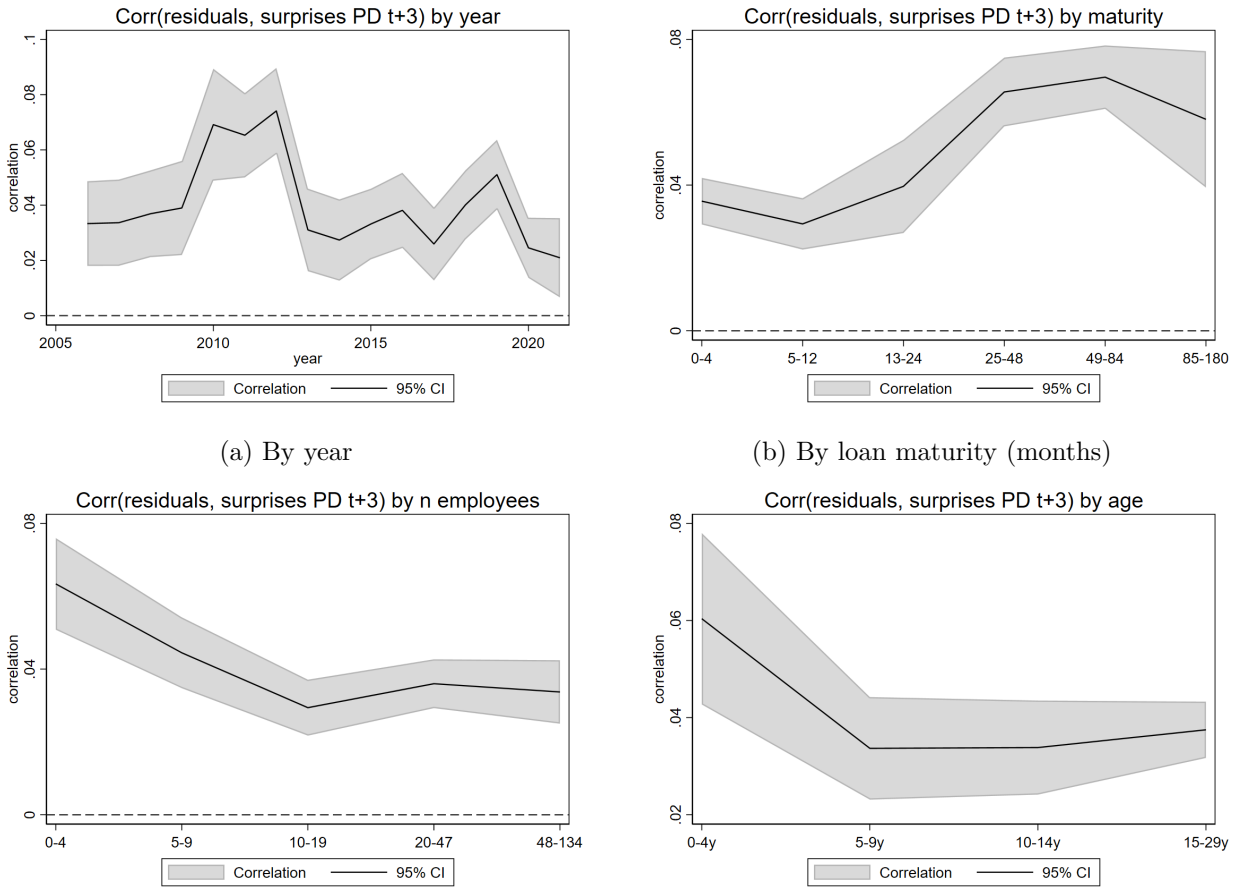
The above methods require many observations, so I rely on a different approach to examine the time dimension and other cross-sectional dimensions: I compute the rate residuals $\tilde{r}_{ilbt}$ and the surprises in rating changes $\widetilde{\Delta P}_{ilbt}$ for the entire population:

$$r_{ilbt} = \alpha^r + \mathbf{x}'_{ilbt}\,\gamma^r + \mu^r_{bt\tau s} + \tilde{r}_{ilbt},$$
$$\Delta P_{i,t\to t+3} = \alpha^P + \mathbf{x}'_{ilbt}\,\gamma^P + \mu^P_{bt\tau s} + \widetilde{\Delta P}_{ilbt},$$

and I extract the pure correlation within each group of firms $\mathrm{Corr}\big(\tilde{r}_{ilbt}, \widetilde{\Delta P}_{ilbt}\big)$.

Fig. 4 shows the results. The correlation varies over **time** (Fig. 4a), peaking around 2009–2012 at the height of the global financial crisis and sovereign debt crisis, a period of heightened uncertainty, and declining during the zero lower bound period. It rises with **loan maturity** up to the 25–84 month range, which also suggests that the value of information tends to be stronger at longer horizons (Fig. 4b). Finally, the correlation is decreasing in firm size, as measured by **number of employees** (Fig. 4c) and age (Fig. 4d): the correlation for firms aged 0–4 years (≈ 0.06) is roughly twice that for firms aged 10 years or more, and similarly higher for firms with fewer than 10 employees relative to larger ones. Younger and smaller firms have shorter credit histories and less public information available.

Overall, these cross-sectional patterns suggest that the informational content of unexplained rate residuals is stronger when information is more valuable: for young and small firms, during the crisis and for longer contracts. In other words, the residuals may reflect *banks' information* about the evolution of firms' creditworthiness. The alternative explanation is that it is the result of a *leverage effect*, with higher rates inducing higher ex-post riskiness through the worsening of firms' fundamentals. Loan-size interactions in Fig. B.19 provide evidence against the leverage hypothesis. However, to rule it out, I adopt a quasi-experimental design, by comparing the effect of a rise in rates between firms borrowing at fixed and variable rates.



(a) By year

(b) By loan maturity (months)

(c) By number of employees

(d) By firm age

Figure 4: *Informational content of rate residuals across borrower and loan characteristics.* Each panel shows the correlation $\hat{\rho}_g$ between the unexplained rate component and the unexpected 3-year change in default probability (ΔPD_{t+3}), both computed in the entire population and correlated by group g. Shaded bands are 95% confidence intervals based on the Fisher z-transformation applied to $\hat{\rho}_g$.

3.5 Test: do higher rates induce rating worsening?

I directly rule out that higher rates cause a credit quality deterioration by testing the transmission of interest rate changes to firms' default probabilities in a quasi-experimental setting, exploiting the interest rate structure of the contract, fixed versus floating. In particular, I compare floating- and fixed-rate loans to test whether firms with variable-rate loans are more likely to transition to a worse rating class when reference rates increase. I find no significant impact of increased debt-servicing costs on subsequent rating worsening.

Baseline. I regress the change in default probability from loan origination t in year 1 to the end of the third year on cumulative increases in reference rates in the first two years from loan issuance. Variation comes from two channels: the change in reference rates is zero for firms borrowing at fixed rates and positive for firms borrowing at variable rates, and for the latter firms it specifically depends on which reference rate the loan is indexed to. In addition, as the intensity of the effect depends on the individual firm's exposure to variable rates, I add interaction terms of the reference rate increase with the loan-to-assets and total bank debt-to-assets ratios, following Gürkaynak et al. (2022) and Core et al. (2025). I restrict the sample to loans with a maturity of at least 30 months,

so that refinancing does not drive the results.

$$\Delta P_{i,t\to t+3} \;=\; \beta\,\Delta\text{Ref}_{l,t\to t+2} \;+\; \delta\left(\Delta\text{Ref}_{l,t\to t+2}\times\text{Share}_{i,t}\right) \;+\; \mathbf{x}'_{ilbt}\,\gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \qquad (3)$$

where $\Delta P_{i,t\to t+3}$ is the change in firm i's default probability over the three years after loan issuance, in percentage points; note that the rating review may happen in any month of year three. $\Delta\text{Ref}_{l,t\to t+2}$ is the overall change in loan l's own reference rate until the end of year two: each floating loan is tied to a specific index (EONIA, Euribor 1m, 3m or 1y, TMO), and a fixed-rate loan is assigned a change of zero. $\text{Share}_{i,t}$ is either loan-over-assets or bank debt-over-assets, proxying the firm's overall exposure to floating rates. $\mathbf{x}_{ilbt}$ holds what the bank observes at issuance: log loan size, loan over assets, maturity, dummies for a variable rate, a personal loan and an individual entrepreneur, firm age and its square, a dummy for a firm younger than three years, and, each scaled by total assets, short-term debt, undrawn credit, pledged guarantees, revenues, bank debt, non-bank debt and mobilized credit, plus region, turnover-class, rating-class and group-status fixed effects. $\mu_{bt\tau s}$ absorbs lender $\times$ month $\times$ loan-type $\times$ two-digit-sector effects. Standard errors are two-way clustered by lender and by month.

The coefficients β and δ compare how default probability varies for the group of firms experiencing a rate increase against the group of firms that did not experience it, depending also on their exposure to variable rates. As shown in Table B.6 (column 5), neither β nor δ is significant. I report seven specifications, using: a floating-rate dummy (column 1), the cumulative rate increase (column 2), the annual increases in reference rates entered separately (column 3), interactions with the share of the loan over assets (columns 4–6) and interactions with the share of the firm's total bank debt over assets (column 7), as a proxy for the firm's total exposure to variable-rate loans.

I check the identifying assumptions in Appendix B.7.1. The main issue is self-selection into fixed- or variable-rate contracts. To mitigate this risk, I take two approaches. First, I show in Table B.7 that the contract choice is predominantly driven by supply-side and sectoral factors: the lender-month dimension alone accounts for approximately 60% of the variation, and the full specification explains up to 91%. Rating transitions for firms taking variable-rate loans do not differ significantly from those of firms taking fixed-rate loans (Fig. B.21). Second, I repeat the exercise on loans issued in 2021q3–q4, at the end of the ZLB period, when firms did not expect a rate hike: I again find no significant effect of reference rate increases on rating worsening (Table B.8), including when the reference rate change accumulates through 2023 (Table B.9).

3.6 From the empirics to the model

Paying a higher interest rate does not worsen the credit rating; this fact, along with the predictive power of residuals being stronger in contexts when information is more valuable, suggests that the correlation between rate residuals and future default documented above reflects information that banks know at issuance. However, this cannot be the sole explanation. Interest rate differentials may be unrelated to firms' risk, and due to other market frictions, as in Amiti et al. (2026), Ferreira and Kozeniauskas (2026). I show indeed that rate dispersion varies with the degree of market concentration, increasing in less concentrated markets. Consequently, I move to a model of loan pricing where information and imperfect competition interact and determine loan rate offers. I model imperfect competition via search frictions, modeled as in Burdett and Judd (1983); Amiti

et al. (2026) use this modeling choice to rationalize interest rate dispersion and quantify search costs. I extend this framework by adding asymmetric information, with firms differing in riskiness. In the context of loan pricing under adverse selection, a common modeling choice is screening via price-quantity menus (Stiglitz and Weiss (1981), Crawford et al. (2018)). Lester et al. (2019) combine search frictions, asymmetric information and screening via menus in a general market setting. In the French loan market, I do not find a positive relationship between rates and quantities[11], so I replace the menu with a public signal of the firm's type; I interpret the signal as the information that banks can learn about a borrower in that market and that they price, which is not observed by the econometrician.

4 Model

In this section, I develop a model of loan pricing that combines search frictions, which generate imperfect competition between banks, with asymmetric information about firms' risk types. I introduce an imperfect public signal of firms' risk types into the price dispersion framework of Varian (1980) and Burdett and Judd (1983), building on the approach of Lester et al. (2019). I model banks as sorting borrowers through the interest rates they offer, the instrument I observe in the data, and a public signal that provides an imperfect indication of firms' types, rather than relying on price-quantity menus. This formulation allows me to incorporate asymmetric information while retaining a framework that can be taken directly to the data. The model generates a mixed-strategy equilibrium in interest rates and allows me to study how search frictions and the public signal jointly shape the relationship between interest rates and default risk. In equilibrium, the positive rate-risk relationship arises for two reasons: a signal precise enough to separate borrowers, or — with no information at all — search frictions that lead banks to randomize between pooling borrowers and inducing self-selection through pricing. In the next section, I bring the model to the data by calibrating it to match the empirical findings.

4.1 Environment

The market consists of a continuum of firms and a continuum of banks, both of measure one. Firms need one unit of capital to undertake projects. A project succeeds with probability p[12] and, when successful, firms get a return $\bar{r}$. Firms repay only when the project succeeds (limited liability). Firms' outside option when not undertaking the project is an infinitely small, strictly positive value $\varepsilon > 0$. Consequently, firms' reservation rate is equal to $\bar{r}$. Banks offer loans of fixed quantity one at an interest rate r. The opportunity cost of bank capital is set by the central bank's policy rate, r^{CB}. Therefore, the competitive break-even rate for banks, c, is defined by the condition that the expected repayment equals the opportunity cost of capital, $c = \frac{r^{CB}}{p}$.

Search Frictions. As in Burdett and Judd (1983) (BJ), each bank meets a fraction π of firms ("shoppers") who compare its offer with one from a competitor bank, while the remaining fraction

[11] I find that, conditional on all observable characteristics in my sample, there is a negative relationship between rates and quantity (Table B.10, column 2); the relationship between rates and loan size relative to assets is not significant (column 1).

[12] It is possible to generalize the model such that a leverage effect is at play, namely $p(r)$ and $p'(r) < 0$, and the main results go through. Given the fixed-versus-floating test of Section 3.5, which supports that higher rates do not raise firms' subsequent default risk, I present the version with constant p.

$1-\pi$ ("captives") only observe its offer. However, banks cannot distinguish shoppers from captives, and the specific number of offers a firm has sampled is private information to the firm. π is constant across firm types, although this assumption can be relaxed. This structure leads to a mixed-strategy equilibrium in interest rates.[13]

Asymmetric Information. The population of firms is divided into two types. A type is denoted as $\theta \in \{g, b\}$, with population shares μ_g and μ_b, where $\mu_b + \mu_g = 1$. Types differ in riskiness, $p_b < p_g$. Consequently, for banks the cost of lending to a bad firm is strictly higher than that of lending to a good firm: $c_g < c_b$, with $c_\theta = r^{CB}/p_\theta$. In the following, I also assume that the reservation rate of good firms is lower than that of bad firms: $\bar{r}^g < \bar{r}^b$. This assumption is consistent with thinking that risky projects are high-risk, high-yield. Finally, I look at the case in which $c_b < \bar{r}^g$; this is a sufficient condition to ensure that the cost of pooling both types is always below $\bar{r}^g$, and that a pooling region exists. As we will see, this structure leads to lending to bad-types only or partially pooling equilibrium in interest rates, depending on the severity of the adverse selection problem.[14]

Signal. Banks observe a public, imperfect signal of firms' type. The signal is binary, $s \in \{s_g, s_b\}$,

$$\Pr(s = s_\theta \mid \theta) = q_\theta \geq 0.5, \quad \forall \theta \in \{g, b\},$$

where s_θ is the signal that matches type θ, and q_θ is the signal precision, the probability of classifying a type θ firm correctly. The value $q = 0.5$ means the signal is completely uninformative, $q = 1$ means it reveals the type. Both precisions are equal, $q_g = q_b = q$. The signal is *public*, so each bank matching with a firm observes the same realization of s.

The shares of firms receiving each signal, σ_g and σ_b, and banks' posterior beliefs about facing a good type, α^{s_g} and α^{s_b}, follow from Bayes' rule:

$$\begin{aligned}
\sigma_g &\equiv \Pr(s = s_g) = q_g\mu_g + (1 - q_b)\mu_b, \\
\sigma_b &\equiv \Pr(s = s_b) = q_b\mu_b + (1 - q_g)\mu_g, \\
\alpha^{s_g} &\equiv \Pr(\theta = g \mid s = s_g) = \frac{q_g\mu_g}{q_g\mu_g + (1 - q_b)\mu_b}, \\
\alpha^{s_b} &\equiv \Pr(\theta = g \mid s = s_b) = \frac{(1 - q_g)\mu_g}{(1 - q_g)\mu_g + q_b\mu_b}.
\end{aligned}$$

By construction, $\sigma_g + \sigma_b = 1$. Note that $\alpha^{s_g} > \mu_g > \alpha^{s_b}$: a good signal increases the belief that the firm is good, while a bad signal decreases it.

Timing. π firms are randomly matched with two banks, $1 - \pi$ with one bank. Firms know their type θ and the number of quotes they receive. Each bank matches with a population of firms of measure one. Banks know all parameters $(\mu_g, p_g, p_b, \bar{r}^g, \bar{r}^b, \pi, q, r^{CB})$, but cannot observe the true type θ, the offers sampled by the firm, nor the quotes of the other bank; each bank reads each firm's signal and quotes a rate, and all banks quote simultaneously. Each firm sees its quotes, accepts the lowest quote if below $\bar{r}^\theta$, rejects otherwise. A shopper with two equal quotes picks each

[13] The same dispersed equilibrium can be obtained with informed and uninformed buyers as in Varian (1980), or under sequential search as in Stahl (1989); see Baye et al. (1992) for the full set of mixed-strategy equilibria in this class of models.

[14] The pooling region and the exit of good types are the counterparts, in my setting, of the two costs of hidden quality identified by Rothschild and Stiglitz (1976) and Stiglitz and Weiss (1981); here, however, separation is obtained through the market's public signal rather than through a contractual screening device as in Bester (1985).

bank with probability 1/2. Finally, projects succeed or fail, and firms repay.

Rate offers and payoffs. Banks maximize their expected profits over rates. $\Delta(\mathbb{R}_+)$ is the set of probability measures on it. The banks' strategy is to offer a (cumulative) distribution over rates for each signal $F = (F^{s_g}, F^{s_b}) \in \Delta(\mathbb{R}_+) \times \Delta(\mathbb{R}_+)$. Firms maximize their expected surplus. A firm of type $\theta \in \{g, b\}$ that borrows at rate r obtains expected payoff $p_\theta(\bar{r}^\theta - r)$, where $p_\theta > 0$ is its project's success probability and $\bar{r}^\theta$ is its reservation rate. A firm that does not borrow obtains zero. Let $R \subseteq \mathbb{R}_+$ denote the set of rate offers received by a firm, with $|R| \in \{1, 2\}$, depending on whether the firm is captive or a shopper. Given R, the firm chooses the lowest offered rate, if lower than their reservation rate, and rejects all offers otherwise. A firm receiving two equal offers chooses each bank with probability one half. A firm indifferent between borrowing and not borrowing accepts the offer.

Definition of Equilibrium. For a given parameter vector $\{\mu_g, p_g, p_b, \bar{r}^g, \bar{r}^b, r^{CB}, \pi, q\}$, a symmetric equilibrium consists of a pair of bank rate distributions $F^* = (F^{s_g,*}, F^{s_b,*}) \in \Delta(\mathbb{R}_+)^2$ and a firm strategy a^* such that:

1. **Firm optimality.** For every firm type $\theta \in \{g, b\}$ and every tuple of received rate offers $R = (r_1, \ldots, r_n)$, where $n \in \{1, 2\}$ and $r_i \in \mathbb{R}_+$, the firm chooses an action that maximizes its expected surplus:
$$a^*(\theta, R) \in \arg \max_{a \in \{r_1, \ldots, r_n\} \cup \{\varnothing\}} U_\theta(a),$$
where $U_\theta(r) = p_\theta(\bar{r}^\theta - r)$ for any received rate r, and $U_\theta(\varnothing) = 0$ denotes the payoff from not borrowing.

2. **Bank optimality.** For each signal $s \in \{s_g, s_b\}$, the bank chooses its rate distribution to maximize its expected profit, taking the rate distributions of competing banks and the firm's strategy as given. In equilibrium, every rate in the support of $F^{s,*}$ maximizes expected profit:
$$\text{supp}(F^{s,*}) \subseteq \arg \max_{r \geq 0} \Pi^s(r; F^*, a^*).$$

Every rate in the support earns the same profit, and no rate outside it earns more. This formulation also embeds firms' choice, as the bank's expected profits are conditional on the firm's acceptance of the offer, which is a function of the firm's type and the number of quotes it has sampled. The equilibrium is symmetric in the sense that all banks post the same distribution for each signal, and all firms follow the same acceptance strategy.

4.2 Properties of Equilibria

Conditional on observing each signal, banks face σ_g firms with good signals and σ_b firms with bad signals, and for each signal the bank updates its beliefs about the shares of types in the population. The bank also takes into account the probability that the offer is accepted, when it is competing over the π shopper firms with another bank. Conditional on receiving signal s, the bank's expected

profits from offering a rate r are:

$$\Pi^s(r) = \begin{cases} [1 - \pi + \pi(1 - F^s(r))]\,(r - c^{avg,s}) & \text{if } r \leq \bar{r}^g, \\ (1 - \alpha^s)\,[1 - \pi + \pi(1 - F^s(r))]\,(r - c_b) & \text{if } \bar{r}^g < r \leq \bar{r}^b, \\ 0 & \text{if } r > \bar{r}^b. \end{cases}$$

Profits are region specific: on the pooling region, below $\bar{r}^g$, both types accept, and the bank's expected cost of lending is $c^{avg,s} \equiv \alpha^s c_g + (1 - \alpha^s)c_b$, where $c_\theta = r^{CB}/p_\theta$. On the separating region, above $\bar{r}^g$, only bad types accept, which the bank expects to meet with probability $(1 - \alpha^s)$.

I show a set of properties of the equilibrium. The equilibrium distributions F^s are continuous, with no mass points (Lemma 4.1), and the highest rate in the support is either $\bar{r}^g$ or $\bar{r}^b$ (Lemma 4.2). Finally, if banks lend on both regions, the support of F^s is disconnected, with a gap between the two regions (Lemma 4.3).

Lemma 4.1 (No mass points)**.** *In a symmetric equilibrium F^s is continuous: no rate is posted with strictly positive probability.*

The intuition behind the lemma is that no bank can optimally post a rate with strictly positive probability. If a rate r were posted with probability $m > 0$, a bank charging r would split shoppers equally with its rival whenever both banks post the same rate. By undercutting r by an arbitrarily small amount ε, the bank would attract an additional fraction $m/2$ of shoppers, while the loss in its margin vanishes as ε goes to zero. Since this deviation increases the probability of acceptance without reducing profits in the limit, a distribution with an atom cannot be an equilibrium. The same logic applies at the lower bound of the separating region: undercutting a rate just above $\bar{r}^g$ brings good types back into the market and reduces the expected cost of lending from c_b to $c^{avg,s}$, making the deviation profitable there as well. Hence, the equilibrium distribution F^s must be continuous. The proof is in Appendix C.1.

Lemma 4.2 (The top of the support is a reservation rate)**.** *In a symmetric equilibrium the highest rate in the support of F^s is either $\bar{r}^g$ or $\bar{r}^b$. Banks earn their equilibrium profit on captive firms alone,*

$$\Pi^s = \max\left\{(1 - \pi)\,(\bar{r}^g - c^{avg,s}),\ \ (1 - \alpha^s)(1 - \pi)\left(\bar{r}^b - c_b\right)\right\}.$$

In a symmetric equilibrium, the highest rate in the support must be one of the two reservation rates, $\bar{r}^g$ or $\bar{r}^b$. At the highest rate in the support, a bank loses every shopper to the rival and earns its equilibrium profit on captive firms alone. If the highest rate were strictly inside either the pooling or the separating region, a bank could raise its rate slightly, continue to trade with the same captive firms, whose reservation rate exceeds the posted rate, and earn a higher margin. Hence, such a rate cannot be optimal. At the two reservation rates, however, raising the rate further entails losing a set of borrowers: above $\bar{r}^g$, good types stop accepting, while above $\bar{r}^b$, no borrowers accept. Therefore, the highest rate in the support must be either $\bar{r}^g$ or $\bar{r}^b$, depending on which reservation rate yields higher expected profits on captive firms alone. The proof is in Appendix C.1.

The top of the support pins down the equilibrium profit, which is earned on captive firms alone. This, with the equal profit condition, pins down the equilibrium distribution on the support.

Lemma 4.3 (The support is disconnected when a bank is lending on both regions)**.** *In a symmetric equilibrium F^s have mass on both regions. In that case, the support of F^s is disconnected. The upper bound of the pooling region is $\bar{r}^g$, and the lower bound of the separating region is $\underline{r_b}^s > \bar{r}^g$. F^s is flat on the gap between them.*

The intuition behind the lemma is that the support of F^s cannot be connected across $\bar{r}^g$. Assume banks offer some rates just above $\bar{r}^g$, where shoppers accept either bank with probability 1/2. A bank could then profitably undercut its rival by offering exactly $\bar{r}^g$: its margin would remain unchanged, while the mass of firms accepting the offer would increase as good firms re-enter the market. Hence, the support of F^s must be disconnected, with a gap between the pooling and separating regions. The proof is in Appendix C.1.

4.3 Construction of equilibrium

I solve the equilibrium in two steps. First, I construct the equilibrium under a perfect signal ($q = 1$), when banks can perfectly recognize types. This case boils down to a Burdett and Judd (1983) per firm type. Second, I outline the equilibrium an imperfect signal ($0.5 < q < 1$), where banks face two subpopulations of firms, one with a good signal and one with a bad signal; banks form posterior beliefs about the composition of each and post a signal-specific distribution of rates. As the signal becomes more precise, the two distributions separate, and at $q = 1$ they are the two type-specific distributions of the perfect-signal case.

4.3.1 Equilibrium under perfect signal (q=1)

The signal matches the type, so the bank recognizes the type of every firm it meets. That is, the bank meets a firm of type θ with probability μ_θ; the bank gets $(r - c_\theta)$ if the firm accepts; the firm accepts with probability 1 if it is captive, and if it observes a higher offer from the other bank when it is a shopper, with probability $(1 - F(r))$. $F(r)$ is the equilibrium rate distribution.

Proposition 4.1. *For $\pi \in (0, 1)$, $q = 1$, and for each type θ, the unique equilibrium $F^\theta(r)$ is such that $F^\theta(r)$ is continuous with compact support $[\underline{r}^\theta, \bar{r}^\theta]$ and satisfies the equal profit condition:*

$$\mu_\theta[1 - \pi + \pi(1 - F^\theta(r))](r - c_\theta) = \mu_\theta(1 - \pi)(\bar{r}^\theta - c_\theta), \quad \forall r \in [\underline{r}^\theta, \bar{r}^\theta].$$

The unique associated equilibrium satisfies:

$$F^\theta(r) = \begin{cases} 0 & \text{if } r < \underline{r}^\theta, \\ 1 - \dfrac{1 - \pi}{\pi}\left[\dfrac{\bar{r}^\theta - r}{r - c_\theta}\right] & \text{if } \underline{r}^\theta \le r \le \bar{r}^\theta, \\ 1 & \text{if } r > \bar{r}^\theta, \end{cases}$$

where the upper bound $\bar{r}^\theta$ is the reservation rate, and the lower bound, from $F^\theta(\underline{r}^\theta) = 0$, is $\underline{r}^\theta = c_\theta + (1 - \pi)(\bar{r}^\theta - c_\theta)$.

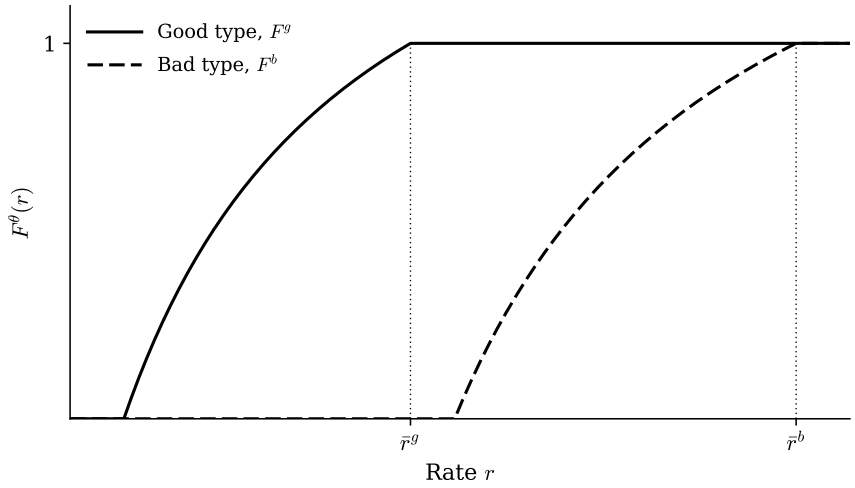


Figure 5: **Equilibrium under a perfect signal.**

Bank optimality requires every rate in the support to earn the same profit. Lemmas 4.1 and 4.2 claim continuity and fix the top of the support at the reservation rate $\bar{r}^\theta$, where the bank loses every shopper and earns $\mu_\theta(1-\pi)(\bar{r}^\theta - c_\theta)$ on its captive firms alone. The equal profit condition pins down the unique equilibrium. Intuitively, no other rate outside the support is more profitable for the bank. Below $\underline{r}^\theta$, acceptance probability is one and profits are linearly increasing in r, so no rate below the lower bound is more profitable than the lower bound itself. Above $\bar{r}^\theta$, firms reject and profits are zero. Fig. 5 shows an example of the rate distribution. The proof of Proposition 4.1 is in Appendix C.3.

Posted rates across types mix the two distributions by population share, $\mu_g F^g(r) + \mu_b F^b(r)$, on the support $\underline{r}^g < r \leq \bar{r}^b$. Rates on the loans actually made follow a different distribution: a shopper pays the lower of two draws, so low rates are over-represented among the loans that get signed. Appendix D.1 derives the map from the posted distribution to the observed one, and the calibration of Section 5 matches moments of the observed one.

In limit cases of monopoly ($\pi = 0$) or perfect competition ($\pi = 1$), the distribution is degenerate. The bank offers to each type the reservation rate $\bar{r}^\theta$ under monopoly, as all firms will always sample only one offer, and the competitive rate c_θ under perfect competition, as all firms will visit both banks and get two rate quotes.

4.3.2 Equilibrium under an imperfect signal ($0.5 < q < 1$)

When $0.5 < q < 1$, the bank reads an informative signal, but not perfect. The signal moves the beliefs, so the bank treats the two signal groups as two separate markets: a share σ_g of firms carries a good signal and a good-type share α^{s_g}, and a share σ_b carries a bad signal and a good-type share $\alpha^{s_b} < \alpha^{s_g}$. Within a group the problem is the one of Section 4.2: the bank posts F^s, lends on the pooling region below $\bar{r}^g$ at the expected cost $c^{avg,s}$, and lends on the bad-only region above $\bar{r}^g$ at the cost c_b, where it meets a borrower with probability $(1-\alpha^s)$.

The adverse-selection index. By Lemma 4.2 the highest posted rate is one of the two reservation rates, and the bank earns its profit there on captive firms alone. It therefore compares the two corners: $\bar{r}^g - c^{avg,s}$ from pooling at the good types' reservation rate, and $(1-\alpha^s)(\bar{r}^b - c_b)$ from

lending to bad types only at their reservation rate. The difference is summarized by the adverse-selection index within each signal group,

$$\phi^s \equiv (\bar{r}^g - c^{avg,s}) - (1-\alpha^s)\left(\bar{r}^b - c_b\right) \ , \tag{4}$$

and the equilibrium profit is $\mathbf{\Pi^s} = (1-\pi)\max\left\{\bar{r}^g - c^{avg,s},\ (1-\alpha^s)(\bar{r}^b - c_b)\right\}$. A positive ϕ^s says adverse selection is mild in the group: the bank earns more by lending to both types at pooling rates. A negative ϕ^s says it is severe, and the bank earns more by lending at high rates to bad types and letting some of the good types exit. A more precise signal raises α^{s_g} and lowers α^{s_b}, so it pushes the good-signal group towards pooling and the bad-signal group towards bad-only lending.

Proposition 4.2 (Equilibrium within a signal group)**.** *For $\pi \in (0,1)$ and the signal realizations $s \in (g,b)$ with precision $q \in [1/2,1]$ the unique symmetric equilibrium $F = \{F^{s_g}(r), F^{s_b}(r)\}$ is such that $F^s(r)$ with $s \in (g,b)$ satisfies:*

$$\begin{aligned} &[1-\pi+\pi(1-F^s(r))](r-c^{avg,s}) = \mathbf{\Pi^s} && \text{if} \quad r \le \bar{r}^g \\ &(1-\alpha^s)[1-\pi+\pi(1-F^s(r))](r-c_b) = \mathbf{\Pi^s} && \text{if} \quad \bar{r}^g \le r \le \bar{r}^b \end{aligned}$$

with $\mathbf{\Pi^s} = (1-\pi)\max\left\{\bar{r}^g - c^{avg,s},\ (1-\alpha^s)(\bar{r}^b - c_b)\right\}$.

The unique associated equilibrium is:

1. Full pooling, $\phi^s > 0$. The support is $[\underline{r}^s, \bar{r}^g]$ and

$$F^s(r) = \begin{cases} 0 & \text{if } r < \underline{r}^s, \\ 1 - \dfrac{1-\pi}{\pi}\left[\dfrac{\bar{r}^g - r}{r - c^{avg,s}}\right] & \text{if } \underline{r}^s \le r \le \bar{r}^g, \\ 1 & \text{if } r > \bar{r}^g, \end{cases}$$

with $\underline{r}^s = c^{avg,s} + (1-\pi)\left(\bar{r}^g - c^{avg,s}\right)$.

2. Hybrid and bad-only, $\phi^s \le 0$. The support is $[\underline{r}^s, \bar{r}^b]$ and

$$F^s(r) = \begin{cases} 0 & \text{if } r < \underline{r}^s, \\ F^s_{pool}(r) & \text{if } \underline{r}^s \le r \le \bar{r}^g, \\ \max\left\{F^s_{pool}(\bar{r}^g),\ F_{bad}(r)\right\} & \text{if } \bar{r}^g < r \le \bar{r}^b, \\ 1 & \text{if } r > \bar{r}^b, \end{cases}$$

where the equal-profit condition on each region gives

$$F^s_{pool}(r) = \max\left\{1 - \frac{1-\pi}{\pi}\left[\frac{(1-\alpha^s)\left(\bar{r}^b - c_b\right)}{r - c^{avg,s}} - 1\right],\ 0\right\},$$

$$F_{bad}(r) = 1 - \frac{1-\pi}{\pi}\left[\frac{\bar{r}^b - c_b}{r - c_b} - 1\right].$$

In the hybrid regime the support starts at $\underline{r}^s = c^{avg,s} + (1-\pi)(1-\alpha^s)\left(\bar{r}^b - c_b\right)$ and has a gap on

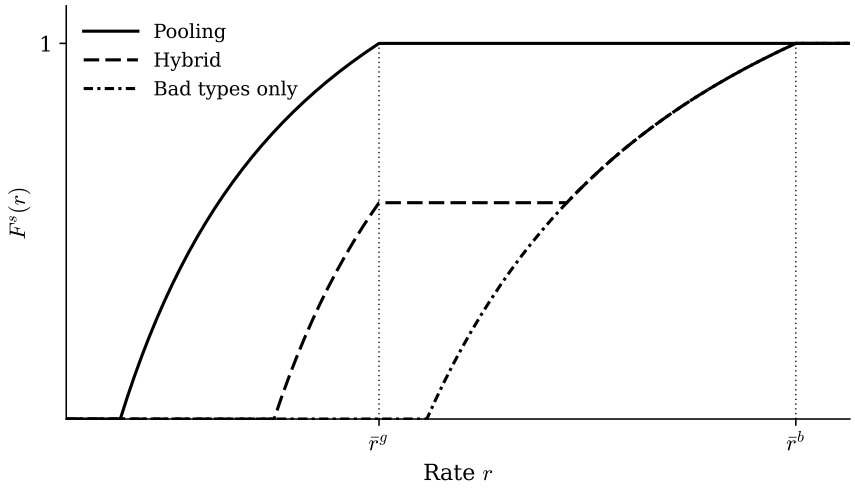


Figure 6: **The three regimes within a signal group.** ϕ identifies the three regimes. Pooling, $\phi^s > 0$. Hybrid, $\underline{\phi}^s < \phi^s \leq 0$. Bad types only, $\phi^s \leq \underline{\phi}^s$.

$\left(\bar{r}^g,\ c_b + \left(\bar{r}^g - c^{avg,s}\right)/(1-\alpha^s)\right)$, *where the first term of the maximum binds (Lemma 4.3). In the bad-only regime* F^s_{pool} *is zero throughout and the support is* $\left[c_b + (1-\pi)\left(\bar{r}^b - c_b\right),\ \bar{r}^b\right]$.

Each branch of the equilibrium is the Burdett and Judd (1983) distribution of a one-type economy, where the belief about the type of the firm determines the cost of lending, and consequently the margin, at the top of the distribution, and the profit level. The hybrid regime is the one case in which the expected margin varies on the two regions, pooling and bad-only, but still satisfies the equal-profit condition. In any case, the shape of the distribution on each region is analogous. Fig. 6 shows an example of the rate under different regimes, which correspond to different values of adverse selection given the signal. Appendix C.3 proves the proposition and rules out deviations in each regime.

Aggregate distribution. Posted rates across the whole population mix the two groups,

$$F(r) = \sigma_g F^{s_g}(r) + \sigma_b F^{s_b}(r).$$

The lower bound of the support is lower when the signal is good, given the higher probability of meeting good types; at the same time, the upper bound is more likely to be higher when the signal is bad, as the lower expected share of good types means a lower ϕ. The mixture adds dispersion between the groups, to the dispersion inside each one, and the rate a firm pays carries information about its type.

Three regimes. The sign of ϕ^s and the size of the search friction determine three possible regimes. When $\phi > 0$, the regime is pooling. The support of $F(r)$ lies below $\bar{r}^g$ and all firms get credit. When $\phi < 0$, two regimes are possible. If $F(\bar{r}^g) > 0$, the regime is hybrid: a fraction of good types $F(\bar{r}^g) > 0$ gets an offer lower than $\bar{r}^g$ and stays in the market, while the remaining offers are in the bad only region. If $F(\bar{r}^g) = 0$, then banks lend only to bad types, and no good firm gets credit. Fig. 7 traces the regime of each group, and the share of good firms it serves, across the index at the prior and the precision of the signal.

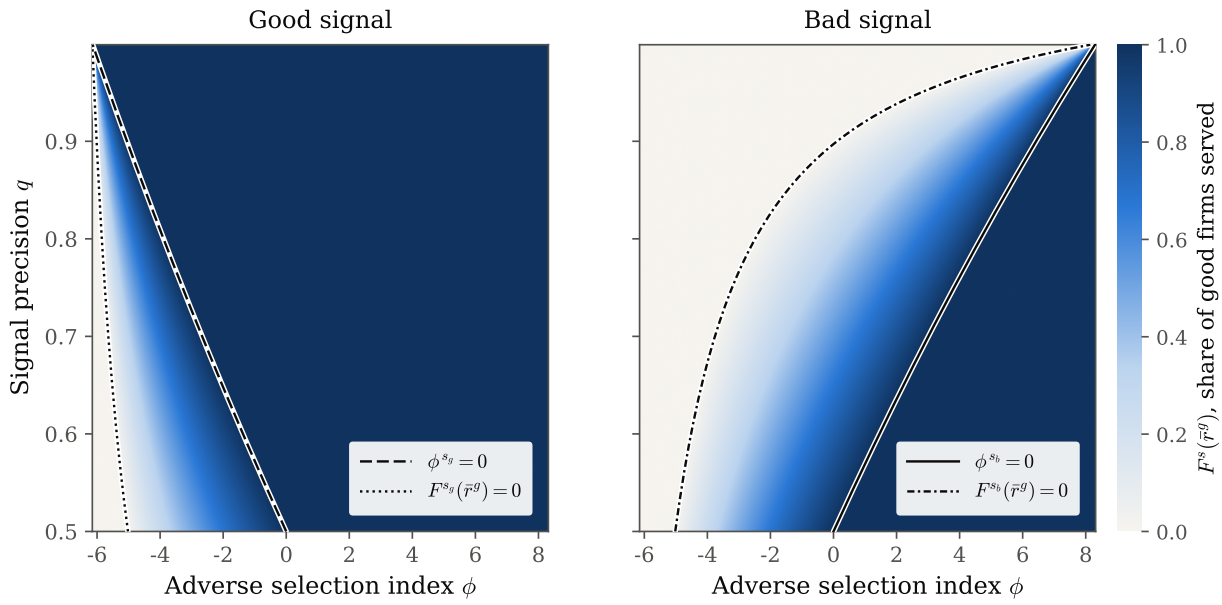


Figure 7: **Share of good firms served, by signal group.** The pooling region lies below the frontier line $\phi^s = 0$, where ϕ is positive. Below the frontier line $F^s(\bar{r}^g) = 0$, no good type gets credit, lending is only on the bad region. Between the two lines, the regime is hybrid.

The two extremes. At $q = 1/2$ the signal carries no information: $\alpha^{s_g} = \alpha^{s_b} = \mu_g$, the two groups merge into one, and ϕ^s becomes the population-wide index ϕ that governs the uninformative case below. As $q \to 1$ the posteriors go to certainty, $\alpha^{s_g} \to 1$ and $\alpha^{s_b} \to 0$, so $\phi^{s_g} \to \bar{r}^g - c_g > 0$ and $\phi^{s_b} \to \bar{r}^g - \bar{r}^b < 0$: the good-signal group pools at the good types' own cost, the bad-signal group lends to bad types only, and the mixture is the perfect-signal equilibrium above. Appendix C.2 derives the two branches and both limits.

4.4 From the model to the data

The model's distribution of rates and its moments map into the observed ones. The calibration in Section 5 matches three moments of the observed distribution of rates: the mean, the variance, and the correlation with the borrower's default probability. First, I derive the distribution of accepted rates in Appendix D.1 from the equilibrium posted rate distributions; then, I write its moments as a function of the model's parameters.

I show that the observed correlation can be the consequence either of information or search. The general formula of the correlation is:

$$\rho(r, d) = \sqrt{n_g n_b}\, \frac{\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g]}{\sqrt{Var(r)}},$$

where $d_\theta = 1 - p_\theta$ is a type's default probability and n_θ the share of loans made to type θ. Default takes two values, so its correlation with the rate depends on the two type means alone. This formula holds in every regime.

The correlation is positive when bad firms are paying a higher interest rate. Fig. 8 shows that both the precision of the signal and the search friction move it, so the correlation alone does not say which one is at work.

4.4.1 Correlation generated by information

Set $\pi = 1$, such that every firm compares two offers and search frictions disappear. Competition drives each signal group to its own expected cost, so the posted distribution collapses to one rate per group,

$$r^s = c^{avg,s}, \qquad c^{avg,s_b} - c^{avg,s_g} = (2q-1)\,\frac{\mu_g\mu_b}{\sigma_g\sigma_b}\,(c_b - c_g)\,.$$

Within each group, all types pay the same rate, and the correlation is zero. However, a bad type is more likely to draw the bad signal,

$$\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g] = (2q-1)\,(c^{avg,s_b} - c^{avg,s_g})\,, \qquad Var(r) = \sigma_g\sigma_b\,(c^{avg,s_b} - c^{avg,s_g})^2\,,$$

and a correlation equal to the correlation between the signal and the type,

$$\rho(r,d) = \rho(S,\theta) = (2q-1)\sqrt{\frac{\mu_g\mu_b}{\sigma_g\sigma_b}},$$

in which S is the indicator of a bad signal. It rises from zero at $q = 1/2$ to one at $q = 1$. Without search frictions the correlation measures the precision of the signal, and nothing else.

Search frictions dilute it. For $\pi < 1$ each group posts a distribution instead of a single rate, and the dispersion inside a group is noise in the relation between the rate and the type,

$$Var(r) = \underbrace{\sigma_g Var(r \mid s_g) + \sigma_b Var(r \mid s_b)}_{\text{search}} + \underbrace{\sigma_g\sigma_b\,(\mathbb{E}[r \mid s_g] - \mathbb{E}[r \mid s_b])^2}_{\text{information}}.$$

Under full pooling the rate depends on the type only through the signal, so the correlation factors into what the rate reveals about the signal and what the signal reveals about the type, $\rho(r,d) = \rho(r,S)\,\rho(S,\theta)$, and $\rho(r,S)^2$ is the information share of the variance above. The first term falls relative to the second as π rises: competition compresses the dispersion inside the groups faster than the gap between them, so $\rho(r,S)$ increases with competition and reaches one in the limit.

4.4.2 Correlation generated by search

Set $q = 1/2$: the signal is worthless, the bank treats the two signal populations as one. Posteriors coincide with priors, $\alpha = \mu_g$, and $c^{avg} = \mu_g c_g + \mu_b c_b$.

The correlation is zero whenever the regime is pooling or lending only on the separating region. In pooling, $\phi > 0$, both types draw from the same distribution, the rate is independent of the type, and $\rho(r,d) = 0$. Also, if $\phi < \underline{\phi}$, the bank offers only rates on the separating region, where good types refuse; as only bad types stay in the market, $\rho(r,d) = 0$.

Yet, there is a case in which the correlation is positive. If $\underline{\phi} < \phi < 0$ the regime is hybrid: the bank posts some rates below and some above $\bar{r}^g$. A bad type always accept offers, while a good type accepts only offers from the pooling region. Write λ for the share of the loans to bad types

made above $\bar{r}^g$, and n_g for the share of loans made to good types,

$$\lambda = \frac{(1-\pi)^2}{\pi(2-\pi)} \left[\left(\frac{\mu_b \left(\bar{r}^b - c_b \right)}{\bar{r}^g - c^{avg}} \right)^2 - 1 \right], \qquad n_g = \frac{\mu_g (1-\lambda)}{1 - \mu_g \lambda}.$$

The gap between the two type means is λ times the distance between the two branches,

$$\mathbb{E}[r \mid b] - \mathbb{E}[r \mid g] = \lambda \left(\mathbb{E}[r \mid r > \bar{r}^g] - \mathbb{E}[r \mid r < \bar{r}^g] \right) > 0,$$

positive because the support is disconnected and the upper branch lies entirely above the lower one (Lemma 4.3), so that

$$\rho(r, d) = \frac{\lambda \sqrt{\mu_g \mu_b (1-\lambda)}}{1 - \mu_g \lambda} \, \frac{\mathbb{E}[r \mid r > \bar{r}^g] - \mathbb{E}[r \mid r < \bar{r}^g]}{\sqrt{Var(r)}}.$$

None of this comes from the bank's information: the firms sort themselves, by refusing the rates they will not pay. A gap between the two types' mean rates, and with it a positive correlation between rates and default, therefore does not require a precise signal.

Search decides how far that sorting goes. The regime is hybrid when $(1-\pi)\mu_b \left(\bar{r}^b - c_b \right) < \bar{r}^g - c^{avg}$, and λ falls from one at that boundary to zero as $\pi \to 1$. Both ends of the range leave nothing to measure: at the boundary no good type borrows, so $n_g = 0$; in the limit no offer lies above $\bar{r}^g$. Search alone therefore moves the correlation only in between, when competition is strong enough to keep some good types in the market and weak enough to leave some offers above their reservation rate.

What the correlation identifies. The two cases bracket the mechanism: information sorts firms across signal groups, search sorts them inside one group. Both carry the same sign, so the correlation on its own pins down neither q nor π, and the calibration separates them by matching the mean and the variance at the same time. One configuration hides the signal altogether: when both groups lend to bad types only they post the same distribution, no good type borrows, and the correlation is zero whatever the precision. Appendix D reports that q is the direction the data pin down least in the classes whose observed correlation is near zero.

The correlation $\rho(r, d)$ can be generated by more information or more search, as shown in Fig. 8. Panel (a) varies the share of firms that compare two offers, π, holding the precision of the signal fixed. The signal is informative, so there is already some sorting irrespectively of search. On top of this, search affects the correlation by affecting the variance, the share of good types served, and crucially the regime: it makes the hybrid regime more likely when adverse selection is severe. Panel (b) varies q, holding π fixed. If starting from pooling when the signal is not informative, the correlation is zero. As the signal becomes more informative, in the pooling region the signal starts separating types, until the bad signal switches to lending in the high-rate region, above $\bar{r}^g$. The discontinuities are the result of a change of regime of the bad signal: first from pooling to the hybrid regime, then to only separating.

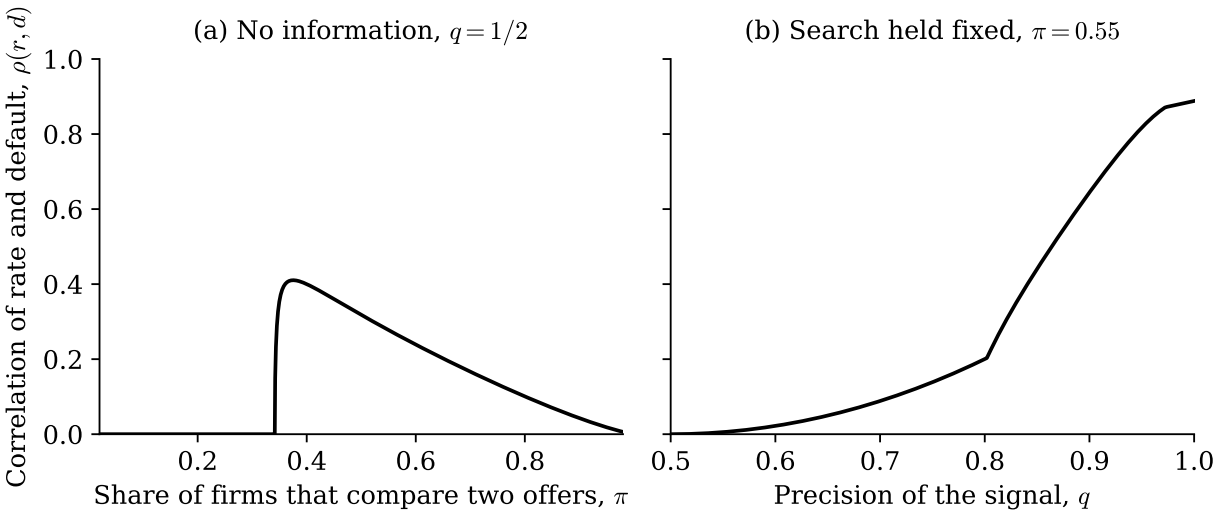


Figure 8: **Information and search can increase the correlation between the rate and default.**

4.5 Implications for surplus sharing

Total surplus (per bank). Under perfect information or perfect signal, per-trade surplus between a bank and a firm of type θ is $(\bar{r}^\theta - c_\theta)$. In the symmetric equilibrium, only half of the π shoppers will effectively accept the bank's offer, while the others will prefer the second offer. The potential surplus attainable per bank if all firms are matched is $TS^{q=1} = (1 - \frac{\pi}{2})(\mu_b(\bar{r}^b - c_b) + \mu_g(\bar{r}^g - c_g))$. However, under asymmetric information total surplus depends on the severity of adverse selection: if adverse selection is severe, at least a fraction of good firms obtains a rate quote beyond their reservation rate, thus refuses and exits the market; the gains from trade remain unrealized.

Proposition 4.3 (Banks' share of total surplus)**.** *Banks' share of total surplus is decreasing in competition and increasing in the precision of the signal.*

Under perfect information, each bank gets a share of realized surplus per trade exactly equal to $1 - \pi$, while firms get the remaining π share. Under asymmetric information, banks' profits are lower: in pooling, the bank charges bad types the pooling rate and loses $(1 - \pi)\mu_b(\bar{r}^b - \bar{r}^g)$ with respect to the perfect information case; under bad-only lending and the mixed equilibrium, each bank loses a share of surplus equal to the unrealized trade with good firms $(1 - \pi)\mu_g(\bar{r}^g - c_g)$.

Proposition 4.4 (Firms' surplus and information)**.** *Information reduces (increases) costs borne by good (bad) types.*

(i) Under no information, when adverse selection is mild ($\phi > 0$), good types trade at the higher pooling rates; when adverse selection is severe ($\phi \leq 0$), they may not trade at all. These gains are increasing in the signal precision q and decreasing in the level of competition.

(ii) On the contrary, bad types benefit from pooling when adverse selection is mild ($\phi > 0$), as they trade at rates below the good types' reservation rate; when adverse selection is severe $\underline{\phi} < \phi \leq 0$, they partially gain from cross-subsidization, but less than in pooling; they do not gain if adverse selection is very severe and banks lend only on the high-rate region. These benefits are decreasing in the signal precision q.

Good types bear two costs of asymmetric information: a cross-subsidy to bad types where the bank pools, and exclusion where the bank lends to bad types only. Both fall with signal precision. Under no information and $\phi > 0$ the bank pools good and bad types. Total cross-subsidy to bad types is $\pi\mu_b(c_b - c_g)$. Under no information and $\phi < 0$, a fraction $1 - F(\bar{r}^g)$ of good types are offered too high a rate, and exit the market. If $F(\bar{r}^g) > 0$, the remaining good types, if any, subsidize bad types.

5 Calibration

I calibrate the model separately in each Banque de France rating class to assess which equilibrium regimes are consistent with the empirical evidence. I treat the rating as a summary statistic of creditworthiness, while allowing for residual heterogeneity in borrower risk within each class.

Four parameters are constructed from the data. I proxy the reservation rate of bad types, $\bar{r}_b$, using the 99th percentile of the observed rate distribution in each rating class. I proxy the population shares of good and bad types using firms' subsequent rating transitions, classifying firms that improve their ratings as good and those that deteriorate as bad. Finally, I construct the success probabilities p_g and p_b from the corresponding three-year-ahead default frequencies. These classifications provide empirical proxies for latent borrower types rather than direct observations of them.

I then calibrate three parameters: the share of firms observing two offers, π, which governs the intensity of competition; the signal precision, q; and the reservation rate of good types, $\bar{r}_g$, which is not directly observed in the data and is important for characterizing adverse selection. The calibration targets three moments of the within-class rate distribution: the mean, the total variance, and the correlation between rates and borrower risk. The calibrated parameters characterize the competition and information environment consistent with these moments and allow me to assess which equilibrium regimes are supported by the model.

The calibrated competition parameter is relatively stable and declines slightly across riskier rating classes. Signal precision, by contrast, varies substantially: it is close to one in safer classes, consistent with a separating equilibrium, and approaches 0.5 in riskier classes, corresponding to an uninformative signal and a pooling equilibrium. These results suggest that the information structure consistent with the data varies systematically with credit quality.

5.1 Calibration Strategy

The calibration proceeds in two stages. First, I conduct a grid search over the parameter space, evaluating 20^3 combinations of parameter values within economically plausible bounds: $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, and $\bar{r}_g \in [0.5\,\bar{r}_{\text{mean}}, 0.98\bar{r}_b]$, where $\bar{r}_{\text{mean}}$ is the class mean rate. For each combination, I compute the model-implied moments - mean rate, variance, and correlation with subsequent default probability changes. Because a fraction π of firms compare offers from both banks and accept the lower one, the equilibrium distribution $F(r)$ derived in Section 5 – each bank's own posted-rate strategy – is not the same object as the distribution of realized, originated rates: a firm that shops around ends up paying the better of two draws from F, not a raw draw from it. I therefore compute all targeted moments under the properly reweighted realized-rate distribution

$g(r) \propto f(r)\,(1-\pi F(r))$, derived in Appendix D.1, rather than under $F(r)$ directly. I keep the combination that minimizes the objective function, which is the weighted sum of squared percentage errors of targeted moments. Second, I refine the best grid point using local optimization (L-BFGS-B algorithm) to minimize the weighted sum of squared percentage deviations between model and data moments. The objective function to minimize is the sum of squared proportional errors across the three moments[15]. With three moments and three free parameters (r_{CB} fixed), the order condition for exact identification is satisfied. The parameters must also generate a feasible equilibrium (costs below reservation rates) and satisfy the mixed-strategy conditions, which impose additional structure beyond the three explicit moments matched. Because the equilibrium is piecewise in $(\pi, q, \bar{r}_g)$, however, a formal rank condition for global identification is not analytically tractable; Section 5.3 reports numerical evidence in its place.

5.2 Calibration Results

Table 1: Search Frictions and Information by Rating Class

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{\bar{r}}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.772	0.997	3.461	4.89	0.41
3	0.759	1.000	3.706	4.56	0.35
4+	0.738	0.999	4.609	3.94	0.15
4	0.757	1.000	5.056	5.06	0.19
5+	0.749	1.000	5.403	3.05	0.20
5	0.729	0.944	6.662	0.04	0.04
6	0.721	0.526	7.057	0.00	0.04
7	0.711	0.526	9.350	0.00	0.00

Notes: $\hat{\pi}$ is the search-friction parameter, $\hat{q}$ the signal precision, $\hat{\bar{r}}_g$ the reservation rate of good borrowers, and "Information Contribution" measures the percentage increase in rate dispersion attributable to screening relative to the no-information benchmark.

The calibration results are showin in Table 1 and reveal three key findings. First, search frictions are relatively low across all rating classes, with $\hat{\pi}$ estimated between $[0.71, 0.77]$, increasing in ex-ante safety. This suggests that the French corporate lending market is relatively competitive, with most firms sampling multiple lenders. This estimate seems realistic when compared to the share of firms with multiple bank relationships in the sample. On average over the time period and across classes, approximately 60% of firms across classes maintain at least two active bank relationships (Table D.11). This number is a lower bound, as the credit registry covers only relationships with an existing stock of credit above 25 thousands euros.

Second, the informativeness of the public signal varies across the credit quality distribution. For good borrowers (ratings 3+ through 5+) the precision of the signal observed by banks is near-perfect ($\hat{q}$ close to 1); banks in these markets are well informed and perfectly separate types. On the contrary, for poor borrowers (ratings 6 and 7) the precision is close to uninformative ($\hat{q} \approx 0.53$); at the calibrated parameters, adverse selection within the class is mild ($\phi > 0$), so uninformed banks pool good and bad types within the class. This result may suggest that, among poorer-rated borrowers, the rating score already captures much of the relevant information about

[15] Proportional errors are used to ensure that moments of different scales are weighted comparably in the estimation.

credit risk, leaving limited additional information for the public signal to convey within each class. Alternatively, the public signal may simply be less informative about differences in borrower quality within these classes.

Third, despite the high informativeness of the signal implied by the data moments for good ratings, information contributes little to observed rate dispersion. I compute the counterfactual no-information equilibria holding the calibrated parameters fixed and I compute the percentage increase in interest rate variance at the calibrated q. The information contribution to variance never exceeds 5.1%, indicating that search frictions account for most of the unexplained dispersion even in segments with near-perfect signal precision. This partly reflects the mild severity of the adverse selection problem within each rating class, for example, the relatively low distance of default probabilities between types mechanically limits between-type variance.

The model fit is reported in Table 2. The correlation moment is matched closely across all classes, and the mean is matched to a reasonable approximation. Variance is matched only partially in every rating class. The objective values in Table 1 are highest for the safest classes and fall with credit quality, from 0.41 at rating 3+ to essentially zero at rating 7. The pattern is mechanical: in the safest classes the two types differ hardly at all in cost - $c_b - c_g$ is about 0.003 at rating 3+, against an observed standard deviation of rate residuals of 0.64 - so between-type dispersion is negligible and the model has to generate nearly all observed variance from search alone. The estimated $\hat{\bar{r}}_g$ sits just below the class upper bound $\bar{r}_b$ in every class, so the pooling region covers most of the support. Appendix E reports the same two tables under the alternative bad-type definition that counts outright defaults as bad.

Table 2: Model Fit: Observed vs. Model-Implied Moments

	Mean		Variance		Correlation	
Rating	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.154	0.025	0.025
3	1.83	2.08	0.443	0.188	0.033	0.033
4+	2.40	2.53	0.553	0.338	0.032	0.032
4	2.37	2.63	0.730	0.424	0.048	0.048
5+	2.57	2.80	0.867	0.492	0.048	0.048
5	3.36	3.44	0.985	0.794	0.017	0.017
6	3.63	3.67	1.137	0.913	0.000	0.000
7	4.79	4.78	1.843	1.725	0.000	0.000

Notes: Mean is the class mean rate and variance the variance of rate residuals within the class, both in percentage points. Correlation is between the rate residual and the three-year-ahead change in the borrower's default probability. For ratings 6 and 7 the estimated correlation is not significantly different from zero ($p = 0.65$ and $p = 0.54$) and is set to zero as a calibration target.

5.3 Robustness of the Estimated Equilibrium

The response to each posterior signal realization depends endogenously on the estimated parameters through the threshold conditions governing ϕ_{good} and ϕ_{bad} - an analytical proof that the three-parameter GMM estimator $(\pi, q, \bar{r}_g)$ is globally identified from the three targeted moments (mean, variance, and the rate-default correlation) is not tractable. I therefore rely on three complementary numerical checks, run separately for each of the eight rating classes, to build confidence

that the parameter vector reported by my estimation procedure is the unique - or at least an isolated, dominant - minimizer of the GMM objective. Appendix D documents each of these three checks, and their complete numerical results, in full.

First, I assess local identification via the standard rank condition (Rothenberg, 1971): at the estimated parameter vector $\hat{\theta}$ I compute the 3x3 Jacobian of the three matched moments with respect to $(\pi, q, \bar{r}_g)$ by numerical differentiation. In every rating class the Jacobian has full rank (3/3), ruling out exact local non-identification, as summarized in Table 3. The associated condition numbers are not uniform, however: they range from roughly 5 to over 400 across rating classes, with the weakest direction consistently loading on q (signal precision) - most pronounced for the safest classes, whose near-perfect estimated precision ($\hat{q}$ close to 1) leaves the moments comparatively insensitive to q. I report this pattern transparently rather than treating identification as uniformly strong: q is comparatively less sharply pinned down than π and $\bar{r}_g$ in some classes, even though the local rank condition is always satisfied.

Table 3: Local Identification Diagnostics: Elasticity-Scaled Jacobian of (mean, var, corr) w.r.t. $(\pi, q, \bar{r}_g)$

Rating	Rank	Condition Number	Weakest Direction
3+	3/3	431.7	q
3	3/3	209.4	q
4+	3/3	134.2	q
4	3/3	82.1	q
5+	3/3	65.9	q (mixed with $\bar{r}_g$)
5	3/3	4.8	π (mixed with $\bar{r}_g$)
6	3/3	21.6	q
7	3/3	14.1	q

Notes: Rank and condition number computed from the singular value decomposition of the elasticity-scaled Jacobian (percentage change in moment per percentage change in parameter) at each rating class's estimated $\hat{\theta}$. "Weakest direction" identifies the parameter with the largest loading on the smallest singular vector; a second parameter is named when its loading also exceeds 0.45 in absolute value.

Second, because local curvature alone cannot rule out a second, disconnected minimum elsewhere in the parameter space, I complement it with a dense search over the full admissible cube of $(\pi, q, \bar{r}_g)$ for each rating class, visualized directly as three-dimensional plots of the parameter combinations whose implied moments come within a chosen tolerance of the data. Within every equilibrium regime realized on the grid, the set of parameter combinations that adequately fit the data forms a single connected neighborhood around one point rather than several disjoint clusters - visual evidence against multiple, unrelated local optima within a given regime.

Third, I exploit the fact that the equilibrium's regime-switching structure (as derived in Section 5) is itself governed by two threshold conditions - one for the pool of borrowers assigned a favorable signal, one for the pool assigned an unfavorable signal - each admitting three possible states (full pooling, hybrid, or bad-only lending), for nine theoretically possible regime pairs. Rather than assume ex ante which pairs are empirically relevant, I determine this data-drivenly: for each rating class I tag every point of the parameter grid with its actually realized (good-pool, bad-pool) regime pair, and find that only a subset - between three and six, depending on the rating class - are ever realized within the admissible parameter region, always respecting a simple

ordering in which the good-signal pool is never in a "less pooled" regime than the bad-signal pool. I re-solve the GMM objective separately within each empirically realized regime and compare the best fit achieved across all of them. In every rating class, the best-fitting parameter vector across all admissible regimes coincides with the parameter vector reported by my baseline, regime-unconstrained estimation. This confirms that the baseline optimizer converges to the true global optimum of the objective function and is not trapped in a locally optimal equilibrium regime.

Taken together, these three checks - local curvature (rank and condition number of the moment Jacobian), a dense global search of the parameter space visualized directly, and an exhaustive comparison across every equilibrium regime the model can produce - provide converging numerical evidence that the reported parameter estimates for each rating class correspond to a well-isolated global optimum of the GMM criterion. They do not constitute a formal identification proof, which the model's regime-switching structure precludes in closed form, but they substantially reduce the concern that my point estimates are an artifact of the numerical optimization routine rather than a genuine feature of the data.

6 Can the calibrated precision be an equilibrium?

The calibration puts signal precision at the two extremes of the admissible range: close to perfect in ratings 3+ through 5, and uninformative in ratings 6 and 7. In this section, I check whether the calibrated precision can be the equilibrium of an information acquisition game, where each bank can buy its own precision at a cost $C_s(q)$, holding its rival at the calibrated $\hat{q}$, and ask whether the estimate is what a bank would choose.

I show that a necessary condition is to introduce an additional punishment to being uninformed. In fact, *a signal that only sorts borrowers does not incentivize information acquisition*, and a full information equilibrium $\hat{q} \approx 1$ is not sustainable. I provide the example of a monitoring effect: a better informed bank not only sorts types, but also reduces their riskiness via monitoring. I find that the estimated full information equilibrium in the safest classes is sustained if the monitoring effect is strong enough, much less in the riskiest classes.

I focus on the two extremes of full and no information and check if they are robust to deviations. I start from full information, $q = 1$, and I assume that one bank deviates to stop acquiring the signal. Its rivals keep posting the two type-specific distributions of Section 4.3, whose supports overlap on $[\underline{r}^b, \bar{r}^g]$. The deviator maximizes its profits by posting a single rate in that overlap; the equal-profit condition holds on the overlap, so it earns exactly the same profits as informed banks: $(1-\pi)(\bar{r}^g - c_g)$ on every good type and $(1-\pi)(\bar{r}^b - c_b)$ on every bad type.[16] While it makes the same profits as informed banks, the deviator saves C_s, so $q = 1$ is not an equilibrium for any strictly positive cost.

I also consider a deviation from the no information equilibrium, $q = 0.5$. A deviating bank that acquires a signal always increases its profits. The deviation is not profitable if the cost of acquiring the signal is higher than the benefit.

[16] The overlap is non-empty when $\pi \geq (\bar{r}^b - \bar{r}^g)/(\bar{r}^b - c_b)$, which holds in all eight rating classes at the calibrated parameters. Where it fails the deviator loses $\mu_b \left[(1-\pi)(\bar{r}^b - c_b) - (\bar{r}^g - c_b)\right] > 0$, and the argument below goes through unchanged.

6.1 Example: monitoring

Suppose that an informed bank does not only recognize types, but also monitors them; for example, an informed bank knows which covenant to write in the contract to incentivize repayment. The clients of an informed bank repay more often, and the cost of lending to them is lower. I specify a functional form:

$$c_\theta(1) = (1 - \delta_j)\, c_\theta(0.5), \qquad \theta \in \{g, b\}, \tag{5}$$

so that full precision cuts the risk-adjusted cost of lending to each type by the fraction δ_j, the *monitoring strength* of rating class j. I assume that the effect is the same for the two types inside a class, while I allow it to vary across classes. Write $\kappa_\theta = \delta_j\, c_\theta(0.5)$ for the saving per unit lent. I derive the two conditions under which $q = 0.5$ and $q = 1$ can be equilibria in Appendix F.

Condition A: $G_j \leq C_s$**.** $\hat{q} = 0.5$ survives if a bank that buys a perfect signal gains less than the cost, $G_j \leq C_s$, where:

$$G_j = \underbrace{\pi\mu_g\mu_b(c_b - c_g)}_{\text{cream-skimming}} + \underbrace{(1-\pi)\mu_b(\bar{r}^b - \bar{r}^g)}_{\text{captive bad types}} + \underbrace{[(1-\pi) + \pi\mu_g]\,(\mu_g\kappa_g + \mu_b\kappa_b)}_{\text{monitoring}}. \tag{6}$$

Condition B: $C_s \leq L_j$**.** $\hat{q} = 1$ survives if a bank that drops the signal loses more than the cost it saves, $C_s \leq L_j$, where:

$$L_j = \mu_g\kappa_g(1-\pi) + \mu_b\kappa_b\,\frac{(1-\pi)(\bar{r}^b - c_b)}{\bar{r}^g - c_b}, \tag{7}$$

where the bank holds a rate on the overlap, so the loss depends on the monitoring penalty. Both G_j and L_j rise with δ_j: stronger monitoring makes the uninformed equilibrium easier to break and the informed one easier to hold. Condition A therefore caps δ_j in ratings 6 and 7, and Condition B puts a floor under it in the other six.

Figure 9 evaluates the two conditions at the calibrated parameters, given a cost C_s. When the cost of acquiring the signal is too low, below $C_s = 0.029$, the gain of being informed is always higher than the cost. As the cost increases, deviating becomes less profitable, and for low enough monitoring effects the equilibrium is stable. For the safest classes, the deviation to the no information equilibrium is not profitable only if the punishment is large enough, that is, if monitoring ensures a considerable reduction in firms' riskiness. The two bands never meet, however. At the cost that brings them closest, $C_s = 0.048$, or between 2 and 8 percent of a bank's profit, the six screening classes need $\delta_j \geq 0.17$ while ratings 6 and 7 admit at most $\delta_j = 0.019$. A monitoring strength of 0.17 raises the bad type's repayment probability by 15 to 17 points; one of 0.019 raises it by one to two points. Feasibility caps the riskiest classes as tightly as Condition A does: their good types already repay with probability 0.98, which leaves monitoring almost nothing to add. The estimated precisions are therefore an equilibrium of the acquisition game in all eight classes at one common signal cost, but only under a technology whose effectiveness is concentrated where borrowers are safest, while it is ineffective for borrowers that are already very risky.

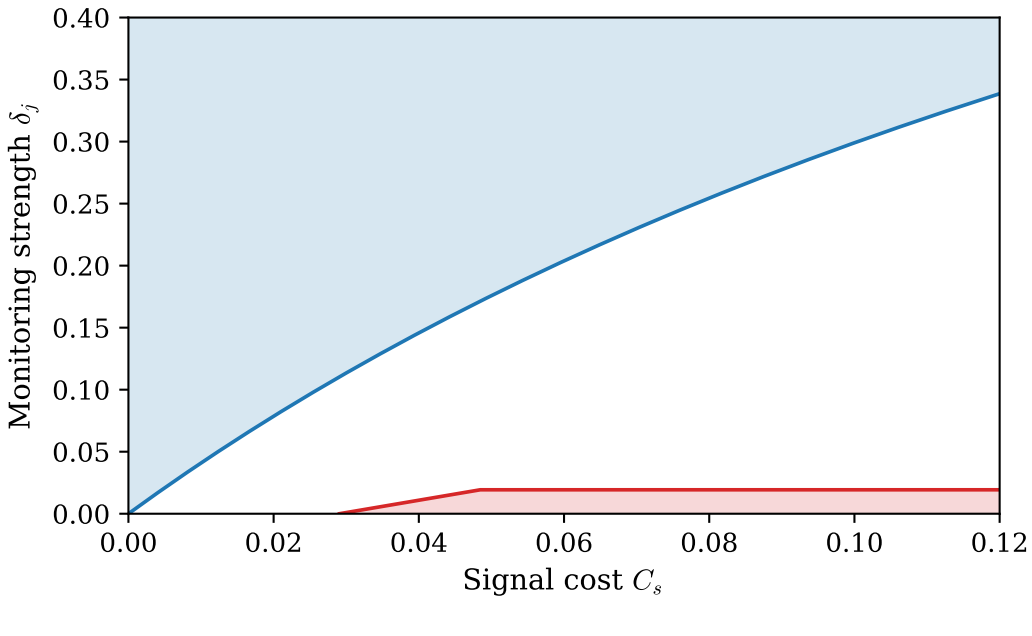


Figure 9: **Monitoring strength that sustains the calibrated precision, against the cost of the signal.** Horizontal axis: the price of a perfect signal over no signal, $C_s \equiv C_s(q{=}1) - C_s(0.5)$, for the mass one of firms a bank screens, in the units of the calibration ($r^{CB} = 1$). Vertical axis: the monitoring strength δ_j of (5), the fraction by which full precision cuts the risk-adjusted cost of lending to each type. The upper blue band is where the full information equilibrium of the safest classes is stable; the lower red band is where the no information equilibrium of the worse classes is stable; the feasibility constraint that monitoring cannot increase the success probability beyond one is binding.

7 Conclusion

This paper studies unexplained interest-rate differentials across firms and the role of information and competition in accounting for them. Using French loan-level data from 2006 to 2024, merged with firm-bank relationships and Banque de France ratings, I document substantial dispersion in loan rates among firms in the same rating class. This dispersion is not only quantitatively important: loan rates also contain information about future changes in firms' default risk. In particular, firms paying higher rates relative to other firms in the same rating class are more likely to experience a subsequent deterioration in their credit rating. This relationship is stronger among firms with safer initial ratings, younger and smaller firms, and loans with longer maturities. It is also more pronounced during the financial crisis. These findings suggest that observed interest-rate differentials reflect not only differences in publicly observed creditworthiness, but also the interaction between residual borrower risk, information, and competition in credit markets.

To rationalize these empirical patterns, I develop a model of loan pricing with search frictions and asymmetric information. Firms differ in their underlying credit risk, while banks observe an imperfect public signal of borrower quality. Search frictions limit the number of offers that firms receive and allow banks to randomize their interest rates, generating dispersion even among observationally similar borrowers. Imperfect information, in turn, affects how banks price different borrower types and how firms sort across offers. The model shows that a positive relationship

between interest rates and borrower risk can arise under different informational and competitive conditions: it may reflect the use of an informative signal, but it can also emerge in a pooling equilibrium in which banks randomize their offers and borrowers self-select. Interest-rate differentials therefore cannot be interpreted independently of the competitive environment in which lending decisions are made.

I calibrate the model separately for each Banque de France rating class to assess which equilibrium regimes are consistent with the observed moments of loan rates and their relationship with borrower risk. The calibration points to substantial differences in information across rating classes: the inferred signal is highly informative for firms in safer classes, while it is close to uninformative in riskier classes. By contrast, the estimated degree of competition varies less across classes. Although the model only partially accounts for the total variance of interest rates, it reproduces their mean and their correlation with borrower risk reasonably well. The quantitative results also indicate that information contributes relatively little to the overall dispersion of interest rates in the calibrated model, particularly in riskier rating classes. This does not, however, imply that information is unimportant. Even when its contribution to rate dispersion is limited, information changes how surplus is shared across borrower types. In particular, more precise information reduces the cross-subsidization of riskier borrowers by safer ones and can improve safer firms' access to credit.

Finally, I extend the analysis to consider whether the inferred levels of information can be sustained when information acquisition is costly and banks may privately acquire signals about borrower quality. In this setting, borrower sorting alone is not sufficient to sustain information acquisition: an uninformed bank may be able to imitate the pricing of informed banks while avoiding the cost of acquiring a signal. Introducing a monitoring benefit, through which information also reduces lending costs, can support the calibrated levels of information across rating classes. The extension highlights that the informational structure of credit markets depends not only on what banks can learn about borrowers, but also on the incentives to acquire and use that information.

Taken together, the results show why explaining interest-rate differentials across firms requires accounting for information and competition jointly. Differences in loan rates among firms with the same public rating are consistent with multiple underlying mechanisms, and the relationship between rates and future credit risk does not by itself identify which mechanism is at work. By combining loan-level evidence with a model of search frictions and asymmetric information, this paper provides a framework for understanding how borrower information and the intensity of competition jointly shape loan pricing, the dispersion of interest rates, and the allocation of credit. The results also suggest that policies aimed at improving information about borrower quality and those aimed at strengthening competition may be complementary: their effects depend on how information and competitive conditions interact in the credit market.

References

Agarwal, S., Grigsby, J., Hortaçsu, A., Matvos, G., Seru, A., and Yao, V. (2024). Searching for approval. *Econometrica*, 92(4):1195–1231.

Agarwal, S. and Hauswald, R. (2010). Distance and private information in lending. *The Review of Financial Studies*, 23(7):2757–2788.

Akerlof, G. A. (1970). The market for lemons: Quality uncertainty and the market mechanism. *The Quarterly Journal of Economics*, 84(3):488–500.

Altavilla, C., Brugnolini, L., Gürkaynak, R., Motto, R., and Ragusa, G. (2019). Measuring euro area monetary policy. *Journal of Monetary Economics*, 108:162–179.

Altavilla, C., Gürkaynak, R., and Quaedvlieg, R. (2024). Macro and micro of external finance premium and monetary policy transmission. *Journal of Monetary Economics*, 147(Supplement):103634.

Amiti, M., Kashyap, A. K., Kovner, A., and Weinstein, D. (2026). Why do firms pay different interest rates on their bank loans? *Working paper*.

Auria, L., Bingmer, M., Caicedo Graciano, C. M., Charavel, C., Gavilá, S., Iannamorelli, A., Levy, A., Maldonado, A., Resch, F., Rossi, A. M., and Sauer, S. (2021). Overview of central banks' in-house credit assessment systems in the euro area. ECB Occasional Paper 284, European Central Bank.

Baye, M. R., Kovenock, D., and de Vries, C. G. (1992). It takes two to tango: Equilibria in a model of sales. *Games and Economic Behavior*, 4(4):493–510.

Bech, M. and Monnet, C. (2015). A search-based model of the interbank money market and monetary policy implementation. *BIS Working Papers No 529*.

Berger, A. N. and Udell, G. F. (1995). Relationship lending and lines of credit in small firm finance. *The Journal of Business*, 68(3):351–381.

Berger, A. N. and Udell, G. F. (2002). Small business credit availability and relationship lending: The importance of bank organisational structure. *The Economic Journal*, 112(477):F32–F53.

Bester, H. (1985). Screening vs. rationing in credit markets with imperfect information. *The American Economic Review*, 75(4):850–855.

Bhutta, N., Fuster, A., and Hizmo, A. (2021). Paying too much? borrower sophistication and overpayment in the us mortgage market. CEPR Discussion Paper 14924, Centre for Economic Policy Research.

Bocola, L., Bornstein, G., Malgieri, C., and Puglisi, F. (2026). Bank market power. *Working Paper*.

Boot, A. W. A. (2000). Relationship banking: What do we know? *Journal of Financial Intermediation*, 9(1):7–25.

Boot, A. W. A. and Thakor, A. V. (2000). Can relationship banking survive competition? *The Journal of Finance*, 55(2):679–713.

Burdett, K. and Judd, K. L. (1983). Equilibrium price dispersion. *Econometrica*, 51(4):955–969.

Cahn, C., Girotti, M., and Salvadè, F. (2024). Credit ratings and the hold-up problem in the loan market. *Management Science*, 7(3).

Carletti, E., Leonello, A., and Marquez, R. (2024). Market power in banking. *ECB Working Paper*, (2886).

Cavalcanti, T., Kaboski, J., Martins, B., and Santos, C. (2023). Financing costs and development. *R&R at the American Economic Review*.

Claessens, S., Ongena, S., and Wang, T. (2025). "if you don't know me by now ..." information collection by banks in lending to private firms. *Working Paper*.

Cloyne, J., Ferreira, C., Froemel, M., and Surico, P. (2023). Monetary policy, corporate finance, and investment. *Journal of the European Economic Association*, 21(6):2586–2634.

Coen, J., Kashyap, A. K., and Rostom, M. (2024). Price discrimination and mortgage choice. NBER Working Paper 31652, National Bureau of Economic Research. July 2024 version.

Core, F., De Marco, F., Eisert, T., and Schepens, G. (2025). Inflation and floating-rate loans: Evidence from the euro-area. *CEPR Discussion Paper*, (20354).

Crawford, G. S., Pavanini, N., and Schivardi, F. (2018). Asymmetric information and imperfect competition in lending markets. *American Economic Review*, 108(7):1659–1701.

Davydenko, S. A. and Franks, J. R. (2008). Do bankruptcy codes matter? A study of defaults in France, Germany, and the U.K. *The Journal of Finance*, 63(2):565–608.

Degryse, H. and Ongena, S. (2005). Distance, lending relationships, and competition. *The Journal of Finance*, 60(1):231–266.

Delis, M., Ferrando, A., Mulier, K., and Ongena, S. (2025). The poor, the rich, and the credit channel of monetary policy. ECB Working Paper 3058, European Central Bank.

Dell'Ariccia, G. (2001). Asymmetric information and the structure of the banking industry. *European Economic Review*, 45(10):1957–1980.

Dell'Ariccia, G. and Marquez, R. (2006). Lending booms and lending standards. *The Journal of Finance*, 61(5):2511–2546.

Diamond, P. A. (1971). A model of price adjustment. *Journal of Economic Theory*, 3(2):156–168.

Drechsler, I., Savov, A., and Schnabl, P. (2017). The deposits channel of monetary policy. *The Quarterly Journal of Economics*, 132(4):1819–1876.

Duffie, D., Gârleanu, N., and Pedersen, L. H. (2005). Over-the-counter markets. *Econometrica*, 73(6):1815–1847.

Farboodi, M., Jarosch, G., Menzio, G., and Wiriadinata, U. (2025). Intermediation as rent extraction. *Journal of Economic Theory*, 227.

Ferreira, M. H. and Kozeniauskas, N. (2026). Beyond risk: Firm financing and interest rates. *Working Paper*.

Flanagan, T. (2025). The value of bank lending. *Journal of Finance*, 80:2017–2061.

Guerrieri, V., Shimer, R., and Wright, R. (2010). Adverse selection in competitive search equilibrium. *Econometrica*, 78(6):1823–1862.

Gürkaynak, R., Karasoy-Can, H. G., and Lee, S. S. (2022). Stock market's assessment of monetary policy transmission: The cash flow effect. *The Journal of Finance*, 77(4):2375–2421.

Haltiwanger, J., Jarmin, R. S., and Miranda, J. (2013). Who creates jobs? small versus large versus young. *The Review of Economics and Statistics*, 95(2):347–361.

Hauswald, R. and Marquez, R. (2006). Competition and strategic information acquisition in credit markets. *The Review of Financial Studies*, 19(3):967–1000.

Ioannidou, V. and Ongena, S. (2010). "time for a change": Loan conditions and bank behavior when firms switch banks. *The Journal of Finance*, 65(5):1847–1877.

Kirti, D. (2020). Why do bank-dependent firms bear interest-rate risk? *Journal of Financial Intermediation*, 41:100811.

Lester, B., Shourideh, A., Venkateswaran, V., and Zetlin-Jones, A. (2019). Screening and adverse selection in frictional markets. *Journal of Political Economy*, 127(1):338–377.

Mazet-Sonilhac, C. (2025). Information frictions in credit markets. *Working Paper*.

Petersen, M. A. and Rajan, R. G. (1994). The benefits of lending relationships: Evidence from small business data. *The Journal of Finance*, 49(1):3–37.

Petersen, M. A. and Rajan, R. G. (1995). The effect of credit market competition on lending relationships. *The Quarterly Journal of Economics*, 110(2):407–443.

Puglisi, T. (2023). Retail credit markets and the passthrough of monetary policy. *Working Paper*.

Rajan, R. G. (1992). Insiders and outsiders: The choice between informed and arm's-length debt. *The Journal of Finance*, 47(4):1367–1400.

Rothschild, M. and Stiglitz, J. (1976). Equilibrium in competitive insurance markets: An essay on the economics of imperfect information. *The Quarterly Journal of Economics*, 90(4):629–649.

Scharfstein, D. and Sunderam, A. (2016). Market power in real estate. *Working Paper*.

Sharpe, S. A. (1990). Asymmetric information, bank lending, and implicit contracts: A stylized model of customer relationships. *The Journal of Finance*, 45(4):1069–1087.

Stahl, D. O. (1989). Oligopolistic pricing with sequential consumer search. *The American Economic Review*, 79(4):700–712.

Stiglitz, J. E. and Weiss, A. (1981). Credit rationing in markets with imperfect information. *The American Economic Review*, 71(3):393–410.

Varian, H. R. (1980). A model of sales. *The American Economic Review*, 70(4):651–659.

Wang, Y., Whited, T. M., Wu, Y., and Xiao, K. (2022). Bank market power and monetary policy transmission. *The Journal of Finance*, 77(4):2135–2193.

Woodward, S. E. and Hall, R. E. (2012). Diagnosing consumer confusion and sub-optimal shopping effort: Theory and mortgage-market evidence. *American Economic Review*, 102(7):3249–3276.

A Additional tables and figures

Back to main ↑

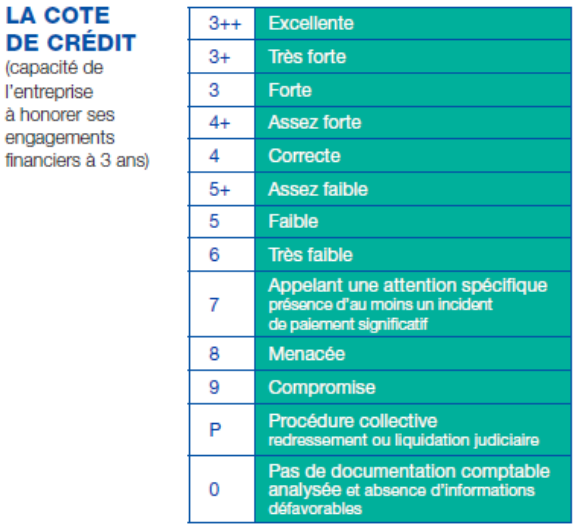


Figure A.1: Source: Banque de France. Credit rating scale until 2022.

Table A.1: Three-year rating transitions, 2021 cohort

	31 Dec. 2023 (% of row)													
1 Jan. 2021	3++	3+	3	4+	4	5+	5	6	7	8	P	0	Exit	Firms
3++	**46.4**	15.3	15.1	9.4	2.6	1.8	0.3	0.7	0.0	0	0.0	8.4	0	9,365
3+	14.8	**20.4**	25.7	22.6	5.8	2.2	0.4	0.6	0.0	0	0.1	7.4	0	16,013
3	5.5	9.9	**26.6**	32.6	11.6	3.9	0.8	0.7	0.0	0.0	0.1	8.1	0	26,543
4+	2.6	3.4	14.8	**35.3**	22.4	8.1	1.4	0.9	0.1	0.0	0.3	10.7	0	35,727
4	0.9	1.2	6.5	26.3	**33.0**	14.5	3.5	1.8	0.3	0.2	0.9	10.9	0.0	53,173
5+	0.3	0.5	1.5	8.5	15.8	**22.1**	7.9	5.0	0.6	0.3	2.6	34.8	0.0	88,043
5	0.0	0.1	0.2	1.0	2.7	4.5	**61.1**	2.8	0.4	0.3	4.3	21.6	1.0	126,013
6	0.1	0.1	0.2	0.7	2.0	4.5	46.0	**6.9**	0.7	0.4	8.4	29.5	0.4	54,905
7	0	0	0.1	0.3	0.8	1.8	5.4	5.3	**7.6**	6.7	27.7	44.0	0.3	4,639
8	0	0	0	0.2	0.8	1.2	3.7	5.1	6.9	**11.1**	33.6	37.1	0.3	2,923
9	0	0	0	0	0.5	0.9	2.7	6.3	4.1	15.8	39.2	30.2	0.5	222

Notes: Firms rated by the Banque de France on 1 January 2021, by credit rating (rows), and their rating on 31 December 2023 (columns). Each cell is the percentage of the row's firms that hold the column's rating at the horizon, so each row sums to 100; the diagonal, in bold, is the share still in the same class. P: the firm is in insolvency proceedings. 0: the firm is in the file but has no rating based on financial statements. Exit: the firm no longer appears in the rating file. Ratings issued after the 2022 scale reform are converted back to the pre-reform scale; the conversion merges the new class 9 into 8, so the horizon has no class 9. Firms: number of firms in the row's class on 1 January 2021.

B Additional facts on rate dispersion

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Table B.2: Data: Nouveaux Credits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Excludes credit lines and undefined maturity products. All credit and balance-sheet variables are expressed in terms of reported assets. Multiple products (dummy) equals one if the firm takes more than one product type in the quarter, counting the excluded products; Multiple loans, one product (dummy) equals one if it takes several loans in the quarter, all of the same type.

	Total effective rate					
log.Loan quantity (K-Eur)	-0.2241***	-0.1558***	-0.1551***	-0.1444***	-0.1475***	-0.0989***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Maturity (months)	-0.0006**	0.0022***	0.0033***	0.0021***	0.0022***	0.0020***
	(0.048)	(0.002)	(0.001)	(0.003)	(0.003)	(0.004)
Age	-0.0139***	-0.0096***	-0.0084***	-0.0074***	-0.0073***	-0.0038***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Age squared	0.0001***	0.0000***	0.0000***	0.0000***	0.0000***	0.0000***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Young firm 0-3y (dummy)	-0.0538***	-0.0414	-0.0432	-0.0198	-0.0169	-0.0677**
	(0.002)	(0.286)	(0.274)	(0.511)	(0.558)	(0.010)
New bank-firm relationship (dummy)	0.1780***	0.0250	0.0190	-0.0139	-0.0148	-0.0339**
	(0.000)	(0.526)	(0.633)	(0.309)	(0.273)	(0.024)
Variable rate (dummy)	0.4331***	0.4889***	0.1677*	0.1176	0.1160	0.1420
	(0.000)	(0.000)	(0.052)	(0.201)	(0.201)	(0.125)
Personal loan (dummy)	-0.2967*	-0.0550	0.0397	-0.0077	-0.0053	0.0581
	(0.051)	(0.701)	(0.626)	(0.932)	(0.951)	(0.466)
Individual entrepreneur (dummy)	0.5594***	0.5740***	0.5365***	0.4364***	0.4237***	0.2432***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.001)
Loan over assets	0.1543**	0.0985*	0.0973*	0.1050*	0.1032*	0.0474*
	(0.021)	(0.060)	(0.079)	(0.077)	(0.074)	(0.083)
Pledged guarantees	0.7897***	0.1116	-0.0221	-0.2582**	-0.2378***	-0.1769**
	(0.003)	(0.137)	(0.683)	(0.014)	(0.010)	(0.024)
Revenues	-0.0451***	-0.0166	-0.0119	0.0033	0.0049	-0.0079
	(0.000)	(0.176)	(0.350)	(0.599)	(0.453)	(0.136)
Bank Debt	0.1596***	0.1232***	0.1202**	0.1132***	0.1249***	-0.0022
	(0.000)	(0.008)	(0.012)	(0.001)	(0.001)	(0.912)
Non-Bank Debt	-0.0179	-0.0601	-0.0469	0.0034	-0.0011	-0.0200
	(0.673)	(0.614)	(0.711)	(0.966)	(0.989)	(0.693)
Mobilised credit	-0.0106**	-0.0021	-0.0012	0.0000	-0.0002	0.0024
	(0.030)	(0.472)	(0.724)	(0.992)	(0.944)	(0.307)
Short Term Debt	0.0157	-0.0018	0.0004	0.0202	0.0201	-0.0126
	(0.541)	(0.897)	(0.978)	(0.154)	(0.153)	(0.276)
Undrawn credit lines	-0.0139*	-0.0057	-0.0043	-0.0071	-0.0071	-0.0024
	(0.063)	(0.227)	(0.337)	(0.119)	(0.110)	(0.447)
Multiple products (dummy)	-0.0256	-0.0091	-0.0368	-0.0327*	-0.0301	-0.0027
	(0.201)	(0.731)	(0.130)	(0.092)	(0.100)	(0.815)
Multiple loans, one product (dummy)	-0.2182***	-0.0656	-0.0836	-0.0427	-0.0421	-0.0073
	(0.002)	(0.257)	(0.143)	(0.216)	(0.215)	(0.786)
Quarter FE	✓					
Lender × Quarter FE		✓				
Lender × Quarter × Loan Type FE			✓			
Lender × Quarter × Loan Type × 2digit Sector FE				✓	✓	✓
Region FE					✓	✓
Group indicator, Rating FE						✓
Observations	457,868	457,868	457,868	457,868	457,868	457,868
R^2	0.55	0.72	0.76	0.82	0.82	0.83

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

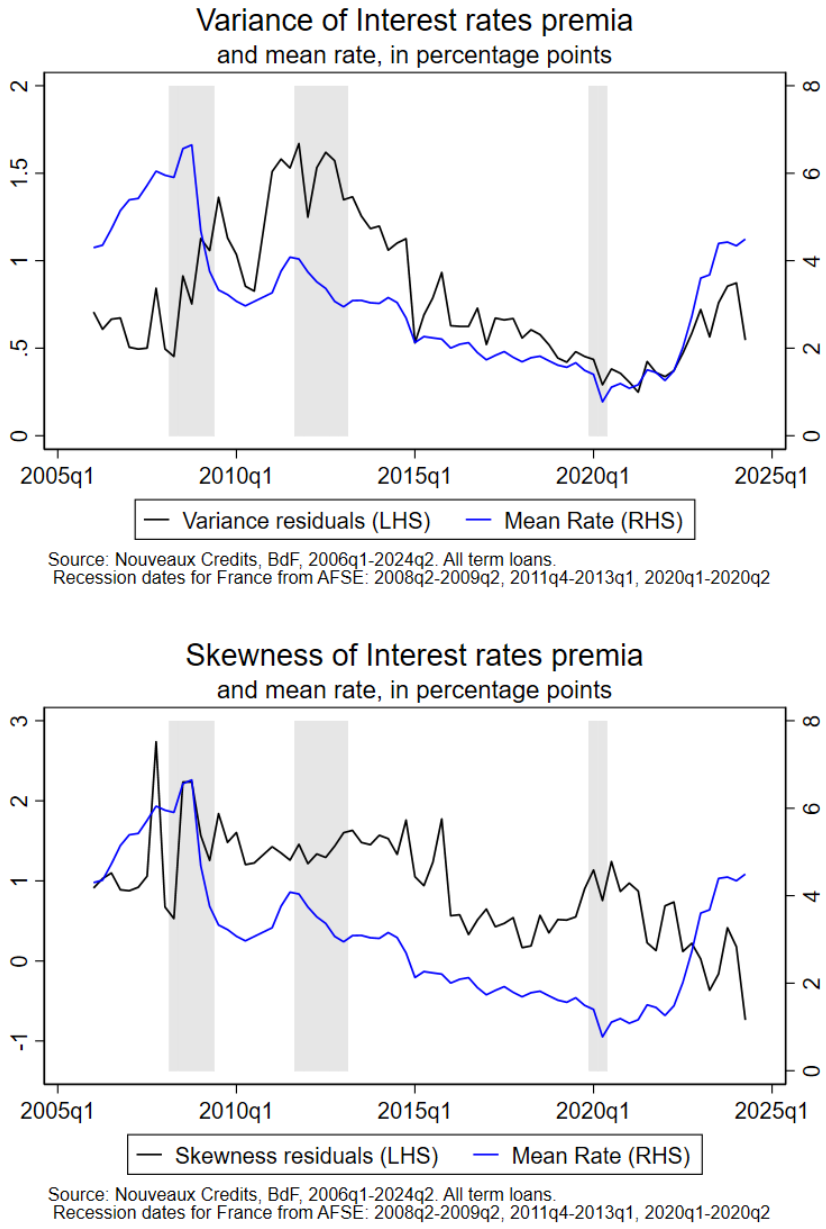


Figure B.2: Variance (top) and skewness (bottom) of interest rate residuals by quarter, black line, left axis, against the mean interest rate, blue line, right axis, both in percentage points. Shaded bands are French recessions (AFSE): 2008q2–2009q2, 2011q4–2013q1 and 2020q1–2020q2. Data: Nouveaux Crédits (BdF), 2006q1–2024q2, all term loans.

Rating Scale	N obs.	rate (mean)	rate (s.d.)	Quantity (mean, K-Eur)	Maturity (mean, months)	Var. Rate (in %)	Age (mean, years)	Loan/Assets (mean, in %)	Bank Debt/Assets (mean, in %)
3++	7948	2.1	1.8	3754.5	59.0	35	39.0	5.4	19.6
3+	22681	2.2	1.8	2919.3	53.5	37	33.7	5.0	20.6
3	43073	2.3	1.8	2271.5	42.4	37	31.7	5.3	22.0
4+	71629	2.8	2.1	1092.2	34.8	39	28.1	6.2	23.0
4	112974	2.8	2.0	670.2	27.9	43	26.1	7.1	25.0
5+	106418	2.9	2.0	545.5	23.9	48	23.4	7.7	30.3
5	46510	3.5	2.2	781.8	17.9	51	23.1	8.3	31.0
6	13625	3.7	2.2	923.4	16.6	52	21.4	10.4	34.2
7	2657	4.2	2.3	134.7	14.8	62	19.7	8.6	32.7
8	1592	5.0	2.6	107.8	13.2	61	18.0	8.4	31.2
9	268	5.5	2.7	134.0	11.9	48	16.5	10.0	31.7
P (liquidation)	3716	5.0	2.0	114.5	3.4	26	21.8	5.7	33.4
0 (no rating)	24777	3.2	2.1	1895.8	45.0	40	16.1	12.4	27.2
Total	457868	2.9	2.1	1098.1	30.5	44	25.8	7.3	26.5

Table B.3: Descriptive table of the loan-level regression sample. Source: Banque de France (Nouveaux Credits, COTA, SCR, FIBEN). The loan-level data refer to the first month of each quarter, so they are effectively monthly. Rate: total effective rate (including all fees).

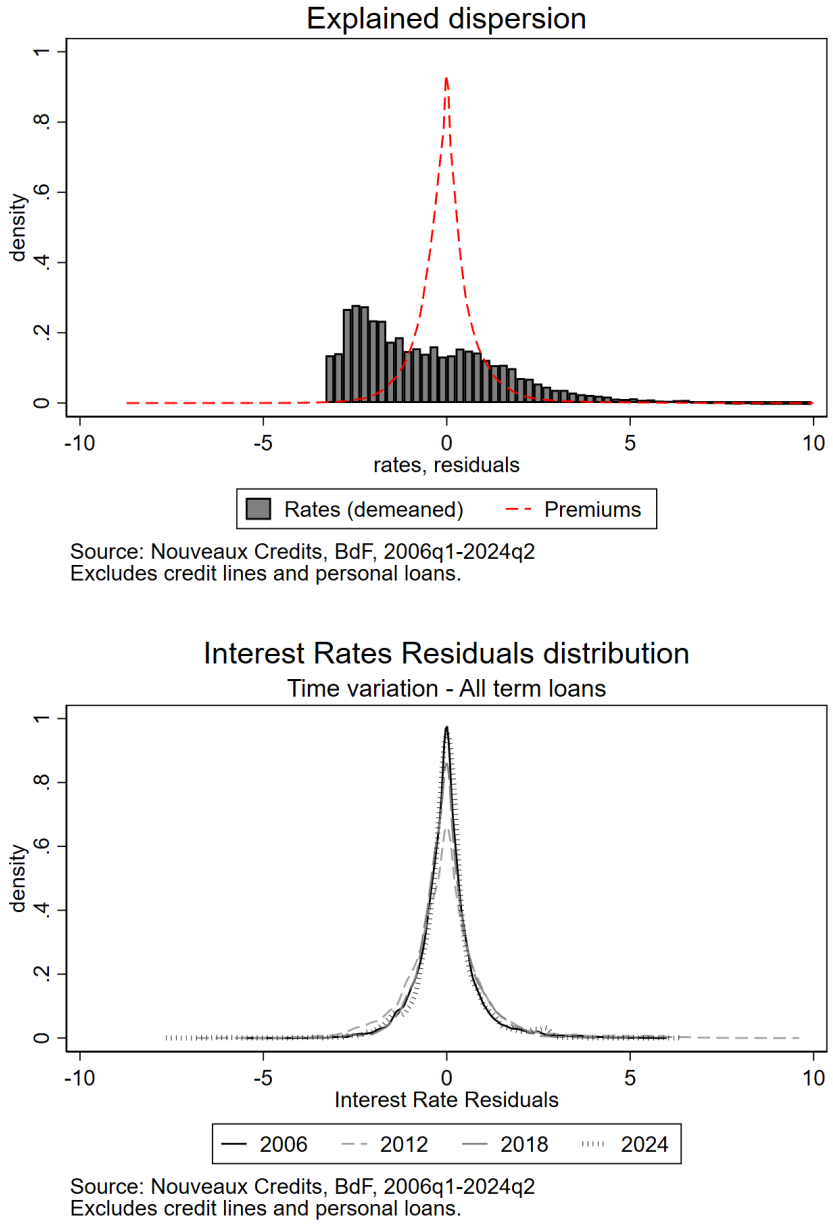


Figure B.3: Regression outcome. Top: residuals versus actual rates, all sample. Bottom: residuals distribution for selected years.

B.1 Additional specification 1: Firm fixed effects

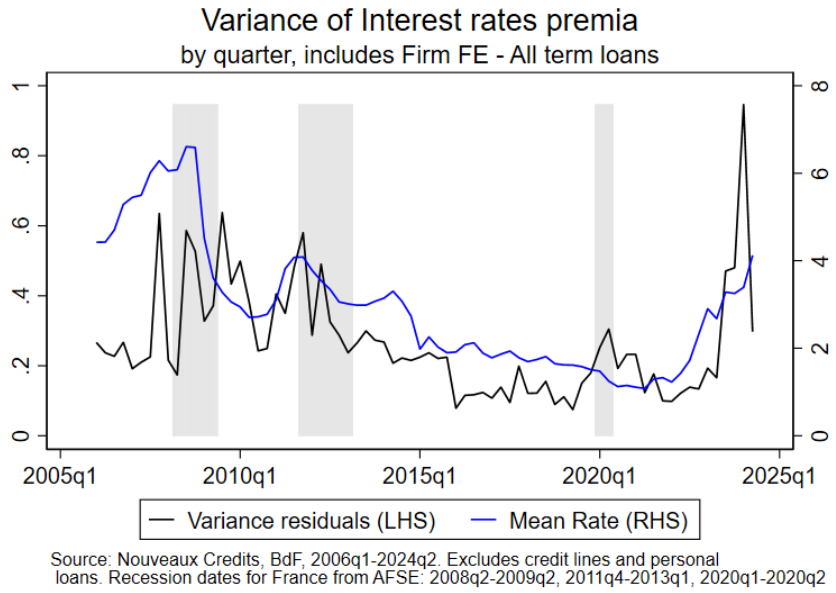
	Total effective rate					
log.Loan quantity (K-Eur)	-0.2337***	-0.0215***	-0.0200***	-0.0183***	-0.0231***	-0.0182***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Maturity (months)	-0.0004***	0.0013***	0.0015***	0.0015***	0.0015***	0.0015***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Variable rate (dummy)	0.4132***	0.1454***	0.2260***	0.2449***	0.2192***	0.2063***
	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)	(0.000)
Loan over assets	0.0809***	-0.0049	-0.0053*	-0.0052*	-0.0047*	-0.0057**
	(0.000)	(0.105)	(0.063)	(0.060)	(0.086)	(0.035)
Quarter FE	✓					
Firm × Quarter FE		✓	✓	✓	✓	
Lender FE			✓			
Lender × Quarter FE				✓	✓	
Firm × Lender × Quarter FE						✓
Loan Type FE					✓	✓
Observations	590,621	197,343	197,328	196,048	196,042	156,211
R^2	0.55	0.92	0.93	0.94	0.94	0.94

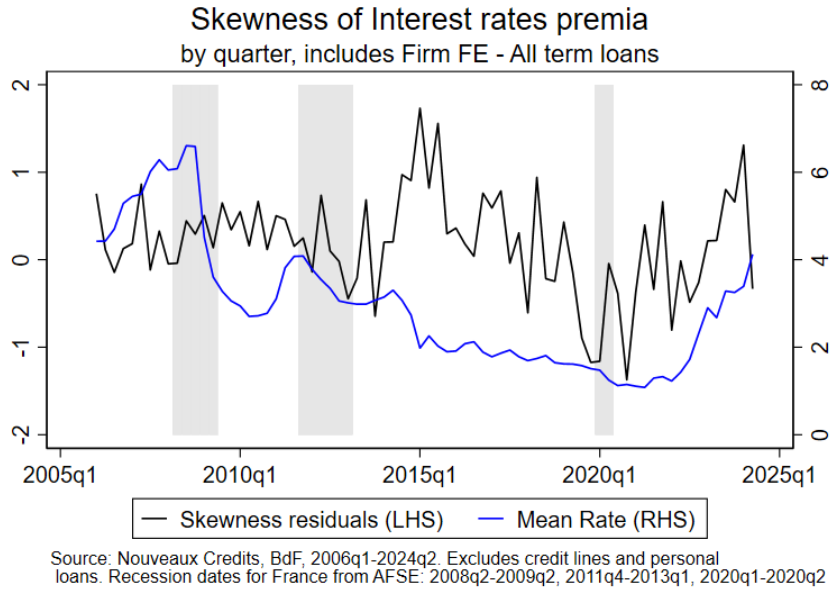
p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.4: Data: Nouveaux Credits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. All credit and balance-sheet variables are expressed in terms of reported assets.

Figure B.4: Variance and skewness of residuals from the firm FE specification.





B.2 Additional specification 2: truncated regression approach

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One possible concern is that it is not possible to observe rates below 0. Regression approaches assuming normality may be downward biased, predicting negative rates even when they are impossible to observe. Two approaches can help address this issue. Firstly, I verify to what extent the OLS is predicting negative values and I find that only 0.4% of predictions fall slightly below zero; this is probably due to the fact that most of the variation is absorbed by fixed effects. Secondly, the same analysis can be repeated using Tobit regression, which accounts for this lower bound; this approach is more limited in the inclusion of many interactions of fixed effects, thus it forces a more parsimonious approach, resulting in a worse performance.

B.3 Rate dispersion and bank heterogeneity

One question that naturally arises is: is this dispersion hiding heterogeneity across banks? The answer is yes. There is consistent heterogeneity between banks in the explained variation, as shown below by repeating the above estimation of rates by bank, and collecting the R^2.

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{LoanChar}_{i,b,t} + \gamma \cdot \text{FirmChar}_{i,t} + \lambda_t + \varepsilon_{i,b,t}. \tag{8}$$

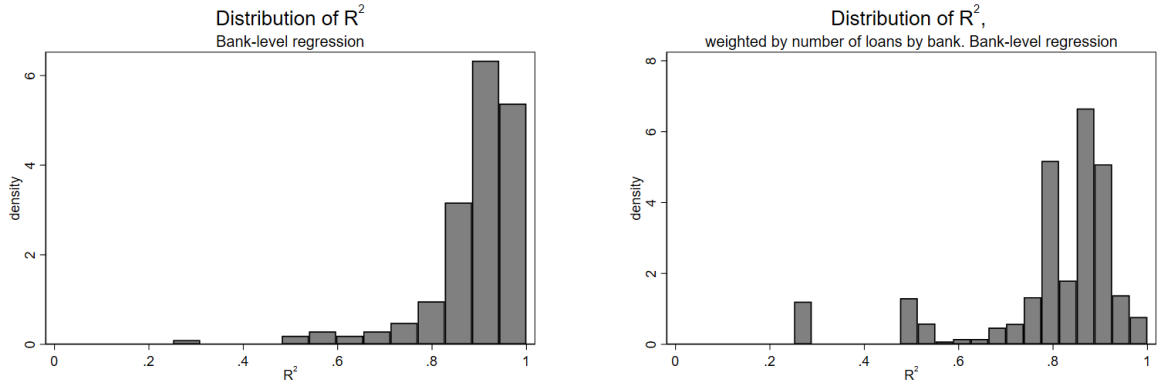


Figure B.5: The right image reports the distribution of R^2 weighted by the number of loans issued.

In the following section the credit rating of firms will be used as a source of public information available to banks and to study the evolution of firms' outcomes in terms of their default probability. Consequently, it is interesting at this stage to report to what extent banks use these ratings. Firstly, the same estimation is repeated by including only the rating as the explanatory variable, along with time and loan type FE and by lender, and collecting the R^2.

$$\text{Rate}_{i,b,t} = \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{i,b,t}, \tag{9}$$

where the only fixed effects kept are by loan type and month.

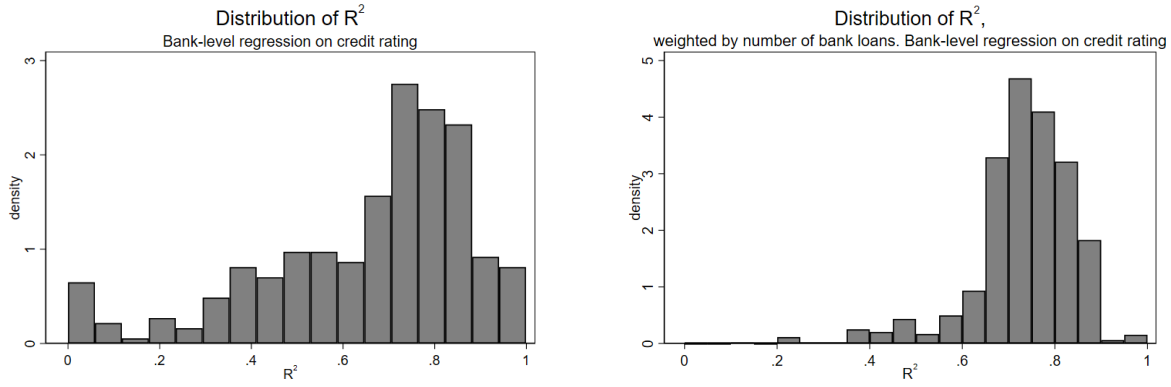


Figure B.6: The right image reports the distribution of R^2 weighted by the number of loans issued.

Secondly, one can look at how much of the variation is explained when including the public rating,

in particular by collecting the increment in the R^2.

$$\begin{aligned} \text{Rate}_{i,b,t} &= \alpha + \lambda_t + \varepsilon_{1,i,b,t}, && \rightarrow \text{Get } R_1^2 \\ \text{Rate}_{i,b,t} &= \alpha + \beta \cdot \text{Firm Rating}_{i,b,t} + \lambda_t + \varepsilon_{2,i,b,t}, && \rightarrow \text{Get } R_2^2 \\ \Delta R^2 &= R_2^2 - R_1^2, \end{aligned}$$

where the only fixed effects kept are by loan type and month.

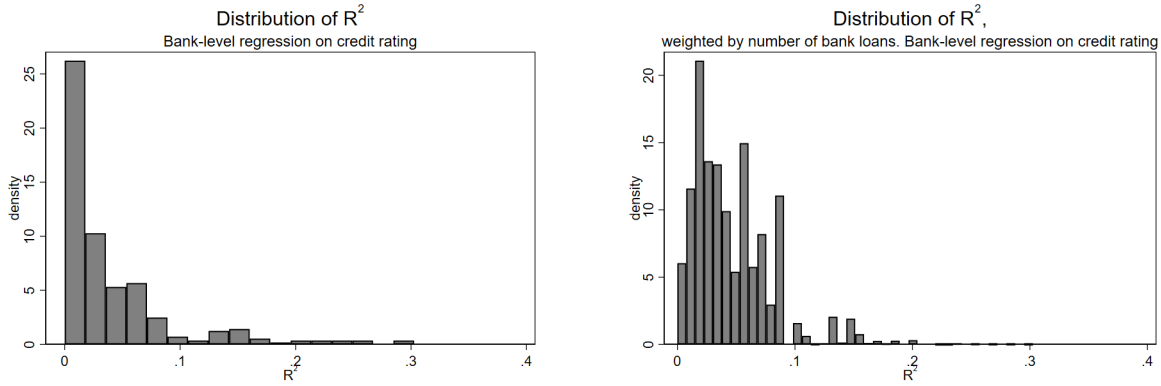


Figure B.7: The right image reports the distribution of R^2 weighted by the number of loans issued.

B.4 Residuals and market competition

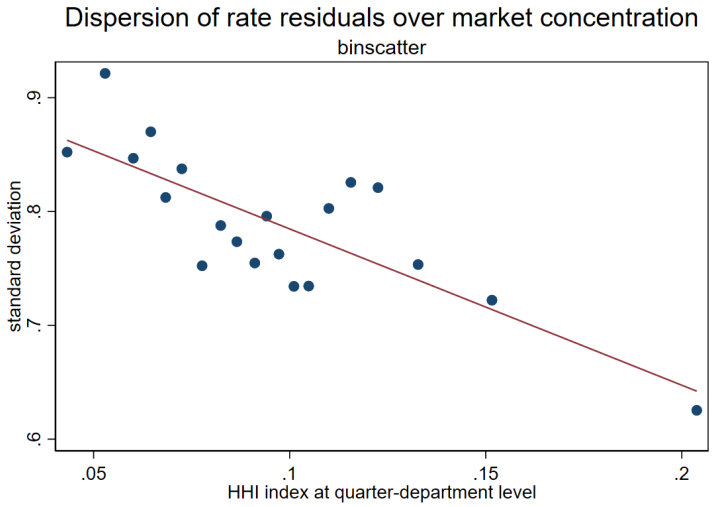


Figure B.8: Unexplained rate residuals' dispersion over HHI by department-quarter.

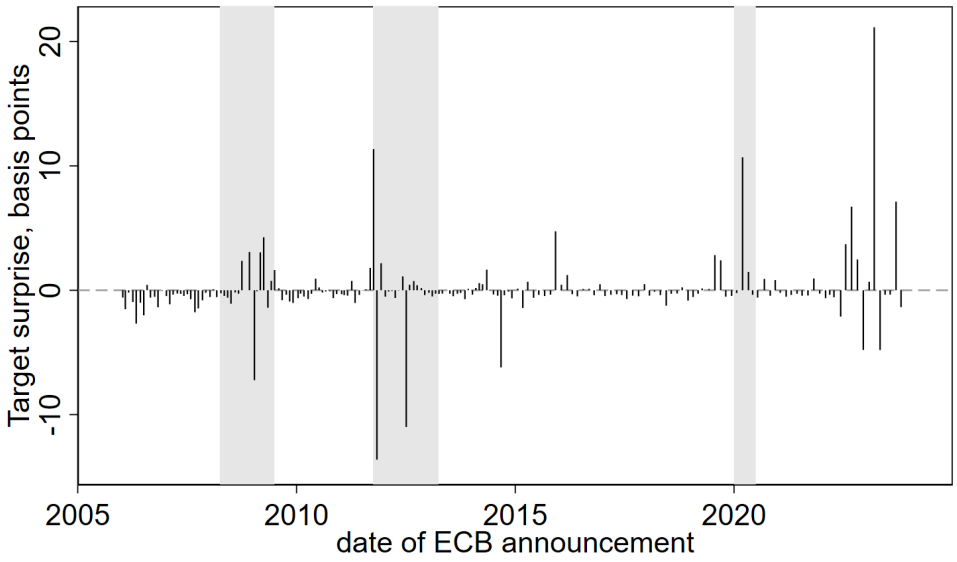
B.5 Rate dispersion and monetary policy

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Monetary policy shocks. I use the monetary policy surprises of Altavilla et al. (2019), updated to October 2023. For each ECB announcement, the Euro Area Monetary Policy Event-Study Database records the change in the 1-, 3- and 6-month and 1-, 2-, 5- and 10-year OIS rates in a narrow window around the press release and around the press conference. In each window I extract three latent factors by principal components and rotate them so that they can be read as policy

surprises: the second and third factors do not load on the one-month OIS rate, and the third has the smallest variance before the 2008 crisis. The identifying assumption is that policy decisions do not respond to asset prices at this frequency, and given the narrow window the movement in asset prices measures the surprise component of the decision. I use the Target factor, from the press-release window, which captures surprises in the current policy rate; it is standardized such that it has a unit effect on the one-month OIS rate. Figure B.9 plots the Target surprise at each announcement.

Figure B.9: Target surprise of Altavilla et al. (2019) at each ECB announcement, January 2006 to October 2023, in basis points: the factor is scaled to have a unit effect on the one-month OIS rate. It is estimated from changes in OIS rates in a narrow window around the ECB press release. Shaded bands are French recessions (AFSE): 2008q2–2009q2, 2011q4–2013q1 and 2020q1–2020q2.



Specification. For each lender and quarter I compute the standard deviation and the skewness of the rate residuals from Eq. (1) across that lender's loans, and regress these moments on the lagged policy surprises, aggregated at the quarterly level. On 7189 lender-quarter observations, I estimate a specification in levels, which controls for the lagged moment because dispersion is persistent,

$$Y_{b,t} = \beta_0 + \beta_1 \cdot \text{MPshock}_{t-1} + \beta_2 Y_{b,t-1} + \text{FE}_b + \varepsilon_{b,t}, \tag{10}$$

where $Y_{b,t}$ is either the standard deviation or the skewness. The coefficient β_1 measures the effect of a surprise on the level of dispersion, holding its own lag fixed. I report it with one lag of the shock and with three.

	Dispersion		Skewness	
	(1)	(2)	(3)	(4)
L1.rate shock	-0.0096***	-0.0085***	-0.0135***	-0.0099*
	(0.000)	(0.003)	(0.010)	(0.093)
L2.rate shock		0.0001		-0.0227***
		(0.979)		(0.000)
L3.rate shock		-0.0017		-0.0208***
		(0.568)		(0.000)
L1.SD / L1.Skew	0.2813***	0.2799***	0.1294***	0.1171***
	(0.000)	(0.000)	(0.000)	(0.000)
Lender FE	✓	✓	✓	✓
Observations	7,189	7,077	7,189	7,077
R^2	0.40	0.41	0.23	0.24

p-values in parentheses

* p<0.10, ** p<0.05, *** p<0.01

Table B.5: Response of the distribution of rate residuals to monetary policy. Each observation is a lender-quarter. Dispersion and Skewness are the standard deviation and the skewness of the lender's rate residuals from Eq. (1) in the quarter. Each column estimates Eq. (10), with one lag of the shock in columns (1) and (3) and three lags in columns (2) and (4). The shock is the Target factor of Altavilla et al. (2019) summed over the ECB announcements in the quarter, in standard deviations of the announcement-level factor (one standard deviation is 2.7 basis points of the one-month OIS rate). Data: Nouveaux Crédits (BdF), 2006q1–2024q2, quarterly shocks. The loan-level data refer to the first month of each quarter, so they are effectively monthly.

Results. Table B.5 reports the estimates. A monetary tightening reduces the dispersion of the unexplained component of loan rates. The effect on the level of the standard deviation is negative and significant at 1% in both specifications (-0.0096 and -0.0085). It is concentrated in the quarter after the shock: the second and third lags are small and not significant. The lagged dependent variable enters at 0.28, confirming that dispersion is persistent, which is why the specification controls for it; with this persistence, the effect of a shock fades within about two quarters. Since one standard deviation of the shock is a surprise of 2.7 basis points in the one-month OIS rate, a surprise of 25 basis points lowers the standard deviation of a lender's rate residuals by about 0.09 percentage points.

The skewness moves in the same direction. A tightening reduces the level of skewness, significantly at 1% with one lag and at 10% with three, and the second and third lags of the shock are individually significant and negative, suggesting that the tail adjusts more slowly than the dispersion does.

One concern is that dispersion falls because the composition of borrowers changes after a tightening. The construction of the residual limits this: it already nets out the loan contract, the firm's observables, its rating class and the lender's schedule in that month, so a shift towards safer or larger borrowers is absorbed by the controls rather than showing up in the residual.

B.6 Alternative specifications: Rate residuals and changes in default probability

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This section reports four alternatives to Eq. (2). The first replaces the rating revision with the default itself. The other three turn the regression around and put the revision in default probability on the left, so that the coefficient of interest is the one on the rate r_{ilbt} itself. The reversal changes the normalization, not the question: a positive coefficient says that firms charged a higher rate at issuance are revised towards a higher default probability afterwards, which is what $\beta_k > 0$ says in Eq. (2). Magnitudes are not comparable across the two directions - one is measured in percentage points of rate per percentage point of default probability, the other the other way round.

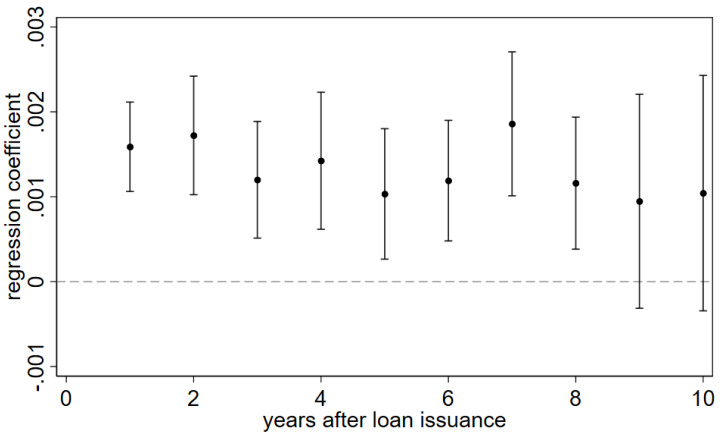
B.6.1 Actual defaults

I repeat the test replacing the rating revision with the default itself; for each horizon k, I run:

$$r_{ilbt} \;=\; \alpha \;+\; \sum_{j=1}^{k} \beta_j^D \, D_{i,t+j} \;+\; \mathbf{x}'_{ilbt}\,\gamma \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{11}$$

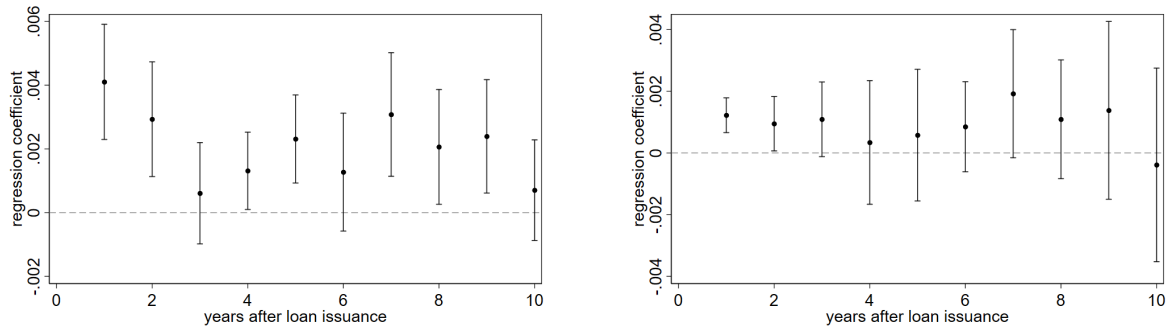
in which $D_{i,t+j}$ equals 100 if firm i defaults in the j-th year after issuance and 0 otherwise. This test uses the annual rating file, which records a default whenever it occurs, so it needs no balanced panel; it adds a dummy for a new bank-firm relationship to $\mathbf{x}_{ilbt}$ and clusters standard errors by lender and by quarter. I estimate Eq. (11) once per horizon $k = 1, \ldots, 10$, each regression carrying the default indicators up to that horizon, and report β_k^D.

Figure B.10: Regression coefficient of actual default, controlling for lags and observables, on interest rates.



Rating revisions on the left-hand side. Three choices distinguish the three specifications that follow: whether the revision is the year-on-year change or the cumulative change since issuance, whether the earlier revisions enter as controls, and whether the sample is balanced across horizons. Everything else is held fixed. All three carry the same controls $\mathbf{x}_{ilbt}$ and fixed effects $\mu_{bt\tau s}$ as Eq. (2), cluster standard errors by lender, drop loans to firms already in default at issuance ($P_{i,t} \neq 1$), and run one regression per horizon $k = 1, \ldots, 10$, reporting the coefficient on the rate. The rating-class

Figure B.11: Correlation computed separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out.



split is the one used in Fig. B.18: classes 3++ to 4+ for the good group and classes 5 to 9 for the bad one, with classes 4 and 5+ left out.

B.6.2 Year-on-year revisions, past revisions controlled for, balanced sample

For each horizon k I estimate

$$\Delta P_{i,t+k} \;=\; \alpha_k \;+\; \theta_k\, r_{ilbt} \;+\; \sum_{j=1}^{k-1} \delta_{kj}\, \Delta P_{i,t+j} \;+\; \mathbf{x}'_{ilbt}\, \gamma_k \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{12}$$

in which $\Delta P_{i,t+k}$ is the change in default probability between years $k-1$ and k after issuance, in percentage points, as in Eq. (2). The sample is balanced: a loan enters only if every revision $\Delta P_{i,t+1}, \ldots, \Delta P_{i,t+10}$ is observed, so the same firms identify every θ_k. This is the closest counterpart to Eq. (2), which is also estimated on the balanced panel. The rate predicts the first-year revision, $\theta_1 = 0.064$ (s.e. 0.017), and the coefficient falls towards zero by the tenth year. The pattern holds for the good rating classes and is too noisy to sign for the bad ones.

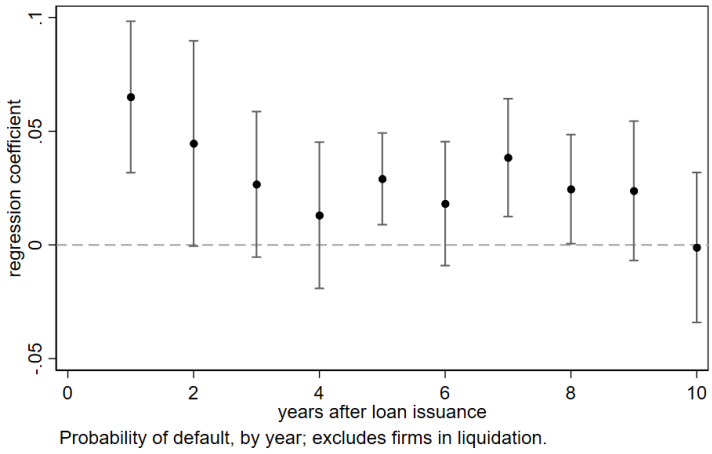
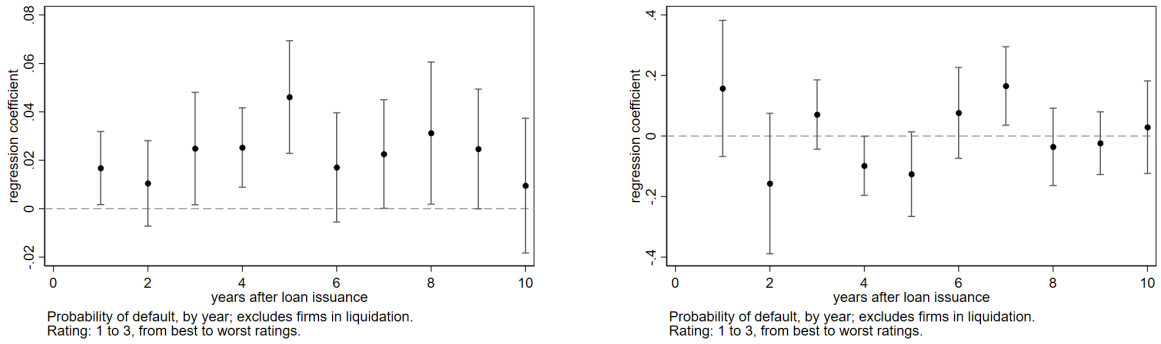


Figure B.12: Coefficient θ_k on the loan rate in Eq. (12), by horizon k. The dependent variable is the year-on-year change in the firm's default probability between years $k-1$ and k after issuance, in percentage points. Balanced sample; past revisions controlled for; bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.13: Coefficient θ_k in Eq. (12), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. Note the different vertical scales.



B.6.3 Year-on-year revisions, past revisions controlled for, unbalanced sample

Eq. (12) keeps only firms followed for the full ten years, which selects on survival in the rating file. Dropping that requirement, I estimate the same equation horizon by horizon on whatever firms are observed at that horizon:

$$\Delta P_{i,t+k} \;=\; \alpha_k \;+\; \theta_k\, r_{ilbt} \;+\; \sum_{j=1}^{k-1} \delta_{kj}\, \Delta P_{i,t+j} \;+\; \mathbf{x}'_{ilbt}\, \gamma_k \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{13}$$

so the sample shrinks with k and differs across horizons. The coefficients are larger than on the balanced panel - $\theta_1 = 0.14$ (s.e. 0.022) - and decline to zero by the tenth year. The gap between the two samples says that the firms that stay in the rating file for ten years are the ones whose rates carry less information about their own revisions.

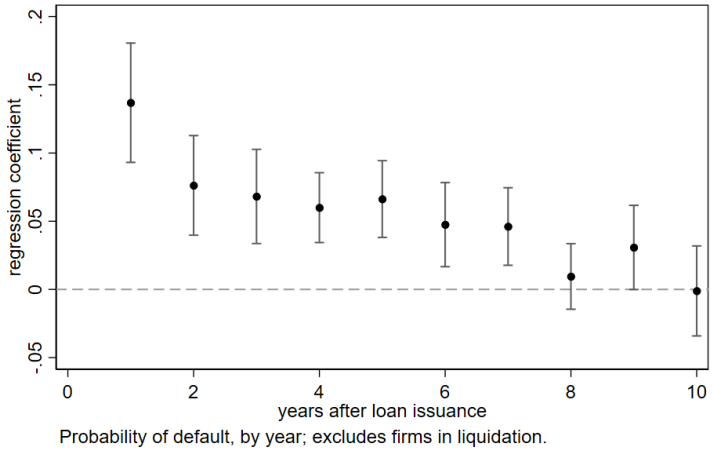
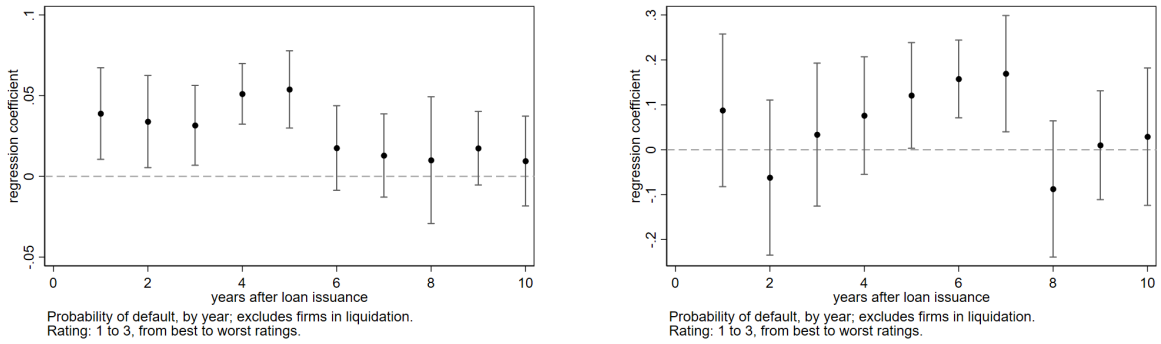


Figure B.14: Coefficient θ_k on the loan rate in Eq. (13), by horizon k. The dependent variable is the year-on-year change in the firm's default probability between years $k-1$ and k after issuance, in percentage points. Unbalanced sample; past revisions controlled for; bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.15: Coefficient θ_k in Eq. (13), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. Note the different vertical scales.



B.6.4 Cumulative revision since issuance

The year-on-year revision at horizon k is an increment, so θ_k in Eqs. (12) and (13) measures how much of the revision arriving in year k alone the rate anticipates. Measuring the revision from issuance instead asks how much of the firm's total drift in default probability the rate anticipates by year k:

$$\Delta P_{i,t\to t+k} \;=\; \alpha_k \;+\; \theta_k\, r_{ilbt} \;+\; \mathbf{x}'_{ilbt}\,\gamma_k \;+\; \mu_{bt\tau s} \;+\; \varepsilon_{ilbt}, \tag{14}$$

in which $\Delta P_{i,t\to t+k} = 100 \times \big(P_{i,t+k} - P_{i,t}\big)$, the same cumulative object used for the demographic splits in Fig. 4. Past revisions no longer enter, since the dependent variable already contains them. At $k = 1$ the two measures coincide and Eq. (14) reproduces Eq. (13), $\theta_1 = 0.14$. Beyond that the coefficient stays positive at every horizon, peaking at $\theta_5 = 0.18$ (s.e. 0.023) and still 0.11 at ten years. The rate anticipates a level shift in the firm's default probability that does not unwind, and the decline in Fig. B.14 reflects the exhaustion of new information, not a reversal of the old.

Splitting by rating class sharpens the reading. For the good classes the cumulative coefficient rises from 0.039 at one year to 0.14 at five and is positive at every horizon. For the bad classes it is two to three times larger at the longer horizons but is not distinguishable from zero in the first three years: rates predict where bad borrowers end up, not when they get there.

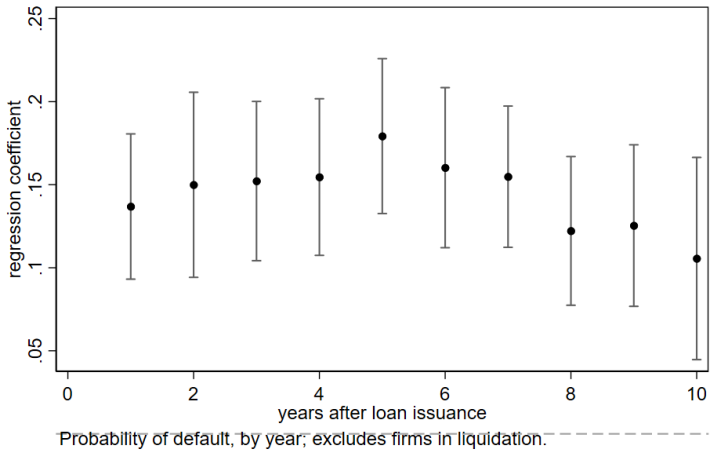
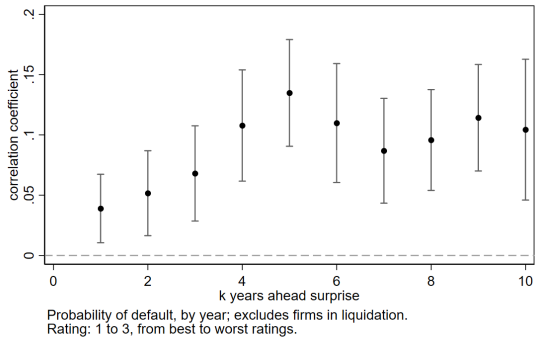
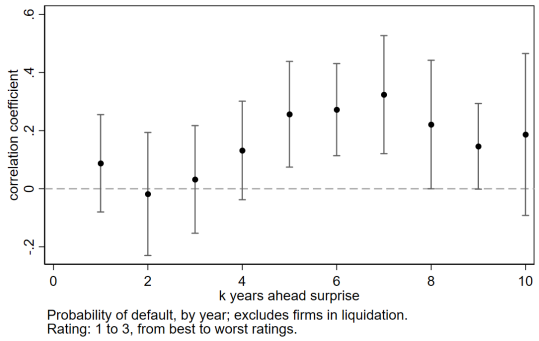


Figure B.16: Coefficient θ_k on the loan rate in Eq. (14), by horizon k. The dependent variable is the cumulative change in the firm's default probability between issuance and year k, in percentage points. Bars are 95% confidence intervals, standard errors clustered by lender.

Figure B.17: Coefficient θ_k in Eq. (14), estimated separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out. The horizontal axis is the horizon k of the cumulative revision. Note the different vertical scales.

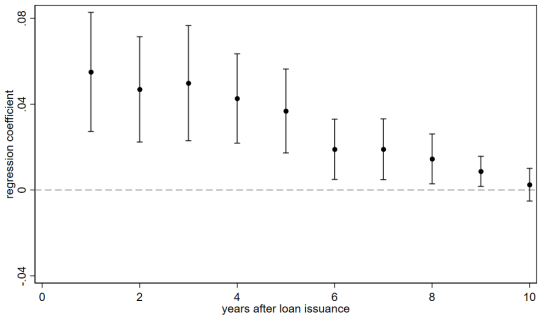




B.7 Correlation across firms' demographics

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Figure B.18: Correlation computed separately for firms with a good ex ante rating, classes 3++ to 4+ (left), and a bad one, classes 5 to 9 (right). Classes 4 and 5+ are left out.



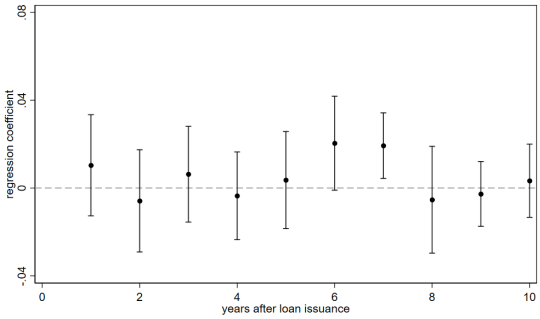
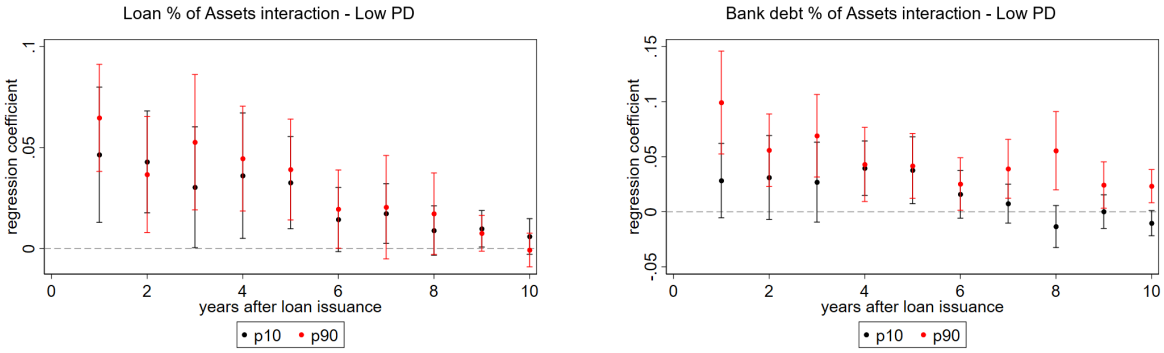


Figure B.19: Interactions of ex-post default probability and loan-size over assets (left) and bank debt over assets (right)



The figure compares the correlation coefficients of residual rates and unexpected changes in ex-post default probability across percentiles of loan-size over assets (left) and size of bank debt over assets (right) for firms with a good ex ante rating (classes 3++ to 4+).

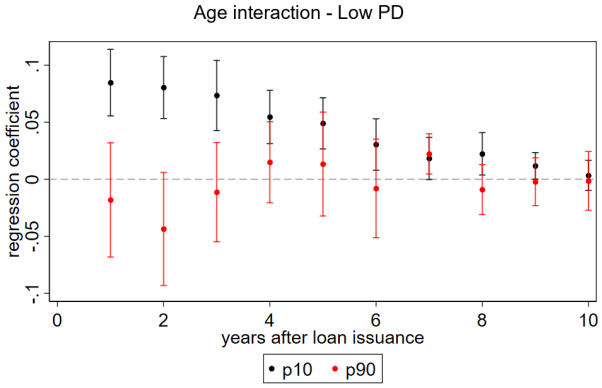


Figure B.20: The figure shows the correlation coefficients of the interactions of ex-post default probability and age (below 10th vs above 90th age percentile). Sample of firms with a good ex ante rating (classes 3++ to 4+).

B.7.1 Fixed vs variable rate test

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	Default Probability						
Variable rate (dummy)	0.1176**		0.1407	0.1039**			
	(0.014)		(0.127)	(0.026)			
cum dRate end y2		0.0433			0.0374		
		(0.229)			(0.254)		
annual dRate end y1			0.0812			0.0524	0.0425
			(0.600)			(0.711)	(0.773)
annual dRate end y2			0.0567			0.0363	0.0128
			(0.330)			(0.551)	(0.879)
Variable rate (dummy) × Loan over assets				0.1932			
				(0.370)			
cum dRate end y2 × Loan over assets					0.0711		
					(0.264)		
annual dRate end y1 × Loan over assets						-0.1330	
						(0.763)	
annual dRate end y2 × Loan over assets						0.0905	
						(0.148)	
annual dRate end y1 × Bank Debt share							0.0058
							(0.980)
annual dRate end y2 × Bank Debt share							0.0986
							(0.623)
Loan over assets	0.2250**	0.2002**	0.2048**	0.1808*	0.1894*	0.1849*	0.1929*
	(0.030)	(0.035)	(0.035)	(0.094)	(0.054)	(0.059)	(0.054)
Bank Debt share	0.0846**	0.1139**	0.1134**	0.0850**	0.1118**	0.1114**	0.1127**
	(0.040)	(0.019)	(0.019)	(0.034)	(0.022)	(0.023)	(0.025)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	102,463	96,414	96,414	102,463	96,414	96,414	96,414
R^2	0.42	0.40	0.40	0.42	0.40	0.40	0.40

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.6: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Term loans, at least 30-month maturity, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

Identifying assumptions. The exercise rests on three assumptions.

(i) *Exogeneity.* Reference-rate changes are set by monetary policy and aggregate conditions, and it is realistic to assume they are not endogenous to expectations about firm i's outcomes.

(ii) *No sorting on private information or rate expectations.* The main concern is that firms may self-select into fixed- or variable-rate contracts based on their expectations about the future evolution of rates. As explained below, I mitigate this issue in two ways.

(iii) *Parallel trends.* Absent the rate increase, default probabilities of fixed- and floating-rate borrowers would have evolved in the same way.

The rest of this section gives evidence on (ii) and (iii), and then repeats the exercise on the sample for which (ii) is least likely to bind.

Self-selection and parallel trends. I mitigate concern (ii) by showing that the contract choice is predominantly determined by supply-side lender preferences rather than firm-specific unobservables. Focusing on firms with a rating at the time of the loan, I estimate a linear probability model of the form:

$$Floating_{loan} = \beta X^{observables} + \delta_{t,b,p,s} + \varepsilon_{loan}, \tag{15}$$

where $Floating_{loan}$ is a dummy for variable-rate loans and $\delta_{t,b,p,s}$ is the fixed effects interaction term for month, bank, loan purpose and firm sector. Following a variance decomposition approach, I gradually saturate the model with fixed effects. As shown in Table B.7, the lender-month dimension alone accounts for 68% of the variation, with the full specification explaining 92% of the total variance. This suggests that the contract choice is predominantly driven by supply-side and sectoral factors rather than idiosyncratic firm risk; in fact, banks may prefer to offer floating rate contracts to match the rate structure of their own liabilities (see Kirti (2020)).

Table B.7: Variance Decomposition of Variable Rate Choice. Data: Nouveaux Crédits (BdF). The loan-level data refer to the first month of each quarter, so they are effectively monthly.

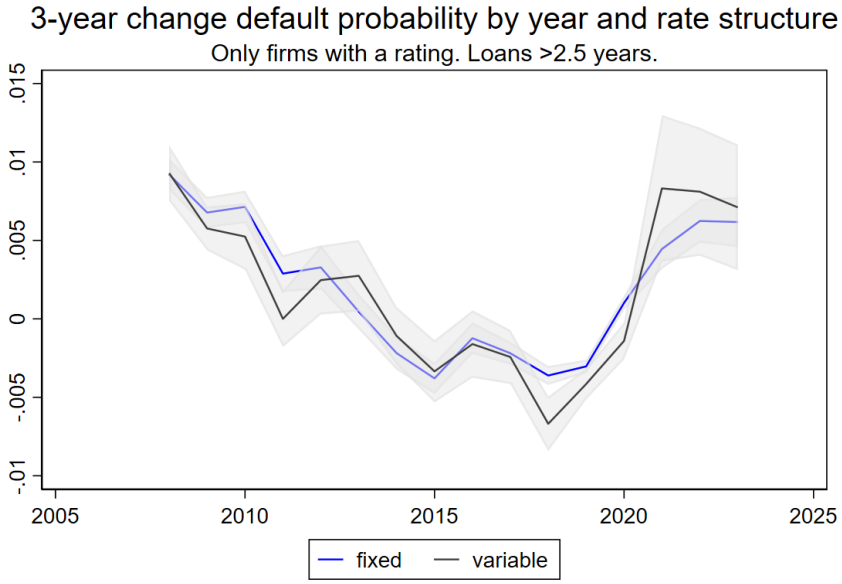
Fixed Effects Included	**Adjusted R^2**
Quarter	0.04
+ Lender	0.68
+ Loan Type	0.89
+ Firm Sector	0.92

I check the balance of observable characteristics between firms taking fixed and variable rate loans. While I find that the two groups differ in some characteristics (variable: fewer firms, larger size, older, riskier, larger financial debt), I include all those characteristics as controls in the following regression, to mitigate the issue of selection bias. I also check the parallel trends assumption by plotting the evolution of rating transitions for firms taking fixed vs variable rate loans. I find that rating transitions for firms taking variable rate loans do not differ significantly from those taking fixed rate loans during the period considered. Fig. B.21 plots the two series.

This evidence overall mitigates concerns about selection bias.

Restrict to loans issued in 2021q3–q4. I repeat the baseline exercise in Eq. (3) on loans issued in the second half of 2021: the policy rate was still at its floor, and the tightening that followed in 2022 and 2023 came as a surprise. The crucial point is that a firm choosing between a fixed- and a floating-rate contract in those two quarters likely was not anticipating a rate increase, so restricting the timing to the end of 2021 mitigates the possibility of sorting on rate expectations. This is the version of the exercise in which the leverage channel, if it operated, would be hardest to hide. Here I lower the maturity cut to 24 months: the 2021q3–q4 window leaves too few loans at the 30-month threshold for the estimator to report standard errors on several coefficients. I run the exercise including the cumulative reference rates increase in the increase to 2022 (Table B.8), and observing the change in probability of default of firms in 2023. For robustness, I also include

Figure B.21: Mean three-year change in the default probability by year, for fixed- and floating-rate borrowers separately. Term loans over 2.5 years, firms with a rating. Shaded: 95% confidence bands.



the rate increase in 2023, although firms' rating revision might have happened well before the end of the year (Table B.9). I again find no significant effect of reference rate increases on rating worsening.

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698		-6.1935	-0.5121			
	(0.169)		(0.193)	(0.186)			
annual dRate end y2			1.8860			-0.3828*	-0.4445
			(0.261)			(0.095)	(0.218)
Variable rate (dummy) × Loan over assets				-0.6271			
				(0.249)			
annual dRate end y2 × Loan over assets						-0.3410	
						(0.228)	
annual dRate end y2 × Bank Debt share							0.0555
							(0.915)
Loan over assets	-0.5329	-1.1142	-1.1243	-0.1584	-0.6316	-0.6318	-1.1395
	(0.374)	(0.168)	(0.216)	(0.740)	(0.224)	(0.223)	(0.177)
Bank Debt share	0.0986	-0.2614	-0.2672	0.1401	-0.2234	-0.2234	-0.2780
	(0.725)	(0.316)	(0.448)	(0.631)	(0.375)	(0.375)	(0.364)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R^2	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.8: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

	Default Probability - 2021q3-2021q4 only						
Variable rate (dummy)	-0.5698		-19.6243	-0.5121			
	(0.169)		(0.479)	(0.186)			
annual dRate end y2			5.0816			0.0935	0.6439
			(0.477)			(0.943)	(0.316)
annual dRate end y3			2.6790			-0.7103	-1.5704
			(0.563)			(0.718)	(0.179)
Variable rate (dummy) × Loan over assets				-0.6271			
				(0.249)			
annual dRate end y2 × Loan over assets						5.0753	
						(0.781)	
annual dRate end y2 × Bank Debt share							-0.6506
							(0.424)
annual dRate end y3 × Loan over assets						-7.6615	
						(0.769)	
annual dRate end y3 × Bank Debt share							0.9643
							(0.155)
Loan over assets	-0.5329	-1.1153	-1.1298	-0.1584	-0.6351	-0.6550	-1.1379
	(0.374)	(0.168)	(0.233)	(0.740)	(0.216)	(0.287)	(0.283)
Bank Debt share	0.0986	-0.2617	-0.2685	0.1401	-0.2242	-0.2431	-0.2795
	(0.725)	(0.313)	(0.569)	(0.631)	(0.369)	(0.431)	(0.674)
Lender × Month × Loan Type × 2digit Sector FE	✓	✓	✓	✓	✓	✓	✓
Region FE	✓	✓	✓	✓	✓	✓	✓
All other observables	✓	✓	✓	✓	✓	✓	✓
Observations	4,324	4,089	4,089	4,324	4,089	4,089	4,089
R^2	0.41	0.41	0.41	0.41	0.41	0.41	0.41

p-values in parentheses

* p< 0.10, ** p< 0.05, *** p< 0.01

Table B.9: Data: Nouveaux Crédits (BdF), 2006q1-2024q2. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Term loans, maturity at least 24 months, only firms with a rating. Dependent variable: change_PD3. Exclude loans with non-specified reference rate.

B.8 Rates and loan quantity

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In Section 4 I deviate from the assumption of menu contracts. Table B.10 shows the evidence behind this choice. Conditional on all observable characteristics in my sample, there is a negative relationship between rates and quantity: a loan twice as large pays about 0.07 percentage points less (column 2). Loan size relative to firm assets has no significant effect on the rate (column 1).

Table B.10: Loan rates and loan size. Each column reruns the baseline regression (Table B.2, column 6) keeping one measure of loan size: loan over assets in column (1), log loan quantity in column (2). Data: Nouveaux Crédits (BdF), 2006q1–2024q2, excluding credit lines and undefined maturity products. The loan-level data refer to the first month of each quarter, so they are effectively monthly. Variables defined as in Table B.2.

	Total effective rate	
	(1)	(2)
log.Loan quantity (K-Eur)		-0.0960***
		(0.000)
Loan over assets	-0.0437	
	(0.735)	
Maturity (months)	0.0012**	0.0020***
	(0.034)	(0.005)
Age	-0.0044***	-0.0038***
	(0.000)	(0.000)
Age squared	0.0000***	0.0000***
	(0.000)	(0.000)
Young firm 0-3y (dummy)	-0.0799***	-0.0653**
	(0.006)	(0.015)
New bank-firm relationship (dummy)	-0.0422***	-0.0379**
	(0.009)	(0.010)
Variable rate (dummy)	0.0907	0.1479
	(0.335)	(0.107)
Personal loan (dummy)	0.1023	0.0741
	(0.136)	(0.309)
Individual entrepreneur (dummy)	0.2617***	0.2434***
	(0.001)	(0.001)
Pledged guarantees	-0.4114***	-0.3477***
	(0.001)	(0.000)
Revenues	0.0054	-0.0030
	(0.464)	(0.630)
Bank Debt	-0.0270	-0.0249
	(0.400)	(0.411)
Non-Bank Debt	-0.0301	-0.0296
	(0.689)	(0.701)
Mobilised credit	0.0413	0.0714**
	(0.158)	(0.013)
Short Term Debt	-0.2244***	-0.1596***
	(0.000)	(0.002)
Undrawn credit lines	-0.1253	-0.0255
	(0.146)	(0.833)
Multiple products (dummy)	0.0016	0.0024
	(0.887)	(0.843)
Multiple loans, one product (dummy)	0.0124	-0.0074
	(0.608)	(0.781)
Lender × Quarter × Loan Type × 2digit Sector FE	✓	✓
Region FE	✓	✓
Group indicator, Rating FE	✓	✓
Observations	449,973	451,305
R^2	0.82	0.83

p-values in parentheses
* p< 0.10, ** p< 0.05, *** p< 0.01

C Model appendix

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C.1 Proofs of the properties of equilibria

This appendix collects the proofs of the three properties stated in Section 4.2.

Proof of Lemma 4.1. Suppose banks post some rate r with probability $m > 0$. A shopper observing two quotes equal to r splits its business between the two banks, so a bank posting r wins that firm half of the time, with probability $m/2$. If one bank reduces the rate by an arbitrarily small amount ε, it wins the remaining $m/2$ fraction of shoppers, at the lower profit for trade $r - \varepsilon$. As ε goes to zero, the deviation raises the acceptance probability by $\pi m/2$, while the profit per trade it gives up vanishes. Therefore, each bank has an incentive to lower the rate by an infinitesimally small ε and gain an additional $\pi m/2$ without giving up profits. Such a distribution cannot be an equilibrium. The argument holds also at the lowest point of the separating region, where lowering the rate just above $\bar{r}^g$ brings the good types back in the market and lowers the expected cost of lending from c_b to $c^{avg,s}$, so the deviation pays there too.

Proof of Lemma 4.2. Write $\bar{r}^s$ for the highest rate in the support. Because no rate is offered beyond it, $F^s(\bar{r}^s) = 1$, and by Lemma 4.1 a bank posting it loses every shopper to the rival: it sells to captive firms only, and earns $(1 - \pi)$ times its margin. Assume $\bar{r}^s$ sits in the interior of one of the regions, pooling or separating. The per-trade margin rises with the rate inside each region, so a bank can raise its rate to $\bar{r}^s + \varepsilon$, trade with the same $(1 - \pi)$ captive firms, that will accept as $\bar{r}^s$ is below the reservation rate, and earn a higher per-trade margin. At $\bar{r}^g$ or $\bar{r}^b$, on the contrary, raising the rate means losing a set of borrowers: $\bar{r}^g$, above which good types stop accepting, and $\bar{r}^b$, above which nobody accepts. The choice between the two reservation rates will depend on banks' profit expectations.

Proof of Lemma 4.3. Assume that the two supports were not disconnected. Take two banks offering $\bar{r}^g + \varepsilon$, with probability of acceptance of a shopper equal to $1/2$. For infinitesimally small ε, each bank could undercut its rival by offering $\bar{r}^g$: the margin on bad types would not change, but the mass of firms suddenly jumps as good firms would re-enter the market. Such deviation would be profitable. Therefore, the support of F^s must be disconnected, with a gap between $\bar{r}^g$ and the lower bound on the separating region $\underline{r_b}^s$, such that $\underline{r_b}^s > \bar{r}^g$.

C.2 Derivations for the signal-group equilibrium

This appendix derives Proposition 4.2: the two branches of the rate distribution within a signal group, the bounds of their supports, and the threshold between the hybrid and the bad-only regime. Throughout, α^s is the share of good types among firms carrying signal s, $c^{avg,s} \equiv \alpha^s c_g + (1 - \alpha^s) c_b$ is the bank's expected cost of lending to such a firm, and

$$A^s \equiv \bar{r}^g - c^{avg,s}, \qquad B^s \equiv (1 - \alpha^s)\left(\bar{r}^b - c_b\right)$$

are the two corner profits per unit of demand, so that $\phi^s = A^s - B^s$ and $\Pi^s = (1 - \pi)\max\{A^s, B^s\}$.

Bank profits within a group. For firms carrying signal s,

$$\Pi^s(r) = \begin{cases} [1-\pi+\pi(1-F^s(r))]\,(r-c^{avg,s}) & \text{if } r \le \bar{r}^g, \\ (1-\alpha^s)\,[1-\pi+\pi(1-F^s(r))]\,(r-c_b) & \text{if } \bar{r}^g < r \le \bar{r}^b. \end{cases}$$

Both types accept a rate in the first region, and the bank lends at the expected cost $c^{avg,s}$; only bad types accept in the second, and the bank lends at c_b to the fraction $(1-\alpha^s)$ of the group that is of the bad type.

The two branches. Equilibrium requires the same profit Π^s at every posted rate. On the pooling region this gives

$$[1-\pi F^s(r)]\,(r-c^{avg,s}) = \Pi^s \qquad \Longrightarrow \qquad F^s(r) = \frac{1}{\pi}\left[1 - \frac{\Pi^s}{r-c^{avg,s}}\right],$$

and on the bad-only region

$$(1-\alpha^s)\,[1-\pi F^s(r)]\,(r-c_b) = \Pi^s \qquad \Longrightarrow \qquad F^s(r) = \frac{1}{\pi}\left[1 - \frac{\Pi^s/(1-\alpha^s)}{r-c_b}\right].$$

Under full pooling the top of the support is $\bar{r}^g$ by Lemma 4.2, so $\Pi^s = (1-\pi)A^s$ and the first expression is the pooling branch of Proposition 4.2. When $\phi^s \le 0$ the top of the support is $\bar{r}^b$ and $\Pi^s = (1-\pi)B^s$; substituting it in the first expression gives F^s_{pool}, and in the second it gives F_{bad}, because $\Pi^s/(1-\alpha^s) = (1-\pi)(\bar{r}^b - c_b)$ no longer involves α^s. The bad-only branch is therefore the same function in both signal groups: above $\bar{r}^g$ the bank prices against bad types only, and the probability of meeting one scales profit and the required margin in the same proportion.

Bounds and the gap. Setting each branch to zero gives the bottom of the support: $c^{avg,s} + (1-\pi)A^s$ under full pooling, $c^{avg,s} + (1-\pi)B^s$ in the hybrid regime, and $c_b + (1-\pi)(\bar{r}^b - c_b)$ in the bad-only regime. In the hybrid regime the bad-only branch starts where it reaches the mass already posted below $\bar{r}^g$, that is at the rate solving $F_{\text{bad}}(r) = F^s_{\text{pool}}(\bar{r}^g)$, which is $r = c_b + A^s/(1-\alpha^s)$. The support therefore has a gap just above $\bar{r}^g$, of width

$$c_b + \frac{A^s}{1-\alpha^s} - \bar{r}^g = \frac{\alpha^s}{1-\alpha^s}\,(\bar{r}^g - c_g) > 0,$$

which closes only as $\alpha^s \to 0$. Raising the rate from $\bar{r}^g$ costs the bank the good types in the group, and it posts again only once the rate compensates for that loss.

The hybrid threshold. The pooling branch carries positive mass if and only if $F^s_{\text{pool}}(\bar{r}^g) > 0$, that is $A^s > (1-\pi)B^s$, which rearranges to

$$\phi^s > -\frac{\pi}{1-\pi}\,(\bar{r}^g - c^{avg,s}) \equiv \underline{\phi}^s,$$

the threshold between the hybrid and the bad-only regime of Section 4.3.2. The friction is what keeps the good types in: a bank posts in the pooling region only because a shopper may bring it the whole surplus of a good type, and as $\pi \to 0$ the threshold rises to zero and the hybrid regime disappears.

Convergence to perfect information. As the signal becomes conclusive ($q \to 1$) the posteriors

converge to certainty,

$$\alpha^{s_g} = \frac{q\mu_g}{q\mu_g + (1-q)\mu_b} \to \frac{1 \cdot \mu_g}{1 \cdot \mu_g + 0 \cdot \mu_b} = 1,$$
$$\alpha^{s_b} = \frac{(1-q)\mu_g}{(1-q)\mu_g + q\mu_b} \to \frac{0 \cdot \mu_g}{0 \cdot \mu_g + 1 \cdot \mu_b} = 0,$$

and the two groups separate:

- **Good signal group:** $c^{avg,s_g} \to c_g$, so

$$\phi^{s_g} \to \bar{r}^g - c_g > 0,$$

  and $F^{s_g}(r)$ converges to $F^g(r)$, the good-type distribution under a perfect signal.

- **Bad signal group:** $c^{avg,s_b} \to c_b$, so

$$\phi^{s_b} \to (\bar{r}^g - c_b) - \left(\bar{r}^b - c_b\right) = \bar{r}^g - \bar{r}^b < 0,$$

  and $F^{s_b}(r)$ converges to $F^b(r)$, the bad-type distribution under a perfect signal.

Since $\sigma_g \to \mu_g$ and $\sigma_b \to \mu_b$, the aggregate distribution converges to

$$F(r) \to \mu_g F^g(r) + \mu_b F^b(r),$$

the perfect-signal equilibrium of Section 4.3. At the other end, $q = 1/2$ gives $\alpha^{s_g} = \alpha^{s_b} = \mu_g$, the two groups merge, and ϕ^s collapses to the population index ϕ of the uninformative case.

C.3 Existence, uniqueness and deviations

Proof of Proposition 4.1. The signal matches the type, so the two markets are separate: the rival's distribution and the firm's acceptance decision both depend on the type alone, and a rate posted to a type-θ firm cannot attract the other type. The check therefore runs type by type. Within a type the distribution exists and is unique: Lemmas 4.1 and 4.2 fix the top of the support at $\bar{r}^\theta$ and the profit level at $\mu_\theta(1-\pi)\left(\bar{r}^\theta - c_\theta\right)$, and given those the equal-profit condition is linear in $F^\theta(r)$ with coefficient $-\pi \neq 0$, so it has one solution at every rate. It is a distribution, zero at the lower bound by its own definition, increasing over the support, and one at the top, where the bank lends to captive firms alone. Assume one bank wants to deviate. If it posts a rate r below the support, the acceptance rate is still one but it earns a lower margin on all borrowers. Thus, profits are linearly increasing in r if $r < \underline{r}^\theta$. If it posts a rate above $\bar{r}^\theta$, demand drops as all types reject, and profits are zero. Hence, no deviation is profitable.

Proof of Proposition 4.2. The three regimes cover the parameter space: the sign of ϕ^s separates pooling from the other two, and $\underline{\phi}^s$ separates hybrid from bad-only lending, so each signal group falls in exactly one of them. In every regime the distribution exists and is unique: Lemmas 4.1 and 4.2 fix the top of the support and the profit level Π^s, and given those the equal-profit condition is one equation with one solution at every rate. It is a distribution, zero at the lower bound by its own definition, weakly increasing over the support, flat across the gap, and one at the top,

where the bank lends to captive firms alone. Assume one bank wants to deviate. In the pooling regime, profits are linearly increasing in r below the lower bound, so any rate below the support is unprofitable. If the bank posts a rate above $\bar{r}^g$ and below $\bar{r}^b$, it gains more profits from bad types, but loses all good types; the net gain is $(1-\alpha^s)(1-\pi)(r-\bar{r}^g)-\alpha^s(1-\pi)(\bar{r}^g-c_g)$, which rises with r and equals $-(1-\pi)\phi^s$ at $\bar{r}^b$, so it is negative on the whole interval because $\phi^s>0$ by assumption. Finally, above $\bar{r}^b$, profits are zero as all firms reject. In the hybrid regime, the same argument holds for any rate below the support and above $\bar{r}^b$; if the deviator posts a rate on the disconnected region, it faces the acceptance probability it already faces at $\bar{r}^g$, because F^s is flat there, and a lower margin on bad types than at the bottom of the bad-only branch, while good ones would still reject. In the bad-only regime the whole pooling region lies below the support, as $\bar{r}^g < c_b + (1-\pi)(\bar{r}^b - c_b)$, so a bank posting there undercuts everyone and lends to both types: it earns at most $\bar{r}^g - c^{avg,s}$, which stays below Π^s precisely because $F^s_{\text{pool}}(\bar{r}^g)=0$ in this regime. Hence, no deviation is profitable.[17]

Details. The steps below remove one degree of freedom at a time, the level of profit, the support, and then the distribution, and the last one gives the algebra behind the rates the support leaves out. Fix a signal s and write Π^s for the profit a bank earns in equilibrium.

Step 1: every rate in the support earns Π^s. By Lemma 4.1 the rival posts no rate with strictly positive probability, so the probability that an offer is accepted, $1-\pi+\pi\,(1-F^s(r))$, is continuous in r, and so is the profit within each region. If two rates in the support earned different profits, a bank would move the probability it puts on the lower one to the higher one. In equilibrium, therefore, profits are equal across the support.

Step 2: the level of profit and the top of the support. By Lemma 4.2 the highest posted rate is $\bar{r}^g$ or $\bar{r}^b$, and the bank trades there with captive firms alone, so Π^s is $(1-\pi)\,(\bar{r}^g - c^{avg,s})$ in the first case and $(1-\alpha^s)(1-\pi)\left(\bar{r}^b - c_b\right)$ in the second. Both rates are available to the bank, so the equilibrium takes the larger of the two profits, which is the first when $\phi^s>0$ and the second when $\phi^s<0$: two equilibria with different tops cannot coexist. The two tie only at $\phi^s=0$, and there the pooling branch of Proposition 4.2 reaches one exactly at $\bar{r}^g$ while the bad-only branch carries no mass, so the two descriptions give the same distribution.

Step 3: the support has no gap inside a region. Take two rates $r_1<r_2$ in the support, in the same region, with no rate of the support between them. Then F^s is constant on $[r_1,r_2]$, and the expected cost of lending is constant within a region, so

$$\Pi^s(r_2)-\Pi^s(r_1)=\kappa\left[1-\pi+\pi\,(1-F^s(r_1))\right](r_2-r_1)>0,$$

with $\kappa=1$ on the pooling region and $\kappa=1-\alpha^s$ above $\bar{r}^g$, which contradicts Step 1. Each region therefore carries a single interval, and with Lemma 4.3 the support is one interval ending at $\bar{r}^g$ and at most one further interval ending at $\bar{r}^b$. This step uses constant lending costs: if the cost rose with the rate, it would need the margin $r-c(r)$ to rise with the rate as well.

[17] No restriction on beliefs off the equilibrium path is needed. The signal is public, so a bank's posterior about the firm it meets follows from Bayes' rule applied to the signal and does not depend on the rate the bank posts, and the share of each type that accepts a rate follows from the reservation rates and the rival's distribution. Profits are therefore defined at every rate, inside the support and outside it.

Step 4: the distribution. Step 1 reads

$$\begin{aligned} \left[1-\pi+\pi\left(1-F^s(r)\right)\right](r-c^{avg,s}) &= \Pi^s \qquad \text{on the pooling region,} \\ (1-\alpha^s)\left[1-\pi+\pi\left(1-F^s(r)\right)\right](r-c_b) &= \Pi^s \qquad \text{above } \bar{r}^g. \end{aligned}$$

Each is linear in $F^s(r)$ with coefficient $-\pi \neq 0$, so it has one solution at every rate, the expression in Proposition 4.2, and the lower bound of the support is the rate at which that solution reaches zero. Any two equilibria share a level of profit, a support and a distribution, and are the same equilibrium.[18]

Step 5: the rates the support leaves out. A rate above the top of the support is refused by every firm the bank meets and earns zero. In the pooling regime a rate between $\bar{r}^g$ and $\bar{r}^b$ meets $F^s(r) = 1$ and is accepted by captive bad types alone, so it earns $(1-\alpha^s)(1-\pi)\,(r-c_b)$, which net of Π^s is

$$(1-\alpha^s)(1-\pi)\,(r-\bar{r}^g) - \alpha^s(1-\pi)\,(\bar{r}^g - c_g)\,.$$

The first term is the gain on bad types, the second the loss of every good type. The sum rises with r and equals $-(1-\pi)\phi^s$ at $r = \bar{r}^b$, so it is negative on the whole interval when $\phi^s > 0$. In the hybrid regime a rate in the gap, above $\bar{r}^g$ and below the bad-only branch, is accepted by bad types alone and faces $F^s(r) = F^s_{\text{pool}}(\bar{r}^g)$, so it earns $(1-\alpha^s)\left[1-\pi+\pi\left(1-F^s_{\text{pool}}(\bar{r}^g)\right)\right](r-c_b)$. This rises with r and reaches Π^s at $c_b + (\bar{r}^g - c^{avg,s})/(1-\alpha^s)$, the bottom of the bad-only branch, so every rate in the gap earns strictly less. In the bad-only regime the whole pooling region lies below the support: $c^{avg,s} \leq c_b$ and the regime condition $\bar{r}^g - c^{avg,s} \leq (1-\pi)(1-\alpha^s)\left(\bar{r}^b - c_b\right)$ give $\bar{r}^g < c_b + (1-\pi)\left(\bar{r}^b - c_b\right)$, the bottom of the support. A bank posting there faces $F^s(r) = 0$: nobody undercuts it, both types accept, and it earns at most $\bar{r}^g - c^{avg,s}$, which the same regime condition puts below $\Pi^s = (1-\pi)(1-\alpha^s)\left(\bar{r}^b - c_b\right)$. The inequality that defines the regime is the inequality that rules out the deviation.

[18] The definition of equilibrium restricts the two banks to the same distribution for each signal. The restriction binds only through the support: if both banks post on the same support, each bank's indifference condition determines the rival's distribution, and the two conditions are the same equation, so the two distributions coincide.

D Numerical Identification Checks for the Calibration Exercise

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D.1 From the offered distribution to the observed distribution

The equilibrium distribution $F(r)$ characterized in Section 5 is each bank's offered distribution. The observed distribution in the data is the one actually accepted by borrowers, after shoppers choose only one of two offers. The probability of acceptance is the probability that a rate is posted $f(r)$ and the probability that the offer is accepted, $w(r) = (1-\pi)\cdot 1 + \pi(1-F(r)) = 1-\pi F(r)$, since a captive always accepts while a shopper accepts only if the rival's draw exceeds r, an event of probability $1-F(r)$. The density of accepted and realized rates is

$$g(r) \;\propto\; f(r)\,w(r) \;=\; f(r)\,(1-\pi F(r))\,, \tag{16}$$

where $f = F'$. The normalizing constant is $1-\pi/2$, because $\int f(r)\,(1-\pi F(r))\,dr = 1-\pi/2$ as, per bank, given symmetry, only half of the π offers will be accepted in expectation. Or, in oher words, the clients effectively served by banks are $1-\pi+\pi/2$. Checking the function at its limits: at $\pi = 0$, $g = f$ exactly (with no search, offered and realized rates coincide); at $\pi = 1$, $g(r) \propto f(r)(1-F(r))$, the density of the minimum of two draws, since every firm then shops.

For any moment $\mathbb{E}[r^k]$, this reweighting can be written as a correction to the moment computed directly under F:

$$\int r^k f(r)\,w(r)\,dr \;=\; \underbrace{\int r^k f(r)\,dr}_{\text{moment under } F} \;-\; \pi\,\underbrace{\int r^k f(r)F(r)\,dr}_{\text{correction term}}, \tag{17}$$

and the correction term admits a closed form in every equilibrium regime of the model (full pooling, hybrid, and bad-only), obtained by symbolic integration and verified against direct numerical quadrature. Throughout the calibration exercise of this section, the three targeted moments (mean, variance, and the rate–default correlation) are computed under $g(r)$ – properly reweighted by $w(r)$ and renormalized by $1-\pi/2$ – rather than under $F(r)$ directly, since it is $g(r)$, not $F(r)$, that corresponds to the distribution of rates actually observed in loan-level data.

Section 5.3 summarizes three complementary numerical checks used to assess whether the parameter vector recovered by my calibration procedure (Table 1) for each rating class is a well-isolated global optimum of the GMM objective, given that the model's regime-switching structure precludes a closed-form identification proof.

D.2 Local identification: rank and condition number of the moment Jacobian

At the estimated parameter vector $\hat{\theta} = (\hat{\pi}, \hat{q}, \hat{\bar{r}}_g)$ for each rating class, I compute the 3×3 Jacobian of the three matched moments (mean, variance, and rate–default correlation) with respect to the three parameters by centered finite differences, with a one-sided fallback near a parameter bound (e.g. when $\hat{q}$ is at or close to 1). I report both the raw Jacobian and an elasticity-scaled version – entry (i,j) rescaled by $\hat{\theta}_j$ divided by the observed value of moment i (or by 0.01 when the observed correlation is within 0.01 of zero, to avoid dividing by a near-zero target) – so that condition

numbers are comparable in units across moments of very different natural scale (correlations of order 10^{-2} against rates and variances of order 1). Table 3 reports, for the elasticity-scaled Jacobian at each rating class's $\hat{\theta}$, the rank, the condition number (ratio of largest to smallest singular value), and the parameter with the largest loading on the singular vector associated with the smallest singular value – the "weakest identified direction."

The Jacobian has full rank (3/3) in every one of the eight rating classes, ruling out exact local non-identification. Condition numbers, however, range from 3.9 (rating 7) to 1097.0 (rating 3+), and the weakest direction loads overwhelmingly on q in six of the eight classes; the exceptions are rating 5, where π dominates, and rating 5+, where the loading is split between q and $\bar{r}_g$. The pattern is most pronounced for rating classes whose observed rate–default correlation is close to zero (ratings 6 and 7, and to a lesser extent the safest classes): correlation is the moment most informative about screening precision q, so a target correlation close to zero leaves q comparatively unpinned by the data, even though the model remains exactly locally identified in the formal rank sense.

D.3 Global search: dense grid visualization of the parameter space

Local curvature alone cannot rule out a second, disconnected minimum elsewhere in the admissible parameter cube $\pi \in [0.05, 0.95]$, $q \in [0.5, 1.0]$, $\bar{r}_g \in [0.5\,\bar{r}_{\text{mean}}, 0.98\,\bar{r}_b]$. I complement the Jacobian check with a dense grid search (tens of thousands of parameter combinations per rating class) evaluating the GMM objective at every point, and visualize, for each rating class, the subset of the cube whose implied moments fall within a chosen multiple of the best-achieved objective, as an interactive three-dimensional scatter colored by objective value, with the reported estimate marked separately. In every rating class, this "acceptable fit" region forms a single connected neighborhood around the reported estimate rather than several disjoint clusters – visual evidence against multiple, unrelated local optima, and a small fraction of the cube in every case (between 0.02% and 0.49% of grid points). Consistent with the Jacobian findings above, the matched region is elongated primarily along the q dimension for the rating classes with high condition numbers (3+ through 4); $\bar{r}_g$ is comparatively tightly pinned in every rating class, while q remains comparatively loosely pinned even for ratings 6 and 7, whose low condition numbers reflect a well-conditioned, but locally flat, objective in the q direction – consistent with their near-zero observed correlation leaving screening precision the least informative parameter to match.

As a complementary exercise, I also compute, for each rating class, the three surfaces in $(\pi, q, \bar{r}_g)$-space along which the model exactly matches one observed moment at a time (mean, variance, or correlation individually). With three free parameters, each single scalar equation generically defines a two-dimensional surface; two such surfaces crossing is generically a one-dimensional curve, and all three crossing is generically an isolated point – the geometric counterpart of the Jacobian's rank condition. This exercise reveals that for rating 3+, the observed variance itself lies outside the model's achievable range anywhere in the admissible cube: the maximum variance the model can generate there, at any $(\pi, q, \bar{r}_g)$, is approximately 0.28, against an observed variance of 0.40. More generally, model-implied variance falls short of the data in every rating class (Table 2), so the three moment-matching surfaces do not intersect exactly anywhere in the admissible region. This is the numerical counterpart of the two-type-versus-continuum discussion in Section 5: with only

two latent borrower types, the maximum rate dispersion the model can generate is mechanically bounded, and this bound binds most severely for the safest rating classes.

D.4 Cross-checking the global optimum across equilibrium regimes

Finally, I exploit the fact that the equilibrium's regime-switching structure is fully characterized by two threshold conditions, ϕ_{good} and ϕ_{bad} – one for the pool of borrowers who receive a favorable screening signal, one for the pool who receive an unfavorable signal – each independently placing that pool in one of three regimes (full pooling, hybrid, or bad-only lending), for nine theoretically possible regime pairs. Rather than assume ex ante which pairs are empirically relevant, I tag every point of the parameter grid with its actually realized $(\phi_{\text{good}}, \phi_{\text{bad}})$ regime pair. Only a subset of the nine pairs is ever realized within the admissible parameter region for a given rating class – six of nine for ratings 3+ through 5+, and only three of nine (all with the good-signal pool fully pooling) for ratings 6 and 7 – and the realized pairs always respect a simple ordering in which the good-signal pool is never in a strictly "less pooled" regime than the bad-signal pool. This is itself a useful by-product of the exercise: the theoretically possible nine-way regime space collapses, empirically, to a much smaller and highly structured set.

For each rating class and each empirically realized regime pair, I re-solve the GMM objective with the search restricted to that regime pair, implemented by penalizing, within the local optimizer, any candidate parameter vector whose realized regime pair differs from the target. In every rating class, the best objective value achieved across *all* admissible regime pairs coincides with the parameter vector reported by my baseline, regime-unconstrained estimation (Table 1), confirming that the baseline optimizer converges to the true global optimum of the objective function rather than a locally optimal point within a single equilibrium regime.

D.5 An alternative specification: fixing π at an observable proxy for search intensity

As a further check on the calibration, I re-estimate q and $\bar{r}_g$ for each rating class holding π fixed at an external, model-free proxy for search intensity, rather than estimating it jointly: specifically, I set π equal to the share of firms in that rating class holding at least two active banking relationships, averaged over the sample period – the same underlying statistic cited in Section 5 as broadly consistent with the freely estimated $\hat{\pi}$ (on average 60% of firms across classes maintain at least two bank relationships). With π fixed at this value and r_{CB} still fixed at 1, I re-estimate $(q, \bar{r}_g)$ by the same grid-search-plus-L-BFGS-B GMM procedure used for the baseline (Table 1).

Table D.11 reports the results. The externally measured π is lower, and more compressed across rating classes, than the freely estimated $\hat{\pi}$ (0.40–0.68 against 0.71–0.77). Fit deteriorates in all eight rating classes: the objective value rises from 0.41 to 0.62 at rating 3+, and from essentially zero to 0.47 at rating 7.

Where the deterioration falls depends on the class. In the five safest classes the variance error barely moves and the extra loss lands on the mean: the model overshoots the class mean rate by 21–33% here, against 5–16% in the baseline. In the three riskiest classes both moments worsen at once - the share of observed variance the model reproduces falls from 81%, 80% and 94% to 67%, 30% and 45%, and the mean overshoots by 33–42%. Because rate dispersion in this model scales

with $\pi \times (\bar{r}_g - c_{\text{avg}})$, a lower π compresses the dispersion the model can generate for a given $\bar{r}_g$, and the optimizer compensates by raising $\bar{r}_g$, which lifts the mean.

The constraint binds hardest at rating 3+, the one class in which the observed variance lies outside the model's achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed π. There $\hat{q}$ falls to its lower bound of 0.5 - no screening at all - against 1.00 in the baseline: at a low π the model buys what dispersion it can from pooling rather than from separation. In the other seven classes $\hat{q}$ stays within 0.05 of the baseline. The observed mean is unreachable in the four safest classes (3+, 3, 4+ and 4).

I read this exercise as informative rather than as grounds to prefer one specification over the other. The two-bank-relationship share is plausibly a lower bound on effective search intensity: maintaining a second banking relationship is necessary but not obviously sufficient for the kind of active rate shopping that π represents in the model, and the freely estimated $\hat{\pi}$ being somewhat higher is consistent with that interpretation. The exercise does, however, sharpen the point made throughout this appendix: the amount of rate dispersion the two-type model can generate is highly sensitive to π, and matching the variance moment closely requires π at or above roughly the level the baseline estimation recovers.

Table D.11: Alternative Specification: π Fixed at the Share of Firms with At Least Two Bank Relationships

Rating	π (fixed)	$\hat{q}$	$\hat{\bar{r}}_g$	Obj. (fixed π)	Obj. (baseline)	Var. reachable?
3+	0.600	0.500	3.055	0.62	0.41	No
3	0.651	0.985	3.709	0.45	0.35	Yes
4+	0.651	0.986	4.609	0.21	0.15	Yes
4	0.675	1.000	5.080	0.25	0.19	Yes
5+	0.670	0.999	5.417	0.25	0.20	Yes
5	0.562	0.991	6.662	0.22	0.04	Yes
6	0.399	0.552	6.347	0.64	0.04	Yes
7	0.426	0.554	8.895	0.47	0.00	Yes

Notes: π is fixed at the share of firms in each rating class with at least two active bank relationships, averaged over the sample period; q and $\bar{r}_g$ are then re-estimated with r_{CB} fixed at 1, as in the baseline. "Obj. (baseline)" reproduces the objective value from Table 1; both runs match the same moments, so the two columns are directly comparable. "Var. reachable?" indicates whether the observed variance falls inside the model's achievable range anywhere in the admissible $(q, \bar{r}_g)$ box at this fixed, lower π.

E Robustness: counting defaults as bad types

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The baseline calibration of Section 5 labels a firm bad when its Banque de France rating deteriorates over the three years following origination. Firms that default outright are not counted as bad. This appendix repeats the exercise under the alternative definition, in which the bad group also contains firms that default. Everything else - the rate regression, the residuals, the targeted mean and variance, $\bar{r}_b$, and p_g - is held fixed, so the two runs differ only through the three quantities the reclassification touches: the share of bad types μ_b, their success probability p_b, and the rate–default correlation.

Table E.12 shows how much those three move. For ratings 3+ through 5+ the reclassification is almost immaterial: μ_b shifts by at most 0.06 and p_b not at all. In the three riskiest classes it is large: μ_b roughly quintuples at rating 6 and sextuples at rating 7, and p_b collapses, because a firm labelled bad on the strength of its own default has, by construction, a default probability of one.

Table E.12: Inputs that change under the alternative bad-type definition

	μ_b		p_b		Rate–default corr.	
Rating	Baseline	With default	Baseline	With default	Baseline	With default
3+	0.846	0.846	0.997	0.997	0.025	0.023
3	0.749	0.751	0.996	0.996	0.033	0.034
4+	0.739	0.745	0.992	0.992	0.032	0.022
4	0.671	0.691	0.981	0.981	0.048	0.037
5+	0.491	0.549	0.965	0.965	0.048	0.044
5	0.268	0.415	0.908	0.630	0.017	0.049
6	0.066	0.344	0.784	0.200	0.000	0.068
7	0.069	0.446	0.630	0.200	0.000	0.000

Notes: "Baseline" is the definition used in Section 5, in which bad types are firms whose rating deteriorates; "With default" adds firms that default. For ratings 6 and 7 the raw data give a default frequency of exactly one in the bad group, which would imply an infinite cost of lending $c_b = r_{CB}/p_b$; p_b is therefore floored at 0.2, so the reported values for those two classes are the floor rather than an estimate.

Table E.13: Search frictions and information by rating class, defaults counted as bad

Rating	$\hat{\pi}$	$\hat{q}$	$\hat{\bar{r}}_g$	Info Contr. (%)	Obj. Value $Q(\theta)$
3+	0.772	0.988	3.461	4.58	0.41
3	0.759	1.000	3.703	4.76	0.35
4+	0.737	0.978	4.609	2.83	0.16
4	0.758	1.000	5.092	3.81	0.19
5+	0.750	1.000	5.416	2.99	0.19
5	0.737	0.757	6.662	1.01	0.06
6	0.767	0.613	7.057	12.79	0.24
7	0.754	0.526	9.350	0.37	0.20

Notes: Columns as in Table 1, which reports the same quantities under the baseline definition.

The estimates are unchanged where the definition is. For ratings 3+ through 5+, which account for the bulk of the sample, the two runs agree almost exactly. Comparing Table E.13 with Table 1, across those five classes $\hat{\pi}$ differs by at most 0.001, $\hat{q}$ by at most 0.021, and the objective value by at most 0.005; the fitted means and variances in Table E.14 reproduce Table 2 to two

Table E.14: Model fit, defaults counted as bad

	Mean		Variance		Correlation	
Rating	Data	Model	Data	Model	Data	Model
3+	1.67	1.94	0.406	0.153	0.023	0.023
3	1.83	2.08	0.443	0.188	0.034	0.034
4+	2.40	2.53	0.553	0.334	0.022	0.022
4	2.37	2.63	0.730	0.426	0.037	0.037
5+	2.57	2.81	0.867	0.494	0.044	0.043
5	3.36	3.51	0.985	0.740	0.049	0.049
6	3.63	4.15	1.137	0.600	0.068	0.070
7	4.79	5.39	1.843	1.064	0.000	0.003

Notes: Moments defined as in Table 2.

digits. The estimate of competition, which is the paper's main quantitative result, is therefore not an artifact of how the bad type is defined: $\hat{\pi}$ lies in $[0.74, 0.77]$ here against $[0.71, 0.77]$ in the baseline.

The three riskiest classes are where the definitions part, and where the alternative is least reliable. At rating 5, $\hat{q}$ falls from 0.94 to 0.76; at rating 6 it rises from 0.53 to 0.61. The fit deteriorates sharply: the objective rises from 0.04 to 0.24 at rating 6 and from 0.00 to 0.20 at rating 7, the model mean overshoots the data by 14% and 13% against 1% and 0% in the baseline, and the share of observed variance the model reproduces falls from 80% and 94% to 53% and 58%.

Two features of the exercise explain this, and both argue for keeping the baseline definition in the text. First, the definition is partly circular in these classes: a firm counted as bad because it defaulted has a default probability of one by construction, so p_b measures the labelling rule rather than the risk of the bad type. Second, the floor imposed on p_b to keep c_b finite binds at exactly 0.2 for ratings 6 and 7, which fixes $c_b = r_{CB}/p_b = 5$ against an upper bound $\bar{r}_b$ of 7.2 and 9.5. That gap is wide enough for information to generate more dispersion than the data contain, which is what produces the 12.8% information contribution at rating 6 - the one figure in either run that exceeds the 5% ceiling reported in Section 5. Excluding rating 6, the largest information contribution here is 4.8%, in line with the baseline. The same floor also makes the implied cross-subsidy at rating 6 unusually large: bad types pay an expected rate of 4.2 against a cost of $c_b = 5$, so the bank runs a loss of 0.8 per bad-type loan, covered by the surplus it earns on the good types pooled with them. I therefore read the rating-6 figure as an artifact of the floor rather than as evidence that screening contributes more to dispersion than the baseline implies.

Table E.15: Redistribution from a perfect signal, defaults counted as bad: percent of the rating class's total surplus

Rating	Bad types	Banks	Good types
3+	-0.893	0.883	0.010
3	-0.881	0.868	0.012
4+	-0.817	0.795	0.020
4	-0.747	0.693	0.052
5+	-0.640	0.547	0.094
5	-1.901	0.430	1.471
6	-12.231	0.396	11.834
7	-9.450	0.506	8.942

Notes: change in each party's expected surplus when signal precision moves from $q = 0.5$ to $q = 1$, as a percentage of the class's maximum total surplus. Rows sum to zero: every class is in full pooling at $q = 0.5$, so no firm is rationed at either end and information redistributes without creating surplus.

Table E.16: Redistribution from a perfect signal, defaults counted as bad: percent of each party's own surplus

Rating	Bad types	Banks	Good types
3+	-1.65	2.43	0.10
3	-1.87	2.29	0.08
4+	-1.83	1.94	0.14
4	-1.73	1.81	0.28
5+	-1.89	1.39	0.35
5	-7.61	1.04	4.35
6	-55.09	1.06	29.28
7	-33.82	1.30	27.02

Notes: the same changes, each divided by that party's own surplus at $q = 0.5$ rather than by the class total. Rows do not sum. This is the denominator relevant to a bank weighing the cost of acquiring precision, and to a borrower judging the size of the repricing.

F Stability of the calibrated precision

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Here I derive the two conditions of Section 6 and I evaluate them numerically in each rating class j.

Let $C_s \equiv C_s(q{=}1) - C_s(0.5)$ be the price of a perfect signal over no signal. A bank is of measure zero, so a deviation leaves unchanged both the rivals' rate distributions and the population pool (μ_g, μ_b). I restrict to deviations to and from the extremes of perfect signal and no signal, as the calibration returns these two corners. Precision affects the cost of lending via borrowers' risk: $c_\theta(1) = (1-\delta_j)c_\theta(0.5)$ as in (5), with saving $\kappa_\theta = \delta_j c_\theta(0.5)$ per unit lent, and $\Delta r \equiv \bar{r}^b - \bar{r}^g$.

F.1 Deviation from the uninformed market

The market sits at $q = 0.5$ and banks lend at the un-monitored pooled cost $c_0^{avg} \equiv \mu_g c_g(0.5) + \mu_b c_b(0.5)$. The equal-profit condition of Section 4.3 gives

$$\bar{\Pi} = (1-\pi)\left(\bar{r}^g - c_0^{avg}\right), \qquad \underline{r} = (1-\pi)\bar{r}^g + \pi c_0^{avg}, \qquad 1 - \pi F(r) = \frac{\bar{\Pi}}{r - c_0^{avg}}.$$

The deviator acquires a signal at cost C_s: it recognizes each type and lends at the monitored cost $c_\theta \equiv c_\theta(1)$. It chooses the rate distribution that maximizes its profits, given the firms' type and the rivals' offers.

Conditional on meeting a good type, the deviator makes maximum profits on the pooling region, where it earns $\Pi_g(r) = \mu_g \bar{\Pi}\,(r - c_g)/(r - c_0^{avg})$. Note that profits on the pooling region are not flat: the deviator has a different cost of lending $c_g < c_g(0.5)$ with respect to the uninformed rivals; I expect a corner solution. The slope in r carries the sign of $c_g - c_0^{avg} = -\left[\mu_b(c_b - c_g) + \kappa^{avg}\right] < 0$, with $\kappa^{avg} \equiv \mu_g\kappa_g + \mu_b\kappa_b$: the deviator undercuts the whole support and posts $\underline{r}$, which every good type accepts.

For bad types, assume the deviator is also posting on the overlapping region, with profits $\Pi_b(r) = \mu_b \bar{\Pi}\,(r - c_b)/(r - c_0^{avg})$. The sign of the slope is ambiguous, so I compute it at the calibrated parameters, and it is positive. The deviator has an incentive to increase the rate and sells to captives only, and posts $\bar{r}^b$.[19] The deviator's profits are:

$$\Pi^{dev}(1) = \mu_g\left(\underline{r} - c_g\right) + \mu_b(1-\pi)\left(\bar{r}^b - c_b\right), \qquad \underline{r} - c_g = (1-\pi)(\bar{r}^g - c_g) + \pi\left[\mu_b(c_b - c_g) + \kappa^{avg}\right],$$

and defining G_j as the net $\Pi^{dev}(1) - \Pi^*(0.5) = \Pi^{dev}(1) - \bar{\Pi}$ gives the stability condition in (6) (Condition A). The gain is made of three terms: the surplus from winning more good types, the surplus lost from pooling bad types with good ones at low rates, and the monitoring saving. The monitoring effect reinforces the gain of the deviator with respect to the non-monitoring case.

In each class j, holding constant the calibrated parameters, Condition A is linear in δ_j:

$$G_j(\delta_j) = G_{0,j} + \delta_j S_j \le C_s \quad \Longleftrightarrow \quad \delta_j \le \frac{C_s - G_{0,j}}{S_j}, \tag{18}$$

[19] The slope is $\mu_g(c_b - c_g) - \kappa^{avg}$. The deviator prefers to undercut if monitoring is strong enough to push the deviator's own bad-type cost below the uninformed market average cost.

where $G_{0,j} = \pi\mu_g\mu_b\Delta c_0 + (1-\pi)\mu_b\Delta r$ is the gain independent of monitoring, $\Delta c_0 = c_b(0.5) - c_g(0.5)$ and $S_j = c_0^{avg}\left[(1-\pi) + \pi\mu_g\right] - \pi\mu_g\mu_b\Delta c_0$.

F.2 Deviation from the informed market

The market sits at $q = 1$: banks are perfectly informed, they price off the monitored costs, so $\bar{\Pi}_\theta = (1-\pi)(\bar{r}^\theta - c_\theta)$ and the type-θ acceptance probability is $A_\theta(r) = 1 - \pi F^\theta(r) = \bar{\Pi}_\theta/(r - c_\theta)$, on $\underline{r}^\theta, \bar{r}^\theta$.

I consider a bank that stops acquiring the signal, and saves the cost C_s. Its clients being riskier, it lends at $c_\theta + \kappa_\theta$, and competes with the offers of the informed banks. If the deviator posts a rate on the overlapping region $\underline{r}^b, \bar{r}^g$, on type θ borrowers it earns:

$$A_\theta(r)\,(r - c_\theta - \kappa_\theta) = \bar{\Pi}_\theta - \kappa_\theta A_\theta(r).$$

The first term is profits before introducing monitoring, which are flat by the equal-profit condition: an uninformed bank posting inside the overlap collects the informed profit. The second is the monitoring penalty, paid on every loan actually made and therefore smallest where acceptance is lowest. The deviator cannot distinguish types and sets one rate for all. Due to the monitoring cost, profits are increasing in r and the deviator chooses either to price at the top of the overlap or in the bad-only region. At the top of the overlap, $r^d = \bar{r}^g$, where $A_g = 1 - \pi$ and $A_b = \bar{\Pi}_b/w$ with $w \equiv \bar{r}^g - c_b$; the deviator's profits are lower by the higher cost of lending. Otherwise, it posts anywhere on the bad-only region, where good types reject, and profits on bad types are the profits of informed banks net of the higher cost penalty. In general

$$L_j = \min\left\{ \mu_g\kappa_g(1-\pi) + \mu_b\kappa_b\frac{\bar{\Pi}_b}{w}\ ,\quad \mu_g\bar{\Pi}_g + \mu_b(1-\pi)\kappa_b \right\}. \tag{19}$$

The former is the loss L_j of (7). With no monitoring both branches collapse to $L_j = 0$, which is the statement in Section 6 that dropping the signal is free.

In each class j, holding constant the calibrated parameters, I aggregate the terms that do not depend on δ_j and I write $t \equiv \delta_j/(1-\delta_j)$, so (19) becomes $L_j = \min\{K_j t,\ M_j + K'_j t\}$ with

$$K_j = \mu_g c_g(1)(1-\pi) + \mu_b c_b(1)\frac{\bar{\Pi}_b}{w}, \qquad K'_j = \mu_b(1-\pi)c_b(1), \qquad M_j = \mu_g\bar{\Pi}_g.$$

The equilibrium is stable if the loss is larger than the cost saving. The calibration parameters are consistent with L_j on the pooling branch, where the constraint is written:

$$C_s \le L_j(\delta_j) \quad\Longleftrightarrow\quad \delta_j \ge \frac{C_s}{K_j + C_s}. \tag{20}$$

This means that a high enough monitoring cost is necessary to satisfy the condition. At the calibrated parameters, as δ_j increases, the bad-only branch overtakes the pooling one - from $\delta_j = 0.68$ in class 3+ and 0.79 in class 4, well above the values below - and both branches rise in δ_j, so for any given cost there is one minimum threshold.

F.3 The two conditions at the calibrated parameters

Table F.17 collects the coefficients of the two conditions above, (18) and (20). The two binding classes read $\delta_7 \leq (C_s - 0.0291)/0.996$ and $\delta_{3+} \geq C_s/(0.234 + C_s)$.

Table F.17: Coefficients of the two stability conditions, by rating class

Rating	Role	c_b	$\bar{\Pi}_b/w$	K_j	M_j
3+	screening	1.003	0.235	0.234	0.087
3	screening	1.004	0.248	0.247	0.164
4+	screening	1.008	0.269	0.269	0.246
4	screening	1.020	0.252	0.252	0.324
5+	screening	1.036	0.258	0.260	0.562
5	screening	1.102	0.278	0.284	1.120
				$G_{0,j}$	S_j
6	not screening	1.275	-	0.014	0.976
7	not screening	1.587	-	0.029	0.996

Notes: computed at the estimates of Table 1 and at the calibrated μ_b, p_g, p_b and $\bar{r}^b$, with $r^{CB} = 1$. "Screening" classes are those the calibration estimates at $\hat{q} \approx 1$, to which Condition B applies; the other two are estimated at $\hat{q} = 0.53$ and Condition A applies to them. c_b is the monitored cost $c_b(1)$ for a screening class and the un-monitored $c_b(0.5)$ for the other two, since each class reports the default probability of its own equilibrium. K_j and M_j are the coefficients of L_j in $t = \delta_j/(1-\delta_j)$; $G_{0,j}$ and S_j the intercept and the slope of G_j in δ_j. K'_j, the slope of the bad-only branch of (19), runs from 0.080 at rating 5 to 0.195 at rating 4+ and never binds in the relevant range.

Each class also faces a feasibility bound: a screening class needs $c_b(0.5) < \bar{r}^g$ for its pooling regime to survive, which caps δ_j at $1 - c_b(1)/\bar{r}^g$, and a non-screening class needs $p_g(1) = \hat{p}_g/(1-\delta_j) \leq 1$, which caps it at $1 - \hat{p}_g$. Table F.18 reports the resulting intervals at $C_s = 0.0483$, the cost at which the two bands of Fig. 9 come closest. Every interval is non-empty, so all eight estimates are stable at that cost, but the floors of the first six lie an order of magnitude above the ceilings of the last two.

Table F.18: Monitoring strength consistent with the estimated precision, at $C_s = 0.0483$

Rating	Role	From	To	Upper end set by
3+	screening	**0.171**	0.710	pooling regime, $c_b(0.5) < \bar{r}^g$
3	screening	0.164	0.729	pooling regime
4+	screening	0.152	0.781	pooling regime
4	screening	0.161	0.798	pooling regime
5+	screening	0.157	0.808	pooling regime
5	screening	0.145	0.835	pooling regime
6	not screening	0	**0.0193**	$p_g(1) \leq 1$; Condition A alone allows 0.0351
7	not screening	0	**0.0193**	Condition A; $p_g(1) \leq 1$ allows 0.0347

Notes: lower ends from (20), upper ends from (18) and from feasibility, at the estimates of Table 1. The binding pair is $\delta_j = 0.171$ in the six screening classes against $\delta_j = 0.0193$ in ratings 6 and 7, a ratio of 8.8. Putting the screening classes at their lower ends and the other two at their upper ends is what minimizes the spread, so these two endpoints answer both questions at once: which condition binds, and how uniform the technology can be. At those values the saving per unit lent is $\kappa_g \in [0.206, 0.210]$ and $\kappa_b \in [0.207, 0.227]$ in the screening classes, raising the bad type's repayment probability by 15.5 to 17.0 points, against $\kappa_g = 0.020$, $\kappa_b \in [0.025, 0.031]$ and a gain of 1.2 to 1.5 points in ratings 6 and 7.

$C_s = 0.0483$ is also where the ceiling on the two non-screening classes changes hands. Figure F.22 resolves that band: class 7's Condition A binds below 0.0483, class 6's probability ceiling above it,

and the ratio between the two bands is smallest exactly at the handover. Raising C_s past it buys no further uniformity, because class 6's ceiling is flat in C_s. Lowering C_s below 0.0291 leaves class 7 with no admissible δ_j at all.

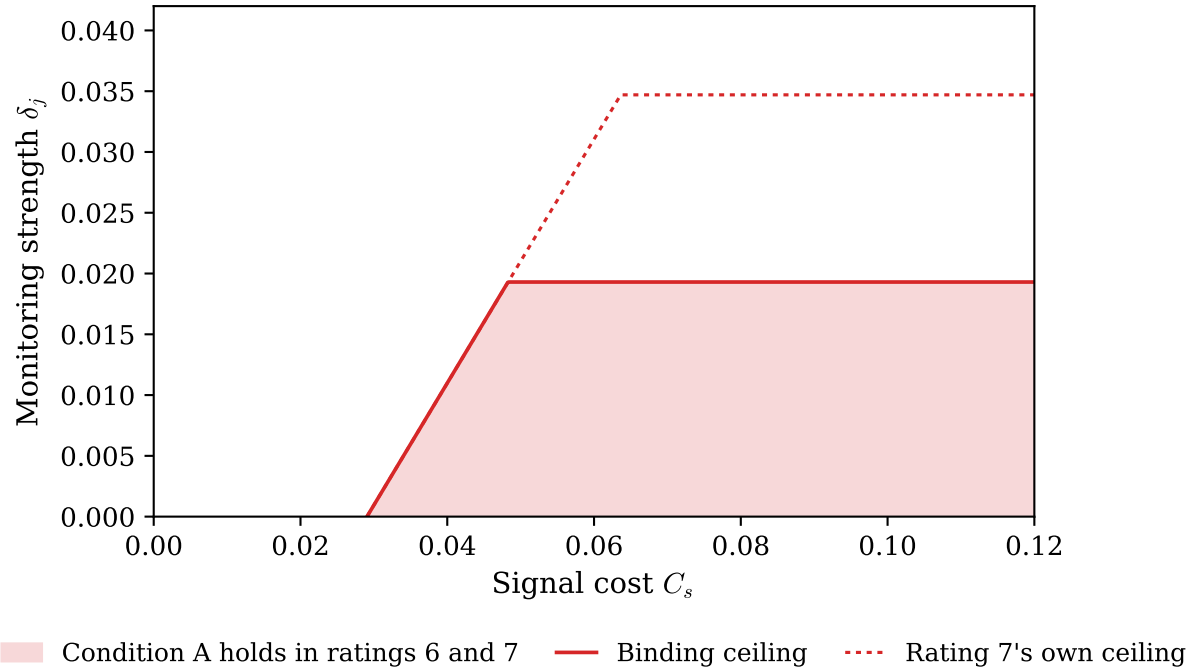


Figure F.22: **Ceiling on monitoring strength in the two non-screening classes, against the cost of the signal.** The lower band of Fig. 9, on a scale that resolves it; axes as there. The solid line is the binding ceiling, the lower of class 7's Condition A and class 6's probability cap $p_g(1) \leq 1$. The dotted line is class 7's own ceiling, which carries on above the band once class 6 takes over at $C_s = 0.0483$ and reaches its own probability cap of 0.0347 at $C_s = 0.0636$.

F.4 Deviations to intermediate precisions

The two conditions compare the corners, which is what the calibration delivers, but a cost *schedule* in the window they define also blocks every deviation in between. Let $\Delta_j(q)$ be the gain to a deviator at precision q against a market at $q = 0.5$, and $L_j(q)$ the loss from dropping to q against a market at $q = 1$, with Δ_j increasing and L_j decreasing. For any C_s satisfying both conditions,

$$C_s(q) - C_s(0.5) \;=\; \max\left\{ \max_{j \text{ not screening}} \Delta_j(q) \;,\;\; C_s - \min_{j \text{ screening}} L_j(q) \;,\;\; 0 \right\}$$

is increasing, equals 0 at $q = 0.5$ and C_s at $q = 1$, and by construction exceeds $\Delta_j(q)$ in every non-screening class while leaving less than $L_j(q)$ on the table in every screening class. One number, and one schedule through it, therefore block every deviation in every class.